Entry into Innovation Areas

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Abstract

This paper investigates the entry decisions of U.S. electronics firms across innovation areas. I apply topic modeling to over 630,000 patents to endogenously define innovation areas and merge those areas with market-implied patent valuations. Firms occupy several areas at once, and the number of firms differs widely across areas. An entry game, estimated by moment inequalities, finds that additional rivals reduce expected profit and that firms gain from occupying more areas at once. In the case study of semiconductor, a payment based on a quarter of predicted value shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

Keywords: Entry; Innovation; Patents; Technological diversification. **JEL:** L25, O31, O34

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1 Introduction

Firms operating in the technology sector face a strategic problem: deciding which specific innovation and technological areas to enter and invest their resources. This paper builds an empirical model to untangle the economic determinants that govern these multi-area entry decisions. Using a panel of publicly listed firms in the United States electronics industry from 1990 to 2019, I quantify the congestion and yearly occupancy cost, measured against the gain from occupying more areas at once.

To analyze these dynamics, I construct a dataset that merges financial information from publicly listed electronics firms with rich, unstructured patent texts. I apply Latent Dirichlet Allocation (LDA) to the text of over 630,000 patent titles and abstracts. Coupled with iterative grid search optimization, this approach allows the data to endogenously define twenty distinct innovation areas based on semantic word co-occurrences, topic uniqueness, and semantic stability. A patent can sit in several of those areas with explicit weights; then, that weights is mapped as the choice set of technological areas to the entry game. I then integrate market-implied patent valuations from Kogan et al. (2017) to assign an expected monetary value to these technological areas.

Using the innovation areas derived from LDA, I document stylized facts regarding the dynamics of the U.S. electronics sector. First, I find that multi-area bundling is a dominant feature of the industry. Innovation rarely occurs in isolated silos; the average patent spans 3.58 technological areas, and the median firm-year occupies five areas, and that firms compete in several distinct areas simultaneously. These empirical patterns suggest that entry is a complex portfolio optimization problem subject to economies or diseconomies of scope.

Second, the number of active competitors is heterogeneous across innovation areas. Areas such as "Integrated Circuit Design, Verification, and Testing" and "Audio and Speech-Signal Processing" are consistently occupied by many players. In contrast, hardware domains such as "Display Panel Tech (LCD/OLED)" are occupied by few firms. Over the same years, the mix of occupied areas moved. Competitor shares fell in "Thin-Film Materials, Deposition and Chemistry" and "Rotational Electromechanics and Magnetic Actuation," and rose in "Wireless Base-Station and Mobile Communications" and "Networking Protocols and Data Infrastructure." Patent value moved with that mix, toward "Mobile UI and Content Services," "Wireless Base-Station and Mobile Communications," and "Power-Supply and Power-Management Electronics."

To rationalize these facts, I build a static, complete-information, simultaneous-move entry game. In this model, a firm chooses a bundle of innovation areas to maximize its total profit in a given year. For each area it occupies, the firm expects to earn the value of its patents in that area, and the number of rival firms in the same area affects how much it earns. Furthermore, the firm has to pay an occupancy cost, i.e., a cost for every year it stays in the area. Since a firm typically occupies several areas at the same time, its total profit also includes a portfolio term. This portfolio term depends on the total count of occupied areas, capturing either the synergistic economies of scope

or the managerial coordination costs of maintaining a broad innovation portfolio. The model estimates answer a contemporaneous portfolio question[1]: which areas the firm occupies in a given year, and how congestion in that year shifts expected profit.

There are several challenges in estimating the model. First, because firms choose a bundle of areas, the strategy space explodes combinatorially. Second, strategic entry games generate multiple equilibria, which forces standard point identification methods to rely on arbitrary equilibrium selection rules. Finally, point identifying models with expected valuations requires the researcher to exactly specify the firm's unobserved internal information set, risking severe model misspecification. To overcome these issues, I abandon point identification and adopt a partial identification approach utilizing moment inequalities.

I construct these inequalities from the necessary conditions for a Nash equilibrium, using revealed preference arguments. Specifically, I evaluate the conditions where a firm unilaterally alters its observed portfolio by adding or dropping a subset of innovation areas, and by swapping one entered area for a non-entered area. By analyzing these specific deviations, the estimation isolates the marginal incentives that drive entry. In order to mitigate the measurement error issue in patent value, I project realized valuations onto several lagged firm covariates, and evaluate the inequalities at those fitted values.

First, I find that additional rivals reduce expected profit. Second, firms gain extra expected profit from occupying more areas at once: occupying an extra area raises expected profit by more when the firm already occupies other areas. I also find that treating extra areas as diminishing synergies is inconsistent with the observed portfolios in the data. Third, firms still leave many areas unentered. I find that the occupancy cost of one more area, counted in the gain from one pair of areas, can be zero and can be as large as about twelve. Take a firm that already occupies five areas and adds a sixth, then the gain from the five new pairs is five. Suppose the occupancy cost equals the gain from three pairs, then this occupancy cost is allowed. The gain exceeds the occupancy cost by two, and those two are a gain. Suppose now the occupancy cost equals the gain from twelve pairs, then this occupancy cost is also feasible. The gain is five and the occupancy cost is twelve, so subtracting the gain from the occupancy cost leaves a cost of seven.

In a numerical simulation, I try to mimic the areas covered by the U.S. CHIPS manufacturing incentive and the EU CHIPS Act. In particular, the coverage of those policies are memory, lithography and fabrication, packaging, and the transistor. A payment equal to one quarter of the predicted valuations shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

Related Literature.—This paper contributes to several strands of literature. First, it relates to the use of text to define markets and product spaces. Gentzkow et al. (2019) survey the methods,

[1] Appendix B.3 discusses what a dynamic treatment of first entry would require.

and Ash and Hansen (2023) review later algorithms. I build on Hoberg and Phillips (2016) and Eisfeld (2025) for continuous market definitions from filings of the listed companies and descriptions of products from startups in the software industry. For patent text, the closest work is topic modeling of recombination and diversification (Kaplan and Vakili, 2015; Teodoridis et al., 2022), breakthrough measurement from the text itself (Kelly et al., 2021), similarity and novelty from claims (Arts et al., 2018, 2021), joint patent-job-earnings text (Kalyani et al., 2025), transformer embeddings (Hain et al., 2022), and text measures of corporate innovation from analyst reports (Bellstam et al., 2021). I translate those textual outputs into state variables for the entry model, and I value each area with the grant-window returns of Kogan et al. (2017).

Second, I estimate an entry game with moment inequalities. This study is related to Bresnahan and Reiss (1991) where they relate the number of firms in a market to the profit of being there. That relation is similar to the congestion term here: profit in an innovation area depends on how many rivals occupy it. Furthermore, in this paper, the firm also chooses its set of areas for the year. Related literature also discuss that several equilibria can fit the same market, so the inequalities have to hold in all of them (Tamer, 2003; Ciliberto and Tamer, 2009). Pakes (2010) and Pakes et al. (2015) derive those inequalities using a revealed preference argument and demonstrate several applications. I use them to check the observed set against adding a few areas, dropping a few, and swapping one area for another. The information structure follows Dickstein and Morales (2018) and Morales et al. (2019): the firm chooses before an area-specific shock is realized. This paper is also related to other literature such as, Eizenberg (2014) where he studies which configurations firms offer in home computers, and Wollmann (2018) where he studies which truck models a manufacturer offers. I use the test statistic following the methods developed by Chernozhukov et al. (2019).

Finally, the findings speak to the limits of the firm and to technology spillovers. Jaffe (1986) and Bloom et al. (2013) measure spillovers in technology space. Aghion et al. (2005) document an inverted-U between competition and innovation, and Hall and Ziedonis (2001) document the semiconductor patenting surge. I find that additional rivals reduce profit. On scope, related work includes pharmaceutical research (Henderson and Cockburn, 1996), product-line dynamics (Klette and Kortum, 2004), recombination (Weitzman, 1998; Fleming, 2001), relatedness and diversity (Breschi et al., 2003; Miller, 2006), and multi-location research (Chikis et al., 2025). The bounds recovered from the model suggest that synergies rise with portfolio size. Rotemberg and Saloner (1994), Seru (2014), and Rawley and Simcoe (2010) stress agency and overload costs. The semiconductor case study also touches on the current debate on industrial policy (Juhász et al., 2024; Goldberg et al., 2024; Barwick et al., 2025; Bloom et al., 2019). In the case study, the rival count in each area is the sum of the other firms' entry, updated until that sum agrees with the portfolios.

2 Data, Methodology, and Stylized Facts

2.1 Data

I construct a firm-level panel covering the U.S. electronics industry from 1990 to 2019. Firm's financial and organizationaal information, such as total assets and R&D expenses, are sourced from the Compustat dataset. I constrain the sample to publicly listed firms categorized under the two-digit Standard Industrial Classification (SIC) code 36 (Electronics and Electrical Components). To ensure the consistency of the data, I require consecutive yearly observations and impute missing R&D and asset values with zero. To proxy for firm size, I use total assets, aligning with standard empirical finance conventions (Seru, 2014). This procedure yields a final sample of 16,653 firm-year observations distributed across 1,373 unique firms. Of these, 755 firms have at least one patent.

To capture the economic value and nature of these firms' innovations, I integrate the patent dataset from Kogan et al. (2017), which provides market-implied patent valuations (derived from stock price movements subsequent to patent issuances), forward citation counts, and primary patent classifications. I link this data to the Compustat utilizing a correspondence table, i.e., from permco/permno to gvkey, provided by Wharton Research Data Services (WRDS)[2]. The data is the intersection of Compustat electronics firms and the Kogan et al. (2017) valuations. Those valuations are only available for a listed firm with a traded return around the patent grant date, so startups and privately held innovators are outside the panel by construction. By focusing on two-digit SIC 36, it also naturally omits firms in another industry code who have electronics patents, and similarly non-electronics firms that enter these topics. Of the 1,373 sample firms, 755 have at least one patent.

Because topic modeling requires rich textual inputs, I augment the Kogan et al. (2017) data by collecting unstructured text, specifically patent titles and abstracts, directly from Google Patents. My final textual corpus consists of 631,369 patents covering the electronics industry across the timeline[3]. Table 1 summarizes the firm characteristics I use.

2.2 Methodology: Topic Modeling

To parse the textual corpus of patents into economically meaningful dimensions, I employ Latent Dirichlet Allocation (LDA; Blei et al., 2003), treating each patent as a probabilistically weighted mixture of underlying "topics" or innovation areas. Conceptually, this mirrors the clustering methodologies frequently utilized in the literature to define product market based on business descriptions (e.g., Eisfeld, 2025), but it applies a sophisticated generative probabilistic model engineered for unstructured text.

[2]For an extensive discussion detailing the nuances of dynamically matching Compustat and USPTO entities, see Arora et al. (2021) and Dyevre and Seager (2023).

[3]A negligible fraction of patents (0.5%) were omitted from the final corpus due to missing records or irretrievable web elements.

Table 1: Summary statistics of firm characteristics

Variable	Statistic	Note
Number of firms (full sample)	1,373 (100%)	
Firms with at least one patent	755 (55%)	
R&D (millions USD, real 2012)	161.39	Average for patenting firms
Assets (millions USD, real 2012)	2663.77	Average for patenting firms
Patents (flow, annual)	54.16	Average annual patents
Patents (stock, cumulative)	4568.41	Average cumulative patents

Administrative patent classes such as the International Patent Classification and the Cooperative Patent Classification assign each patent based on the judgement of patent-examiner. Furthermore, the assigned-class for each patent have the properties of discrete, historically path dependent, and do not supply weights when the patent-examiner lists several classes for a patent. Meanwhile, LDA recovers a probability distribution over areas from the patent text, so a single invention can span several domains with explicit weights.[4] I select the number of topics and the Dirichlet priors by a five-fold iterative grid search that balances several metrics, e.g., perplexity, NPMI coherence, exclusivity, and decade-to-decade stability.[5] Appendix A.1 through Appendix A.5 detail the preprocessing, the interative grid search, and the metrics being used to select the hyperparameters

To assign labels, I extract the most heavily weighted terms for each area and iterate between two large language models (OpenAI's GPT-o3 and xAI's Grok 3 Think) until the proposed names converge. Table 2 lists the twenty labels, and top 25 words can be found in Appendix A.6.

Translating the raw LDA results into tractable inputs for the model requires aggregating from the patent level to the firm-year level. By dropping trivial topic exposures or outliers (i.e., weights below a certain threshold $\bar{w}$. See Appendix A.3 for further details), I construct the following key variables: (i) A monetary valuation of the patent and of the firm within each innovation area, A_{ipt} and v_{ijt}, along with the LDA weights, w_{pjt}, at the patent level. (ii) A binary indicator of firm i's entry into innovation area j at time t, d_{ijt}, which equals 1 if the firm has sufficient topic presence, and 0 otherwise. (iii) A congestion metric capturing the aggregate number of competing firms residing within area j, $N_{jt,-i} \equiv \sum_{i' \neq i} d_{i'jt}, \forall j, t$. (iv) The number of areas the firm occupies in each period, $k_{it} := |J_{it}| = \sum_j d_{ijt}$. (v) Predetermined firm covariates used as instruments: lagged R&D, RD_{it-1}, total assets, AT_{it-1}, and patent flow, PAT_{it-1}. These variables serve as the inputs for the model estimation introduced in the subsequent section.

[4] Appendix A.7 shows how many class codes examiners list on a grant in each decade, and which subclasses receive the most mass in each area (Tables 16 and 17). Certain important and crucial topics, such as, lithography, packaging, and transistor device physics all concentrate in subclass H01L. In the patent text those are different processes: etch and photoresist, die and bond pads, and gate and drain. A market defined as H01L would lump all of them together.

[5] From the optimization procedure, I find that twenty is the value at which those held-out metrics stop improving.

Table 2: Innovation areas

Topic	Label	Topic	Label
1	Semiconductor Memory Architecture	11	Power-Supply and Power-Management Electronics
2	Audio and Speech-Signal Processing	12	Networking Protocols and Data Infrastructure
3	Semiconductor Lithography and Fabrication Processes	13	Transistor Gate Structures and Semiconductor Device Physics
4	Integrated Circuit Design, Verification, and Testing	14	Imaging Sensors and Camera Pipelines
5	PCB and Electronic Hardware with Connectors	15	Processor Architecture, Caches, Instruction Execution
6	Video Recording, Encoding, and Media Processing	16	Display Panel Tech (LCD/OLED)
7	Semiconductor Packaging and Interconnect Bonding	17	Rotational Electromechanics and Magnetic Actuation
8	Thin-Film Materials, Deposition and Chemistry	18	Optics, Lasers, and Photonics
9	Wireless Base-Station and Mobile Communications	19	Analog and Differential Amplifier Circuits
10	Clock Generation, Timing Circuits, and Signal Processing	20	Mobile UI and Content Services

Notes: Labels are assigned from the twenty-topic LDA fit in Section 2.2. Top words for each area are in Appendix A.6.

2.3 Stylized Facts

I document two stylized facts: multi-area bundling and large differences in how many firms occupy each area. I then describe how competitor shares and patent value moved across decades.

Table 3: Patent-level summary statistics

Variable	Min	Q25	Q75	Max	Median	Mean	SD
w_{pjt} (LDA weight)	0	0	0.001	1	0	0.05	0.154
A_{ipt} (Patent value, million USD)	0	0.006	0.169	39.947	0.052	0.184	0.506
k_{ipt} (Number of areas per patent)	0	3	4	11	3	3.583	1.387

Stylized Facts 1 (On Multi-Area Bundling and Portfolio Scope) *Firms occupy several innovation areas in the same year. The average patent sits on 3.58 areas, and the median firm-year occupies five areas.*

Tables 3 and 4 demonstrate that multi-area bundling is a dominant feature of innovation in the electronics industry. Table 3 shows that the average patent spans 3.58 innovation areas (k_{ipt}), and at least three quarters of patents span three or more areas. That is, a single invention often combines knowledge from several areas. As for the firms, the median firm-year occupies five areas (k_{it}), and a quarter of firm-years occupy ten areas or more (Table 4). In the model, I therefore treat entry as the choice of a bundle of areas, in which occupying one area can make another area more or less profitable for the firm, i.e., economies or diseconomies of scope.

Stylized Facts 2 (On Competitive Congestion and Market Structure) *The number of firms differs*

Table 4: Firm-year summary statistics

Variable	Min	Q25	Q75	Max	Median	Mean	SD
d_{ijt} (Entry indicator)	0	0	1	1	0	0.269	0.443
$N_{jt,-i}$ (Competitor entry count)	20	83	141	218	105	110.255	39.913
k_{it} (Number of areas firm enters)	0	0	10	20	5	5.373	5.429
$\sum_p w_{pjt} A_{ipt}$ (Patent value sum, million USD)	0	0	0.721	112263.324	0	51.205	743.379

widely across innovation areas. Some areas are occupied by many firms, and others by few, throughout the sample.

Table 5: Composition of competitors by topic and decade

Topic	Label	1990s (%)	2000s (%)	2010s (%)	Mean (%)	SD
1	Semiconductor Memory Architecture	3.76	3.37	2.94	3.41	0.41
2	Audio and Speech-Signal Processing	6.97	7.19	7.1	7.08	0.23
3	Semiconductor Lithography and Fabrication Processes	4.39	4.33	4.27	4.34	0.32
4	Integrated Circuit Design, Verification, and Testing	7.49	7.68	7	7.44	0.41
5	PCB and Electronic Hardware with Connectors	6.97	6.47	6.73	6.72	0.4
6	Video Recording, Encoding, and Media Processing	3.7	3.73	3.35	3.62	0.31
7	Semiconductor Packaging and Interconnect Bonding	3.77	4.61	5.18	4.44	0.64
8	Thin-Film Materials, Deposition and Chemistry	6.31	5.14	4.66	5.45	0.79
9	Wireless Base-Station and Mobile Communications	3.62	4.35	4.74	4.18	0.53
10	Clock Generation, Timing Circuits, and Signal Processing	6.87	6.83	6.4	6.74	0.34
11	Power-Supply and Power-Management Electronics	5.97	6.14	6.27	6.11	0.28
12	Networking Protocols and Data Infrastructure	3.54	4.36	4.45	4.08	0.54
13	Transistor Gate Structures and Semiconductor Device Physics	3.6	3.78	3.72	3.7	0.2
14	Imaging Sensors and Camera Pipelines	3.12	3.3	4.22	3.46	0.54
15	Processor Architecture, Caches, Instruction Execution	5.1	5.47	5.53	5.35	0.33
16	Display Panel Tech (LCD/OLED)	1.86	1.81	1.92	1.85	0.19
17	Rotational Electromechanics and Magnetic Actuation	7.39	5.82	5.95	6.43	0.87
18	Optics, Lasers, and Photonics	3.84	3.78	3.91	3.84	0.22
19	Analog and Differential Amplifier Circuits	6.24	6.12	5.9	6.11	0.28
20	Mobile UI and Content Services	5.51	5.73	5.78	5.66	0.26
	Total	100.02	100.01	100.02	100.01	–

Table 5 and Table 6 reports the share of active firms and share of value in each area, respectively. Foundational areas, such as "Integrated Circuit Design, Verification, and Testing" (Topic 4) and "Audio and Speech-Signal Processing" (Topic 2), has around 7 percent share of the industry. "Display Panel Tech (LCD/OLED)" (Topic 16), however, remains sparse, under 2 percent throughout.

The results also show that legacy hardware such as "Rotational Electromechanics and Magnetic Actuation" (Topic 17) and "Thin-Film Materials, Deposition and Chemistry" (Topic 8) experienced a decline in share from the 1990s to the 2010s, while "Semiconductor Packaging and Interconnect Bonding" (Topic 7), "Wireless Base-Station and Mobile Communications" (Topic 9), and "Networking Protocols and Data Infrastructure" (Topic 12) experienced a gain. In the 1990s, "Rotational Electromechanics and Magnetic Actuation" (Topic 17) and "Thin-Film Materials, Deposition and Chemistry" (Topic 8) together accounted for more than 23 percent of patent value. By

Table 6: Composition of patent valuation by topic and decade

Topic	Label	1990s (%)	2000s (%)	2010s (%)	Mean (%)	SD
1	Semiconductor Memory Architecture	2	2.12	1.62	1.94	5.15
2	Audio and Speech-Signal Processing	6.42	7.1	6.99	6.84	10.98
3	Semiconductor Lithography and Fabrication Processes	3.27	3.65	3.74	3.55	7.91
4	Integrated Circuit Design, Verification, and Testing	5.66	6.12	5.14	5.7	8.43
5	PCB and Electronic Hardware with Connectors	10.2	8.73	8.19	9.08	16.36
6	Video Recording, Encoding, and Media Processing	2.17	2.16	2	2.12	5.38
7	Semiconductor Packaging and Interconnect Bonding	2.42	4.4	4.96	3.88	9.98
8	Thin-Film Materials, Deposition and Chemistry	9.42	7.2	5.89	7.6	16.49
9	Wireless Base-Station and Mobile Communications	2.34	3.02	4.14	3.1	7.45
10	Clock Generation, Timing Circuits, and Signal Processing	7.76	7.85	6.79	7.53	11.91
11	Power-Supply and Power-Management Electronics	4.71	5.31	6.09	5.32	10.53
12	Networking Protocols and Data Infrastructure	3.4	5.2	5.19	4.58	10.87
13	Transistor Gate Structures and Semiconductor Device Physics	2.59	3.38	3.73	3.21	8.51
14	Imaging Sensors and Camera Pipelines	1.98	2.06	3.06	2.31	5.83
15	Processor Architecture, Caches, Instruction Execution	4.57	4.46	5	4.64	9.42
16	Display Panel Tech (LCD/OLED)	1.07	1.35	1.33	1.25	4.83
17	Rotational Electromechanics and Magnetic Actuation	14.14	8.85	8.39	10.52	18.68
18	Optics, Lasers, and Photonics	3.4	3.98	4.08	3.81	9.45
19	Analog and Differential Amplifier Circuits	7.19	7.38	6.22	7	12.44
20	Mobile UI and Content Services	5.28	5.68	7.45	6.03	12.5
	Total	99.99	100	100	100.01	–

the 2010s their combined share had fallen, and value had moved toward "Mobile UI and Content Services" (Topic 20), "Wireless Base-Station and Mobile Communications" (Topic 9), and "Power-Supply and Power-Management Electronics" (Topic 11). Topic 20's share rose from 5.28 percent in the 1990s to 7.45 percent in the 2010s. Appendix A.8 reports the corresponding average count and dollar levels.

3 Model and Implementation

To explain the bundling and congestion patterns in Section 2.3, I develop a simple entry model following Pakes (2010), Pakes et al. (2015), Dickstein and Morales (2018), and Morales et al. (2019). For empirical implementation, I adopt the moment inequalities approach of Canay and Shaikh (2017), Chernozhukov et al. (2019), and Canay et al. (2023).

3.1 Information Structure and The Game

I model the firm's decision to enter innovation areas as a static, simultaneous-move, aggregative game under symmetric uncertainty, or complete information over observable state variables. When deciding to enter area j in period t, firm i uses its *ex-ante* information set $\mathcal{F}_{ijt}$ to evaluate expected profits. For example, the firm knows its own lagged R&D stock, its lagged total assets, and where it has patented in the past, and it also knows some characteristics of its rivals. I observe only part of this information. Further, I assume that unobservable area-specific market shocks u_{ijt} are realized only after entry, such that $E[u_{ijt} \mid \mathcal{F}_{ijt}] = 0$. Firms therefore play a simultaneous-move

Nash equilibrium over expected payoffs.

This information structure follows Dickstein and Morales (2018) and Morales et al. (2019): firms do not select on shocks they have not yet seen, which yields me linear moment inequalities. On the static game, this setup answers how a firm composes its contemporaneous portfolio and how rival counts in the same period shift expected profit.[6]

3.2 Firm Valuation and Expected Profit

Each patent p filed by firm i in period t has a dollar value A_{ipt}. Using topic modeling from the prior section, weights w_{pjt} (where $0 \leq w_{pjt} \leq 1$ and $\sum_j w_{pjt} = 1$) allocate each patent across innovation areas. As such, I can define the area-specific value as $A_{ipjt} = w_{pjt} A_{ipt}$ [7].

I define the *realized ex-post* firm valuation as follows

$$v_{ijt} = b\left(\sum_{p \in \mathcal{P}_{it}} w_{pjt} A_{ipt}\right) + e_{ijt},$$

where $\mathcal{P}_{it}$ is firm i's patent set in period t, and $b(\cdot)$ is a monotonically increasing transformation function. This function maps the raw aggregated patent values into a tractable scale, which manages the high skewness of valuation data. The term e_{ijt} captures valuation measurement error, absorbing the noise inherent to LDA topic assignments and the stock market's reaction in the event studies of Kogan et al. (2017). Firms do not observe the exact realized error e_{ijt} prior to entry, meaning their ex-ante expectation is strictly deterministic, i.e., $E[e_{ijt} \mid \mathcal{F}_{ijt}] = 0$. Then, the expected valuation is simply $\check{v}_{ijt} \equiv E[v_{ijt} \mid \mathcal{F}_{ijt}]$.

The true realized profit for firm i of entry into area j in period t can then be written as

$$\pi_{ijt} = v_{ijt} + \alpha h(N_{jt,-i}) + \beta f(k_{it}) - \kappa + u_{ijt}$$

where both the second and third terms reflect competitive and organizational factors, respectively. By substituting the expected valuation, $v_{ijt} = \check{v}_{ijt} + e_{ijt}$, into the true profit function, the estimating equation of the profit can be written as

$$\pi_{ijt} = \check{v}_{ijt} + \alpha h(N_{jt,-i}) + \beta f(k_{it}) - \kappa + \epsilon_{ijt}$$

where $N_{jt,-i}$ is the number of competitors (excluding the firm itself) in area j, and $k_{it} \equiv |J|_{it}$ is the total number of innovation areas firm i is active in during period t. $\check{v}_{ijt} \equiv E\left[b\left(\sum_p w_{pjt} A_{ipt}\right) \mid \mathcal{F}_{ijt}\right]$

[6] I treat rival counts as given for two reasons. First, each area has many rivals: about 110 on average, and at least 20 in every area-year. As such, the entry of one more firm raises the number of rivals in an area by less than 1% on average, and by at most 5%, so it changes the payoffs of the other firms only a little. Second, rival counts change little from one year to the next, with an autocorrelation of 0.97. That is, the number of rivals a firm expects to face is close to the number it actually observes. Appendix A.10 reports how persistent occupancy is in the panel.

[7] The construction of A_{ipt} is in Appendix B.1. The construction of the LDA weights w_{pjt} is in Appendix A.2.

is the expected firm valuation, and $h(\cdot)$ and $f(\cdot)$ denote functional forms for both rival counts and portfolio size, respectively. As for ϵ_{ijt}, the unobserved error term in the estimating equation, $\epsilon_{ijt} \equiv e_{ijt} + u_{ijt}$, can be thought of as a *composite ex-post forecasting error*. It absorbs both the valuation measurement noise (e_{ijt}) and the market shocks (u_{ijt}). Because both components are mean-zero conditional on the firm's information set, the conditional expected value of ϵ_{ijt} is zero, i.e., $E[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$.

Parameter α captures congestion effects: positive values indicate spillovers (e.g., net-agglomeration effect), while negative values reflect crowding out. Parameter β measures the impact of breadth of innovation portfolio, with positive values suggesting benefits from diversification (e.g., economies of scope) and negative values indicating costs of overextension (e.g., diluted focus). As for parameter κ, it is a per-area fixed cost paid for each occupied area in year t. Consequently, a firm that drops an area in that year stops paying it. Let $\theta = (\alpha, \beta, \kappa)$ be the vector of parameters of interest.

Let d_{ijt} be an indicator whether firm i enters into area j at time t. The firm chooses a bundle of areas to maximize expected total profit:

Result 1 (Firm's Expected Total Profit) *Because $E[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$, the expected total profit simplifies to its deterministic components:*

$$\begin{aligned}\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}] &= \sum_{j=1}^{J} d_{ijt}\mathbb{E}[\pi_{ijt} \mid \mathcal{F}_{ijt}] = \sum_{j=1}^{J} d_{ijt}\left(\check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta f(k_{it}) \sum_{j=1}^{J} d_{ijt} \\ &= \sum_{j \in S}\left(\check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta g(k_{it})\end{aligned}$$

where $S = \{j \mid d_{ijt} = 1\}$, $k_{it} = |J|_{it}$, and $g(k_{it}) \equiv k_{it} f(k_{it})$. Result 1 highlights the importance of the bundling problem: the presence of the portfolio term $\beta g(k_{it})$ creates interdependencies across choices[8].

Firm i selects the bundle $\mathbf{d}_{it} = (d_{ijt})_j \in \{0,1\}^J$ to maximize expected profit $\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}]$ in a simultaneous-move game. With $J = 20$ areas, a firm can choose among 2^{20} bundles in every year, which is more than one million, so computing the full equilibrium of the game is very difficult. However, in any Nash equilibrium, the bundle a firm chooses must be its best response to the areas its rivals choose, i.e., to the rival counts $N_{jt,-i}$.

Since the portfolio term depends only on how many areas the firm occupies, k_{it}, the firm's problem becomes much simpler once the rival counts are given. That is, once the firm knows how many areas to occupy, it only has to rank the areas by the expected profit of each area. This yields a characterization of the firm's optimal bundle:

[8] A standard independent single-agent discrete choice framework is ill-posed in this setting because the portfolio term $\beta g(k_{it})$ renders the entry decisions non-separable; the marginal payoff of entering area j explicitly depends on the endogenous decision to enter area m.

Result 2 (Characterization of the Optimal Bundle) *The expected profitability index for area j is:*

$$\zeta^e_{ijt} = \check{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa$$

For a fixed portfolio size k_{it}, the firm maximizes profit by choosing the k_{it} areas with the largest ζ^e_{ijt}. That is, the areas are sorted such that: $\zeta^e_{(1)} \geq \zeta^e_{(2)} \geq \cdots \geq \zeta^e_{(J)}$. Then, the optimized expected profit for a portfolio of size k is:

$$U^e_{k_{it}} = \sum_{m=1}^{k_{it}} \zeta^e_{(m)} + \beta g(k_{it})$$

An optimal size is any $k^*_{it} \in \arg\max_{k_{it}=0,1,\ldots,J} U^e_{k_{it}}$. Any optimal bundle is a set of k^*_{it} areas with the largest ζ^e_{ijt}, allowing ties at the cutoff, and $d_{ijt} = 1$ if and only if j belongs to that bundle.

Proof. See Appendix B.2. ■

Assuming that the portfolio cost $\beta g(k_{it})$ depends only on the count k_{it}, it simplifies the firm's optimization problem: the firm ranks areas by ζ and selects a top-k^*_{it} set. However, this simplification does not solve the problem faced by the econometrician. The decision to enter area j is still not separable from area m[9]. Furthermore, although Result 2 provides a clean characterization of the problem, the idea of point identification of θ remains challenging due to multiple equilibria and combinatorial explosion. Which configuration materializes depends on an unobserved equilibrium-selection rule (as shown by Tamer, 2003; Ciliberto and Tamer, 2009).

Instead, I pursue partial identification following Pakes (2010) and Pakes et al. (2015). This replaces unattainable moment equalities with inequalities satisfied by every equilibrium. Based on Result 2, any deviation from the observed choice S^* must yield a lower (or equal) expected profit. To operationalize this notion, I consider 'step and swap deviations': (i) entering M additional areas, (ii) exiting M current areas, or (iii) swapping the entry portfolio while keeping k_{it} constant.

Let $Z_{ijt} \in \mathbb{R}^n$ be a vector of instruments observable to both the econometrician and the firm, and that this instrument is assumed to be a subset of $\mathcal{F}_{ijt}$. Let $l : \mathbb{R}^n \to \mathbb{R}_+$ map Z_{ijt} to a non-negative scalar. Following Canay et al. (2023), these instruments are weights that partition the sample and tighten the inequalities. Note that these instruments do not instrument rival counts $N_{jt,-i}$ or portfolio size k_{it} in the two-stage least squares sense. As for the common area-time attractiveness, it is absorbed in the first-stage projection of expected value.

First, I will derive the moment functions for adding M areas.

Result 3 (Moment Functions for Adding M Areas (Non-Entrant Deviation)) *Let $\mathcal{M}$ be an exogenously drawn subset of M areas ($M = |\mathcal{M}|$). I consider the event where the firm chose to enter none of*

[9] That is, the marginal cost of entering j depends on k_{it}, which is endogenously determined by the entry decisions for all other areas. Because the choices are jointly determined, those decisions cannot be estimated as J independent single-choice models (e.g., in a typical logit). The outcome is a single chosen bundle S^*.

the areas in $\mathcal{M}$, indicated by $D^0_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}}(1-d_{ijt}) = 1$. The difference in expected profit between the deviation (adding $\mathcal{M}$) and the observed choice is, $\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}]$:

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = \sum_{j\in\mathcal{M}} \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) + \beta\Delta g^+_M(k_{it}) = \eta^+_{i,\mathcal{M},t}$$

where $\Delta g^+_M(k_{it}) = g(k_{it}+M) - g(k_{it})$. If the firm chose not to enter any area in $\mathcal{M}$, the expected deviation profit must be non-positive: $\eta^+_{i,\mathcal{M},t} \leq 0$. Multiplying this by the choice indicator and the non-negative instrument yields the unconditional moment function as follows:

$$\mathbf{M}_1(Z_{i,\mathcal{M},t};\theta) = E\left[D^0_{\mathcal{M},it} \cdot \eta^+_{i,\mathcal{M},t} \cdot l(Z_{i,\mathcal{M},t})\right] \leq 0 \tag{1}$$

Proof. See Appendix B.2. ■

Now, I will derive the moment functions for dropping M areas.

Result 4 (Moment Functions for Dropping M Areas (Entrant Deviation)) *Let $\mathcal{M}$ be an exogenously drawn subset of M areas ($M = |\mathcal{M}|$). I consider the event that the firm entered every area in $\mathcal{M}$, indicated by $D^1_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}} d_{ijt} = 1$. The expected profit difference of dropping $\mathcal{M}$ is:*

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = -\sum_{j\in\mathcal{M}} \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i}) - \kappa\right) - \beta\Delta g^-_M(k_{it}) = -\eta^-_{i,\mathcal{M},t}$$

where $\Delta g^-_M(k_{it}) = g(k_{it}) - g(k_{it}-M)$. If the firm chose to enter all areas in $\mathcal{M}$, dropping them cannot yield strictly higher expected profit. Thus, $-\eta^-_{i,\mathcal{M},t} \leq 0$. I can then define the unconditional moment function as follows:

$$\mathbf{M}_2(Z_{i,\mathcal{M},t};\theta) = E\left[D^1_{\mathcal{M},it} \cdot (-\eta^-_{i,\mathcal{M},t}) \cdot l(Z_{i,\mathcal{M},t})\right] \leq 0 \tag{2}$$

Proof. See Appendix B.2. ■

These M-step deviations $\mathbf{M}_1$ and $\mathbf{M}_2$ moments are all valid and necessary conditions, yet they do not fully exploit the sorting characterization outlined in Result 2. To avoid endogenous selection bias, I define swap moments over exogenously chosen pairwise deviations. Because the portfolio term depends solely on the count k_{it}, any profitable swap of an entered area for a non-entered area, while keeping k_{it} fixed, is ruled out. I can then derive this 'swap' moment function as follows.

Result 5 (Moment Functions for Pairwise Swap Deviation) *Let (j,m) be an ordered pair where firm i chose to enter j but not m: $D^{swap}_{j,m,it} \equiv d_{ijt}(1-d_{imt}) = 1$. The expected profit of deviating by substituting m for j leaves the portfolio size constant ($\Delta g = 0$):*

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] = \left(\breve{v}_{imt} + \alpha h(N_{mt,-i})\right) - \left(\breve{v}_{ijt} + \alpha h(N_{jt,-i})\right) = \mu_{ijmt}(\theta)$$

The optimality condition requires that the expected swap profit is non-positive: $\mu_{ijmt}(\theta) \leq 0$. Then, the unconditional linear moment function can be written as follows:

$$\mathbf{M}_3(Z_{ijmt};\theta) = E\left[D^{swap}_{j,m,it} \cdot \mu_{ijmt}(\theta) \cdot l(Z_{ijmt})\right] \leq 0 \tag{3}$$

Proof. See Appendix B.2. ■

The swap moment, $\mathbf{M}_3$, isolates the congestion effect α. Now consider a firm that occupies area j and does not occupy area m. A swap asks whether that firm would be better off occupying m instead of j. Since the firm still occupies the same number of areas, the gain from occupying more areas at once does not change, and the firm still pays occupancy cost for the same number of areas. However, what can still differ is expected valuations and how crowded each area is. Suppose the two areas look similar in expected valuations. If the firm occupies the less crowded of the two, then the ranking identifies a congestion effect.

3.3 Empirical Implementation

I define entry as $d_{ijt} = \mathbf{1}\{\frac{1}{|\mathcal{P}_{it}|}\sum_{p\in\mathcal{P}_{it}} w_{pjt} \geq \bar{w}\}$ using the weights from the LDA model with a threshold $\bar{w}$. The reason to choose the threshold is that weights below this threshold are the long tail of the LDA model weights[10]. As for constructing $\check{v}_{ijt}$, I apply the inverse hyperbolic sine to realized v_{ijt} and project onto lagged R&D stocks, lagged assets, and lagged patent flow, together with firm and year effects, in a separate regression for each area, so that the fitted value carries firm-by-area and year-by-area effects. As such, the fitted value is the proxy that enters the estimating profit equation.

On the congestion effect, I define the congestion function as $h(N) = \text{asinh}(N)$, which is steep in thin markets and flatter once an area is crowded. As for the functional form for the portfolio term, I define it as $g(k) = \binom{k}{2}$, which counts pairs among entered areas and delivers the sorting in Result 2. I use pair and triplet add and drop deviations, which vary how many areas enter the portfolio term. Add and drop change how many areas the firm occupies. Under assumed pairwise interactions for the portfolio term, those comparisons can only discipline the occupancy-to-scope ratio $\rho = \kappa/\beta$[11]. Furthermore, I can only identify the ratio, I have to assume that β cannot be zero under the pairwise interactions. As such, I can only identify $\tilde{\theta} = (\alpha, \rho)$ instead of $\theta = (\alpha, \beta, \kappa)$.

To operationalize the model estimation, I first rank candidate add and drop subsets from predicted values $\check{v}_{ijt}$ before looking at realized d_{ijt}. For drop moments I keep the subsets with the lowest

[10]The brief explanation of the threshold $\bar{w}$ is the following: at the patent level, keep only the positive shares, i.e., $w_{pjt} > 0$. Let f be the density of $\ln(w_{pjt})$ on that set, and let $m_L < m_R$ be its two modes. The minimum point on the log scale then can be defined as $x^* = \arg\min_{x\in(m_L,m_R)} f(x)$. The threshold in the entry is simply $\bar{w} = \exp(x^*)$. I find that $\bar{w} = \exp(x^*) = 0.0094 \approx 1\%$. Further details and motivation can be found in Appendix A.3.

[11]See Appendix B.4 for further discussion on the identification of ρ

average predicted value, where the decision to stay is most informative about the cost. Similarly, for add moments I keep those with the highest average predicted value, where the decision not to enter is most informative. For swaps I rank the 380 ordered pairs by divergence in $\check{v}$ and draw from the top, the bottom, and the middle of that ranking to cover a representative sample of the ranked pairs. The reported sets use samples of 20 or 40 per deviation type, and the specification with all three instruments uses the largest of those samples. The test statistic is the max-type studentized sample moment of Chernozhukov et al. (2019), with their two-step self-normalized critical value at significance level 0.01. As for the model estimation, I first draw parameter vectors at random inside a bounded region and keep those for which the test statistic does not exceed the critical value. If the best draw still exceeds it, I minimize that excess and use the resulting vector as a starting value. The main reason to conduct this first step is that the criterion from the moment inequalities is noncovex, flat in many places, and non-differentiable. From those starting values I search the same region by simulated annealing a la Grieco (2014) and keep the vectors that satisfy the criterion, and I further profile each parameter in $\tilde{\theta} = (\alpha, \rho)$ by its lowest and highest value subject to that same restriction. Appendix B.3 details the search over a bounded parameter region and the profiled bounds.

4 Results

I first present the first-stage projection that replaces noisy realized patent values with the component that is predictable from several lagged covariates. I then show a reduced-form linear probability model of entry. Finally, I present the estimated identified sets from the moment inequalities.

4.1 First-stage Projection: Extracting Expected Valuations

The raw aggregated patent value, $v_{ijt} = b\left(\sum_{p\in\mathcal{P}_{it}} w_{pjt} A_{ipt}\right)$, is an *ex-post* realization inherently plagued by measurement errors. There are three main sources of these errors. First, LDA can assign part of a patent's text to the wrong area, which adds noise to the weights w_{pjt}. Since these mistakes are likely random across patents, I treat this noise as classical. Second, the patent values A_{ipt} of Kogan et al. (2017) come from the stock market reaction around the grant date, and that reaction also reflects news that is unrelated to the patent. Third, I add up the weighted patent values to the firm, year, and area, so both errors carry over to v_{ijt}, and a single patent can move the value of an area when the firm has few patents in that year. If I use v_{ijt} directly in the entry model, these errors enter the inequalities together with the value itself. To form a proxy for expected private value, I project the observed v_{ijt} onto the predetermined instruments $Z_{ijt} = (\mathrm{RD}_{it-1}, \mathrm{AT}_{it-1}, \mathrm{PAT}_{it-1})$ and firm and year-by-area effects. This exercise yields a fitted value $\check{v}_{ijt}$. The idea is that the fixed effects absorb the systematic part of the value, and the lagged covariates capture the part that the firm can predict. The remaining noise then averages out in the fitted value. Here, the assumption being made is that $E[\epsilon_{ijt} \mid Z_{ijt}] = 0$, i.e., the composite forecast error is mean independent of the

instruments used as weights.

As discussed in the previous section, the variables Z_{ijt} are discretized into binary indicators to manage the extreme skewness of the data. Specifically, past R&D stocks (RD_{it-1}), past total assets (AT_{it-1}), and past patent flow (PAT_{it-1}) take a value of 1 for firm-year observations in the upper tier (medium-to-high levels). The lower tier (zero or bottom-quartile) serves as the baseline[12].

Table 7: First Stage: Expected Profit Projections

Dependent Variable:	$v_{ijt} = b(\sum_p w_{pjt} A_{ipt})$		
Model:	(1)	(2)	(3)
Variables			
Constant	-0.464***		
	(0.006)		
RD_{it-1}	0.320***	0.072***	0.072***
	(0.022)	(0.021)	(0.021)
AT_{it-1}	0.206***	0.109***	0.109***
	(0.021)	(0.021)	(0.021)
PAT_{it-1}	1.90***	0.837***	0.837***
	(0.130)	(0.072)	(0.072)
Fixed-effects			
Firm		Yes	Yes
Year		Yes	
Area		Yes	
Year × Area			Yes
Fit statistics			
Observations	225,220	225,220	225,220
R^2	0.502	0.709	0.710
Within R^2		0.063	0.063

Clustered (Firm) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is realized patent value after asinh and a global standardization.

Table 7 reports the first-stage projection. Focusing on the specification with firm and year-by-area fixed effects (Column 3), these covariates exhibit significant predictive relationships. Because the dependent variable v_{ijt} is transformed via an inverse hyperbolic sine and then standardized, the estimated coefficients translate to standard deviation units. Moving from a zero or near-negligible patenting history to a medium-to-high past patent flow (PAT_{it-1}) shifts the firm's expected area valuation upward by a 0.837 standard deviation. Similarly, total assets (AT_{it-1}) and R&D stocks (RD_{it-1}) provide significant premiums of 0.109 and 0.072 standard deviations, respectively. Across the twenty areas, the mean absolute deviation of the fitted valuation is clustered around 0.34. Comparison of predictive performance across specifications can be found in Appendix A.9.

[12] As discussed in the implementation of the instrument Z_{ijt}, transforming these covariates into non-negative binary scalars allows them to function efficiently as weights $l(\cdot)$.

4.2 Reduced-Form Evidence on Firm Entry

Before implementing the moment inequalities, it is useful to assess firm's entry likelihood via standard tools. Table 8 estimates a reduced-form linear probability model on entry by regressing the entry indicator d_{ijt} on the fitted-value of valuation valuation $\check{v}_{ijt}$, competitive congestion $\text{asinh}(N_{jt,-i})$, and the portfolio scale variable $(k_{it}-1)/2$. Focusing on the specification with fixed effects (Column 2), I find the following results: First, valuations matter, i.e., higher expected profits increase the likelihood of entry. Second, the coefficient on $\text{asinh}(N_{jt,-i})$ is positive and remains that way after expected valuation and portfolio size enter the same regression.

Table 8: Entry, expected valuation, rival counts, and portfolio size

Dependent Variable:	d	
Model:	(1)	(2)
Variables		
Constant	-0.8239***	
	(0.0334)	
$\tilde{v}_{ijt} = b(\hat{v}_{ijt})$	7.43×10^{-6}**	0.0003***
	(3.72×10^{-6})	(3.9×10^{-5})
$\text{asinh}(N_{jt,-i})$	0.1643***	0.0799***
	(0.0063)	(0.0140)
$(k_{it}-1)/2$	0.0996***	0.0995***
	(0.0002)	(0.0001)
Fixed-effects		
Firm		Yes
Year		Yes
Area		Yes
Fit statistics		
Observations	225,220	225,220
R^2	0.3974	0.4075

Clustered (Firm) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The same OLS specification also generates a positive coefficient on the portfolio measure (0.0995). However, recall that the regressor is $(k_{it}-1)/2$, so a coefficient near 0.1 is the adding-up identity of the occupancy rate. Furthermore, unobserved firm capability can also raise both k_{it} and the probability of entering any given area, which is a reason the linear probability model cannot solely identify and isolate scope.

4.3 Model Estimates

I partially identify congestion, α, and the occupancy-to-scope ratio, ρ, with the add, drop, and swap comparisons ($\mathbf{M}_1, \mathbf{M}_2, \mathbf{M}_3$) derived in the previous section. Table 9 reports 99% confidence sets under the two-step self-normalized procedure of Chernozhukov et al. (2019). The procedure constructs a confidence set using a max-type studentized statistic and a two-step critical value that recenters slack moments. The table shows three cases: (i) no instruments with $|M| = 20$, (ii) lagged patent flow as an instrument with $|M| = 20$, and (iii) all three instruments with $|M| = 40$. Appendix C provides the complete results from using no instruments, a single instrument, and all three instruments.

Table 9: Estimated parameter bounds

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1084, -0.1111]$	—
PAT_{it-1}	20	$[-1.1106, -0.0966]$	$[0, 12.6250]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$

Notes: The table reports 99 percent confidence sets for congestion α and occupancy-to-scope ratio ρ under pairwise interactions among entered areas, add and drop deviations of size two and three, and swap moments. A dash means that ratio is not identified. I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$, which also implies that $\beta < 0$ is not possible. Table 24 reports the full instrument and subset-size grid. Table 25 repeats that grid with add and drop deviations of size two only. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

At the specification that uses all three instruments and the largest subset sample, congestion α lies in $[-1.1160, -0.1847]$. Meanwhile, the occupancy-to-scope ratio ρ lies in $[0, 12.3750]$. First, take a look at the result without instrument, each inequality is then simply a single average over all firm-years. In that average, congestion α is already negative, and the occupancy-to-scope ratio ρ is not identified. Moving on to using lagged patent flow as a single instrument already provides a non-negativeset for ρ. Finally, using all three instruments and the largest subset sample, the set is bounded above by 12.375, a tighter bound than the one using only the lagged patent as an instrument. Furthermore, from further profiling of the parameters, the 99% sets in Table 9 do reject the possibility that $\alpha \geq 0$ and $\rho < 0$, which implies that $\beta < 0$ is not possible. I also explore alternative portfolio functions, and the results of such exercises are reported in Table 10.

Competitive congestion. The baseline identified set for congestion α is $[-1.1160, -0.1847]$. The economic interpretation is that firms require compensation to enter a more crowded area. Raising the inverse hyperbolic sine of the rival count by one lowers expected profit by between 0.18 and 1.12 units of predicted value. Since rival counts enter through the inverse hyperbolic sine, one more rival matters more in an area with few rivals than in an area with many. That is, in the thinnest area-year of the sample, with 20 rivals, one more rival changes the congestion index by about $\text{asinh}(21) - \text{asinh}(20) \approx 0.05$. At the same time, adding the 166th rival to an area with 165, about the average for integrated circuit design, changes it by about $\text{asinh}(166) - \text{asinh}(165) \approx 0.006$. Take, for example, two areas with the same predicted value: display, with about 41 rivals,

and integrated circuit design, with about 165. Then the firm expects between 0.26 and 1.55 units less profit in integrated circuit design. As expected, the result from the moment inequalities put congestion in negative, as opposed to the positive coefficient from the reduced-form linear probability model. As usual, since I am using an aggregative rival counts, I can only interpret α as a net effect of business stealing, litigation, and any residual unobserved area attractiveness that the first-stage projection did not absorb.

Occupancy cost relative to scope. Under pairwise interactions among entered areas, add and drop comparisons recover $\rho = \kappa/\beta$. At all three instruments and the largest subset sample, that set is $[0, 12.3750]$. To see what this ratio means, take a firm that already occupies five areas, A, B, C, D, and E, and that is deciding whether to also occupy a sixth area, F. Under pairwise interactions, F forms a new pair with each of A, B, C, D, and E, which gives the firm five new pairs and an extra scope benefit of five times β. Meanwhile, occupying F costs the firm $\kappa = \rho\beta$ in every year, i.e., a fee of ρ pairs' worth of scope. The five new pairs therefore exactly cover the fee when $\rho = 5$. Suppose now that $\rho = 3$. This means that the five new pairs more than cover the fee, and the firm gains two pairs' worth of scope by occupying F. If instead $\rho = 12$, the five new pairs fall short of the fee, and occupying F costs the firm seven pairs' worth of scope.

The data allow both cases, since the 99% set contains $\rho = 3$ as well as $\rho = 12$. However, a value such as $\rho = 20$ lies outside the set. From further profiling of the parameters, I also find that the data allow the yearly occupancy cost κ to be zero, and they allow it to be as large as about twelve pairs' worth of scope. The intuition of this result is that the data reject a fee so large that a firm would need to occupy more than about twelve areas already before the new pairs from one more area cover it. Take the example of the median firm that occupies five areas. Because the set includes both $\rho < 5$ and $\rho > 5$, that median firm may already have enough pairs to match the fee, or it may not. To conclude, rival congestion and the ratio of occupancy-to-scope bound together rationalize both the observed multi-area bundling and the fact that even large electronics firms leave many topics unentered[13].

Alternative portfolio functions. Table 10 reports congestion and the occupancy object under alternative portfolio functions, at all three instruments and the largest subset sample. When the portfolio function is linear in the number of occupied areas, take again as an example the firm that already occupies five areas (A, B, C, D, E), and is deciding whether to occupy a sixth area, F. Again, the extra scope benefit of F is β. It is also the case that the extra occupancy cost of F is κ. Similar to the pairwise interactions case, add and drop moment comparisons recover $\delta = \beta - \kappa$. Suppose now that $\delta = 2$, it means that the extra scope exceeds the fee by 2. Similarly, if $\delta = 3$, then extra scope exceeds the fee by 3. From the table, the 99% set is $[1.6250, 3.3750]$. That is, values such as $\delta = 2$ and $\delta = 3$ are feasible since they remain inside the set. However, values such as $\delta = 1$ or $\delta = 4$ do not. From the table, congestion α on this row lies in $[-1.1160, -0.8500]$.

Let's now move to the case of multiplying the portfolio size by its inverse hyperbolic sine, $kasinh(k)$,

[13]The same pattern holds when I consider restricting $M = 2$ (Table 25 in Appendix C).

Table 10: Identified sets under alternative portfolio functions

$g(k)$	$\lvert M\rvert$	α	ρ or δ
$g(k) = \binom{k}{2}$	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$
$g(k) = k$	40	$[-1.1160, -0.8500]$	$[1.6250, 3.3750]$
$g(k) = \text{asinh}(k)$	40	—	—
$g(k) = k\,\text{asinh}(k)$	40	$[-1.1160, -0.2000]$	$[0, 4.1250]$
$g(k) = \sqrt{k}$	40	—	—
$g(k) = \ln(1+k)$	40	—	—

Notes: Each row uses all three instruments and the largest subset sample ($|M| = 40$). Each row uses all three instruments, lagged R&D, assets, and patent flow, and add and drop deviations of size two and three together with the swap moments. On the pairwise rows $g(k) = \binom{k}{2}$ and on $g(k) = k\,\text{asinh}(k)$, add and drop moments depend on occupancy cost and scope through their ratio $\rho = \kappa/\beta$. On the linear row $g(k) = k$, one more area always adds the same scope benefit, so add and drop moments restrict only $\delta = \beta - \kappa$. On $g(k) = \text{asinh}(k)$, $g(k) = \sqrt{k}$, and $g(k) = \ln(1+k)$, the inequalities found no jointly feasible point, so both columns are empty and neither ρ nor δ is reported. Those empty rows are why I do not use a concave portfolio function as the main specification. A dash in the last column means that object is not identified. I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$, which also implies that $\beta < 0$ is not possible. Inference for the last column uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

where this remains feasible. First, I find that the congestion α lies in $[-1.1160, -0.2000]$, and the occupancy-to-scope ratio ρ lies in $[0, 4.1250]$. Again, take the example of the firm that already occupies five areas (A, B, C, D, E), and is deciding whether to occupy a sixth area, F. In this case, the extra scope benefit of F is $\beta\big(6\text{asinh}(6) - 5\text{asinh}(5)\big) \approx 3.39\beta$. If $\rho = 2$, then the fee equals 2, and the extra scope benefit of 3.39 more than matches the fee. If $\rho = 4.125$, then the fee equals 4.125, and the extra scope benefit falls short of the fee. Finally, when extra areas enter as diminishing synergies, through the inverse hyperbolic sine, the square root, or the logarithm of portfolio size, the 99% set is empty.

Fixed and sunk costs. The occupancy cost above is a fee paid in every year the firm occupies an area. The same comparisons can also allow a separate fee paid only in the year the firm enters, i.e., a sunk cost. To measure that sunk cost separately from the fixed cost, I extend the add, drop, and swap comparisons by one period, borrowing the idea outlined by Morales et al. (2019). The deviation changes the portfolio only in one year and then follows the path the firm actually chose, so continuation values cancel except for the sunk cost that this year's choice brings forward or postpones[14]. Table 23 in Appendix B.5 shows the results of this exercise for both fixed-cost ratio ρ_f and the sunk-cost ratio ρ_s. Both ratios measure the fee against the gain from occupying more areas at once. Each row holds congestion α at one value from Table 9. Take for example the case at the midpoint of that set, when firms absent from the sample next year are dropped (Panel A), the accepted fixed-cost ratio runs from 0 to 9.2 and the accepted sunk-cost ratio from 0 to 7.4. From this exercise, a zero sunk cost is still accepted. Suppose the sunk cost is zero, then the firm only pays the fixed cost, as in the static model. At the weaker end of congestion $\alpha = -0.1847$, the fixed-cost ratio runs from 3.5 to 9.4 and the sunk-cost ratio from 0 to 6.8, still the zero sunk cost is

[14] Appendix B.5 derives the step by step derivation of my exercise here.

feasible. In this case, when an extra rival reduces expected profit only a little, the results call for a larger fixed cost. Treating firms that are absent next year as having left (Panel B) yields similar interpretation. These exercise provide lessons on how far off the above static results when we decompose ρ into the sunk-cost ratio ρ_s and the fixed-cost ratio ρ_f.

Alternative entry thresholds. In the baseline results, I define entry whenever a firm enters an area when, on average, its patents in that year exceeds the threshold $\bar{w} \approx 1\%$ of their text there (see Appendix A.3). For sensitivity analysis, I re-estimate using the same specification of the third row of Table 9 at $\tilde{w} = 0.5\%$ and $\tilde{w} = 2\%$. These two points are not the minimum point in the density distribution of LDA weights, but represent the beginning points of the left and right modes, respectively (see Figure 1). Table 26 in Appendix C.2 reports the results. At $\tilde{w} = 0.5\%$, an extra rival still negatively affects expected profit, and from further profiling of α I reject the possibility that $\alpha \geq 0$. However, I find that the occupancy-to-scope ratio ρ is not identified at $\tilde{w} = 0.5\%$. Among the chief reason of unidentified ρ at $\tilde{w} = 0.5\%$ is that 'unimportant' patent texts or texts that do not cleanly define an area are now being included. At $\tilde{w} = 2\%$, I observe that this cutoff lies above the fourth-largest share and below the third (see Figure 2). Therefore, based on this cutoff, the average firm then occupies only 4.34 areas in a year, against 5.37 at $\bar{w} \approx 1\%$. Furthermore, for this specification, I find no combination of congestion and occupancy-to-scope ratio that satisfies the inequalities, so the set is empty.

5 Case Study: Semiconductor

In these numerical exercises, I try to mimic a targeted subsidy for certain areas as outlined in the U.S. CHIPS incentive[15] and the EU Chips Act[16]. I map the areas listed in these regulations to memory, lithography and fabrication, packaging, and the transistor. There are two numerical exercises: (i) a subsidy equal to one quarter of predicted value in these four areas, and (ii) a subsidy that lowers the occupancy-to-scope ratio ρ by 25% in those areas. For each of these two exercises, firms can either keep the number of areas in their portfolios ("fixed areas") or add an extra area ("extra areas")[17]. For all the exercises, I use the result in the third row of Table 9, and the rival count in each area is the sum of the other firms' entry, updated until that sum agrees with the portfolios. The details of my numerical exercises, along with additional exercises on halving rival

[15]The CHIPS Act of 2022, Public Law 117–167, Division A, 136 Stat. 1366 (August 9, 2022), funds assistance under 15 U.S.C. § 4652 for facilities and equipment in the United States for the fabrication, assembly, testing, advanced packaging, production, or research and development of semiconductors, of materials used to manufacture semiconductors, or of semiconductor manufacturing equipment. Public Law 119–21, 139 Stat. 72 (July 4, 2025), § 70308, raises the advanced manufacturing investment credit in Internal Revenue Code § 48D from 25 percent to 35 percent for property placed in service after December 31, 2025. That credit remains the credit for a facility whose primary purpose is manufacturing semiconductors or semiconductor manufacturing equipment.

[16]Regulation (EU) 2023/1781 of the European Parliament and of the Council of 13 September 2023, OJ L 229, 18.9.2023, pp. 1–53, Article 2.

[17]For the "extra areas" exercises, I use several values of b. Since the estimates of the model can only recover α and ρ, the level of $\beta = b$ is a scale that has to be chosen. Note that the occupancy cost is $\kappa = b\rho$, so a higher b also means a higher occupancy cost in units of predicted value. Furthermore, the inequalities accept different values of α and ρ at different values of b, and for each b I only use the values accepted at that b.

congestion (Table 29), applying a uniform subsidy (Table 30), and the simulation where I hold the rival counts (see Tables 27 and 28), can be found in Appendix D.

5.1 Case I: a 25% subsidy based on predicted value

For this exercise, I add a quarter of predicted value, $0.25\check{v}$, for each of the four semiconductor areas. In the first column, firms can only reshuffle the areas within the portfolio they actually occupy ("fixed-size"). I find that the four semiconductor areas all gain share. Memory gains between 0.16% and 0.46%, packaging between 0.26% and 0.54%, lithography between 0.24% and 0.53%, and the transistor between 0.14% and 0.37%. Meanwhile, every other area loses share. Video moves between a loss of 0.06% and a gain of 0.01%. For example, imaging falls by 0.01% to 0.08%, optics and photonics by 0.02% to 0.07%, and analog circuits by 0.10% to 0.24%.

As for the remaining columns, the "extra-area" columns indicate that firms are allowed to add empty areas to their portfolios. I again find that the four semiconductor areas all gain share, and that the gain is smaller at a higher value of b. At $b = 0.25$, memory gains between 0.04% and 0.06%, the transistor gains between 0.05% and 0.06%, packaging gains between 0.04% and 0.05%, and lithography gains between 0.03% and 0.04%. At $b = 1$, memory gains between 0.02% and 0.03%, the transistor gains 0.02%, packaging gains between 0.00% and 0.01%, and lithography gains 0.01%. At $b = 4$, each of the four areas moves by at most 0.01%. I also find that integrated circuit design, audio, and clock circuits lose their shares.

5.2 Case II: a 25% occupancy-to-scope ratio reduction in semiconductor areas

For this exercise, I lower the occupancy-to-scope ratio by 25% in each of the four semiconductor areas. In the fixed-area columns, firms can only reshuffle the areas within the portfolio they actually occupy. I find that the four semiconductor areas gain share, and that the gain is larger at a higher value of b. At $b = 0.25$, the gains run from 0% to 1.06%. At $b = 1$, lithography gains between 5.06% and 5.86%, packaging between 5.07% and 5.80%, the transistor between 5.15% and 5.98%, and memory between 5.18% and 6.10%. At $b = 4$, memory gains between 6.36% and 7.11%, the transistor between 6.30% and 6.89%, lithography between 6.20% and 6.46%, and packaging between 6.06% and 6.28%. Meanwhile, every other area loses share at $b = 1$ and at $b = 4$. For example, at $b = 1$, display falls by 1.36% to 1.47%, imaging by 1.34% to 1.50%, and integrated circuit design by 1.38% to 1.69%.

As for the remaining columns, the extra-area columns indicate that firms are allowed to add empty areas to their portfolios. I again find that the four semiconductor areas gain share, and that the gain is larger at a higher value of b. At $b = 0.25$, each of the four areas moves by at most 0.03%. At $b = 1$, the gains range from 0.11% to 0.15% for packaging to 0.26% to 0.35% for memory, with the transistor and lithography in between. At $b = 4$, the gains are about twice as large, and memory again gains the most, between 0.48% and 0.61%. The main reason is that a 25% cut in the occupancy-to-scope ratio lowers the occupancy cost $\kappa = b\rho$ by more when each pair is worth

more. I also find that integrated circuit design, audio, and clock circuits lose their shares. At $b = 4$, integrated circuit design falls by 0.52% to 0.60%, audio by 0.46% to 0.54%, and clock circuits by 0.38% to 0.45%.

6 Conclusion

This paper develops a data-driven framework to analyze the strategic entry of U.S. electronics firms into multiple innovation areas. I define those areas from patent text, so that a single invention can sit in several domains with explicit weights. Applying Latent Dirichlet Allocation (LDA) to a large corpus of titles and abstracts yields twenty innovation areas. The resulting areas display two stylized facts: a multi-bundle innovation in the electronics industry, and the heterogeneous congestion effect. To identify the mechanisms behind these patterns, I construct an entry model and estimate the parameters by partial identification, using moment inequalities to handle bundling, equilibrium multiplicity, and unobserved firm heterogeneity.

A reduced-form linear probability model of whether a firm occupies an area reveals three patterns: (i) a higher predicted patent valuation raises the probability of entry, (ii) a larger number of rivals also raises that probability, and (iii) there is a positive relationship between portfolio size and the probability of entry. However, crowded areas could be crowded because they are attractive, so this positive relation mixes a congestion effect with the unobserved attractiveness of those areas. Similarly, the coefficient on portfolio size is about one tenth. With twenty areas, a coefficient of this size follows mechanically from how portfolio size is constructed. Furthermore, more capable firms may both occupy more areas and enter any given area more often. As such, the reduced form cannot measure a gain from scope.

The model estimates separate the above concerns. Taking predicted patent value as the unit of profit, I find that additional rivals reduce expected profit. The yearly occupancy cost, measured against the gain from occupying more areas at once, can be zero, and it can be large. Negative values of that comparison are rejected. The penalty from rivals is steep when an area has few rivals and flattens once the area is crowded. Firms also gain from occupying more areas at once, through the pairwise interactions among the areas they enter. Treating extra areas as diminishing synergies is inconsistent with the data.

In a numerical simulation, I try to mimic the areas covered by the U.S. CHIPS manufacturing incentive and the EU CHIPS Act. In particular, the coverage of those policies are memory, lithography and fabrication, packaging, and the transistor. A payment equal to one quarter of the predicted valuations shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

The paper has three limitations. First, I model the choice of areas as a static game. That is, the model answers how a firm composes its portfolio in a given year, and how the number of rivals in

Table 11: Case I: a 25% subsidy based on predicted value

	Fixed size	Extra areas		
Area		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.16, 0.46]	[0.04, 0.06]	[0.02, 0.03]	[−0.00, 0.01]
Audio and Speech-Signal Processing	[−0.09, −0.06]	[−0.03, −0.03]	[−0.02, −0.02]	[−0.02, −0.00]
Semiconductor Lithography and Fabrication Processes	[0.24, 0.53]	[0.03, 0.04]	[0.01, 0.01]	[−0.00, 0.01]
Integrated Circuit Design, Verification, and Testing	[−0.18, −0.08]	[−0.03, −0.02]	[−0.03, −0.02]	[−0.02, −0.00]
PCB and Electronic Hardware with Connectors	[−0.20, −0.09]	[−0.02, −0.02]	[−0.02, −0.01]	[−0.01, 0.00]
Video Recording, Encoding, and Media Processing	[−0.06, 0.01]	[0.00, 0.01]	[0.00, 0.02]	[0.00, 0.01]
Semiconductor Packaging and Interconnect Bonding	[0.26, 0.54]	[0.04, 0.05]	[0.00, 0.01]	[0.00, 0.01]
Thin-Film Materials, Deposition and Chemistry	[−0.15, −0.05]	[−0.01, −0.00]	[−0.01, 0.00]	[−0.00, 0.00]
Wireless Base-Station and Mobile Communications	[−0.10, −0.02]	[−0.01, −0.00]	[0.00, 0.01]	[0.00, 0.01]
Clock Generation, Timing Circuits, and Signal Processing	[−0.13, −0.04]	[−0.03, −0.03]	[−0.02, −0.01]	[−0.01, −0.00]
Power-Supply and Power-Management Electronics	[−0.13, −0.06]	[−0.03, −0.02]	[−0.02, −0.01]	[−0.01, −0.00]
Networking Protocols and Data Infrastructure	[−0.08, −0.01]	[−0.01, 0.00]	[0.00, 0.01]	[0.00, 0.01]
Transistor Gate Structures and Semiconductor Device Physics	[0.14, 0.37]	[0.05, 0.06]	[0.02, 0.02]	[−0.00, 0.01]
Imaging Sensors and Camera Pipelines	[−0.08, −0.01]	[0.01, 0.02]	[0.01, 0.02]	[−0.00, 0.01]
Processor Architecture, Caches, Instruction Execution	[−0.11, −0.03]	[−0.02, −0.01]	[−0.01, −0.00]	[−0.01, 0.00]
Display Panel Tech (LCD/OLED)	[−0.07, −0.03]	[0.01, 0.02]	[0.03, 0.03]	[0.00, 0.02]
Rotational Electromechanics and Magnetic Actuation	[−0.15, −0.09]	[−0.02, −0.01]	[−0.01, −0.01]	[−0.01, 0.00]
Optics, Lasers, and Photonics	[−0.07, −0.02]	[−0.01, −0.01]	[0.01, 0.01]	[0.00, 0.01]
Analog and Differential Amplifier Circuits	[−0.24, −0.10]	[−0.03, −0.02]	[−0.02, −0.02]	[−0.01, −0.00]
Mobile UI and Content Services	[−0.11, −0.03]	[−0.02, −0.02]	[−0.01, −0.00]	[−0.01, −0.00]
Firm-years adding an area	0	[0.76, 0.83]	[0.34, 0.43]	[0.07, 0.19]

Notes: Each area row reports the change, in percentage points, in that area's share of occupied slots, and the last row reports the change in the share of firm-years that add at least one area. Under "Fixed size", every firm keeps the number of areas it occupies in the data, while under "Extra areas" a firm keeps the areas it already occupies and may add new ones. Rival counts are those implied by the portfolios.

Table 12: Case II: a 25% occupancy-to-scope ratio reduction

	Fixed size			Extra areas		
Area	$b = 0.25$	$b = 1$	$b = 4$	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 1.00]	[5.18, 6.10]	[6.36, 7.11]	[−0.00, 0.03]	[0.26, 0.35]	[0.48, 0.61]
Audio and Speech-Signal Processing	[−0.20, 0.00]	[−1.51, −1.28]	[−1.73, −1.57]	[−0.02, 0.00]	[−0.29, −0.24]	[−0.54, −0.46]
Semiconductor Lithography and Fabrication Processes	[0.00, 1.04]	[5.06, 5.86]	[6.20, 6.46]	[0.00, 0.02]	[0.15, 0.21]	[0.32, 0.41]
Integrated Circuit Design, Verification, and Testing	[−0.23, 0.00]	[−1.69, −1.38]	[−2.18, −1.73]	[−0.01, 0.00]	[−0.32, −0.25]	[−0.60, −0.52]
PCB and Electronic Hardware with Connectors	[−0.20, 0.00]	[−1.53, −1.30]	[−1.83, −1.62]	[−0.01, 0.00]	[−0.23, −0.17]	[−0.35, −0.30]
Video Recording, Encoding, and Media Processing	[−0.34, 0.00]	[−1.47, −1.30]	[−1.55, −1.47]	[−0.00, 0.00]	[0.11, 0.14]	[0.18, 0.21]
Semiconductor Packaging and Interconnect Bonding	[0.00, 1.02]	[5.07, 5.80]	[6.06, 6.28]	[0.00, 0.03]	[0.11, 0.15]	[0.23, 0.31]
Thin-Film Materials, Deposition and Chemistry	[−0.23, 0.00]	[−1.33, −1.16]	[−1.56, −1.42]	[−0.00, 0.00]	[−0.02, 0.01]	[−0.02, −0.00]
Wireless Base-Station and Mobile Communications	[−0.32, 0.00]	[−1.55, −1.36]	[−1.66, −1.59]	[−0.00, 0.00]	[0.02, 0.05]	[0.04, 0.08]
Clock Generation, Timing Circuits, and Signal Processing	[−0.18, 0.00]	[−1.38, −1.15]	[−1.63, −1.48]	[−0.02, 0.00]	[−0.24, −0.19]	[−0.45, −0.38]
Power-Supply and Power-Management Electronics	[−0.23, 0.00]	[−1.58, −1.30]	[−2.08, −1.67]	[−0.01, 0.00]	[−0.20, −0.17]	[−0.34, −0.30]
Networking Protocols and Data Infrastructure	[−0.30, 0.00]	[−1.42, −1.28]	[−1.52, −1.48]	[−0.00, 0.00]	[0.05, 0.08]	[0.09, 0.11]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 1.06]	[5.15, 5.98]	[6.30, 6.89]	[0.00, 0.03]	[0.21, 0.28]	[0.41, 0.52]
Imaging Sensors and Camera Pipelines	[−0.35, 0.00]	[−1.50, −1.34]	[−1.58, −1.49]	[−0.00, 0.01]	[0.13, 0.15]	[0.19, 0.24]
Processor Architecture, Caches, Instruction Execution	[−0.23, 0.00]	[−1.41, −1.21]	[−1.60, −1.49]	[−0.01, 0.00]	[−0.07, −0.06]	[−0.16, −0.11]
Display Panel Tech (LCD/OLED)	[−0.30, 0.00]	[−1.47, −1.36]	[−1.53, −1.34]	[−0.00, 0.01]	[0.29, 0.35]	[0.52, 0.60]
Rotational Electromechanics and Magnetic Actuation	[−0.23, 0.00]	[−1.50, −1.26]	[−1.83, −1.61]	[−0.01, 0.00]	[−0.15, −0.11]	[−0.25, −0.21]
Optics, Lasers, and Photonics	[−0.35, 0.00]	[−1.46, −1.31]	[−1.53, −1.39]	[−0.01, 0.00]	[0.05, 0.07]	[0.17, 0.20]
Analog and Differential Amplifier Circuits	[−0.20, 0.00]	[−1.47, −1.20]	[−1.78, −1.53]	[−0.02, 0.00]	[−0.17, −0.14]	[−0.27, −0.23]
Mobile UI and Content Services	[−0.25, 0.00]	[−1.47, −1.27]	[−1.71, −1.54]	[−0.01, 0.00]	[−0.13, −0.10]	[−0.26, −0.20]
Firm-years adding an area	0	0	0	[0.00, 0.34]	[4.10, 5.51]	[8.84, 10.43]

Notes: Each area row reports the change, in percentage points, in that area's share of occupied slots, and the last row reports the change in the share of firm-years that add at least one area. Under "Fixed size", every firm keeps the number of areas it occupies in the data, while under "Extra areas" a firm keeps the areas it already occupies and may add new ones. Rival counts are those implied by the portfolios.

that year shifts its expected profit. As such, the occupancy cost I recover is a cost that the firm pays in every year it occupies an area. The game also leaves out learning, option value, preemption, and sunk costs of R&D, and these can push the estimated cost in either direction. For example, if a firm occupies an area in part because it makes a later expansion cheaper, the occupancy cost I recover is net of that benefit and understates the true yearly cost. When entry is sunk and waiting is valuable, however, the same inequalities overstate the cost. To check the role of sunk costs, I extend the revealed-preference comparisons by one period, so that the firm pays a yearly fixed cost and, in addition, a sunk cost in the year it enters an area (Appendix B.5). In this exercise, a zero sunk cost is still consistent with the data, in which case the firm only pays the yearly cost, as in the static game. Second, the sample consists of listed U.S. electronics firms from 1990 to 2019, so the findings and the simulations should be read as a case study of that industry. In particular, startups and privately held innovators are not in the sample, since market-implied patent values require a traded stock return. Third, the simulations are local exercises. They start from the portfolios that firms actually occupy, and most of the effects depend on the value of one pair of areas, which the data do not pin down and which I therefore have to choose. In the main text, the number of rivals in each area adjusts until it agrees with the new portfolios, and Appendix D reports the same exercises with the number of rivals held at its observed value.

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A Data, Methodology, and Stylized Facts

A.1 Latent Dirichlet Allocation (LDA)

Before performing LDA, I first combine the title of the patent and its abstract[18]. Then, I perform preprocessing on the combined text[19]. I proceed to tokenize the combined text, create a set of 'vocabs' consisting of unigrams (a single word) and bigrams (a combination of two words) that will be considered (V), and perform vocab-pruning[20]. This process leaves me with 7,588 vocabs V with other information such as how many times the term appears in the whole corpus (term-frequency) and how many documents have the term (document-frequency)[21]. From here, I construct the document-term matrix of dimension $P \times V$, where P is the number of patents, that will be used as an input for fitting the LDA.

I then fit LDA on this document-term matrix. The idea of LDA is that each patent is a mixture of a few latent topics, such as semiconductors or wireless communication, and that each topic is a distribution over words. For example, a patent on a base-station receiver would put most of its weight on a wireless topic, in which words such as "base station" and "uplink" have a high probability. The model therefore gives me two outputs: the share of each topic in each patent (the document-topic distribution), and the probability of each word in each topic (the topic-word distribution). Besides the document-term matrix, LDA needs three hyperparameters: the number of topics K and two Dirichlet priors. The prior τ on the topic shares of a patent governs how many topics a patent tends to cover, while the prior ω on the word probabilities of a topic governs how concentrated its vocabulary is. In the notation of Blei et al. (2003), τ is their document-topic prior and ω is their topic-word prior.

Note that a Dirichlet prior with parameters below one produces sparse mixtures, i.e., most of the weight goes to a few topics or a few words, while parameters above one produce more even mixtures. Formally, each patent p draws its topic shares, and each topic j draws its word probabilities,

$$w_{p\cdot t} \sim \text{Dirichlet}(\tau), \qquad \psi_j \sim \text{Dirichlet}(\omega).$$

[18]The results of my topic modeling are robust if I include title, abstract, and claims, at the cost of higher RAM usage and a significantly longer computation time. Including description is not feasible on my personal machine; that is, the data is about 33.6 GB, and it is difficult to perform topic modeling given the data size and the machine limitation.

[19]First, I translate the combined text into lower case, remove any punctuation and stopwords from the combined text, and I remove all standalone numbers (one or more digits) or letters from the combined text. Finally, I lemmatize the combined text. Example of sentence "Activity 1: I bought 2 coffee makers and the machine is amazing!". After the preprocessing, it becomes "activity buy coffee maker machine amazing".

[20]I set the following restriction: 1. Both unigrams and bigrams must appear at a maximum of 10% of my patent sample. This reduces the redundant words, such as 'invent', being considered in the final set of vocabs V, 2. Both unigrams and bigrams must appear at least 0.1% of my patent sample. The idea of this restriction is to limit overtly technical words, such as 'Fe_3O_4' or 'SiC', being considered in vocabs V.

[21]Using title, abstract, and claims, I have 12,788 vocabs.

The words of a patent are then drawn by first sampling a topic and then sampling a word from that topic. To estimate the model, I use the WarpLDA sampler of Chen et al. (2016), which assigns each word token to a topic by Metropolis-Hastings. The posterior means combine the priors with the counts: the topic shares of a patent mix τ with the number of its tokens assigned to each topic, and the word probabilities of a topic mix ω with how often each word is assigned to that topic. Richer versions of LDA, such as hierarchical, correlated, and dynamic topic models (Teh et al., 2004; Blei and Lafferty, 2006), have the same structure, but they add hyperparameters, and I do not use them here.

Among the chief challenges of LDA is to identify the optimal hyperparameters, in particular, the number of topics K. One might also argue that different initial priors of τ and ω can yield different results. I try to mitigate these concerns by computing several measures, summarized in Table 13, that guide the selection of the optimal hyperparameters. The first one is perplexity, which measures how well the model predicts patents that it did not see during fitting, i.e., a held-out test set. Perplexity is computed from the likelihood of the held-out words, and a lower perplexity means that the model predicts them better. However, Chang et al. (2009) show that a model with a lower perplexity does not always give topics that are easier for a human to interpret, so I do not choose the hyperparameters on perplexity alone. For this measure, I will be looking for an 'elbow', an inflection point where the decrease in perplexity is marginal compared to the increasing number of topics being considered.

Table 13: LDA Evaluation Metrics

Measures	**Purpose**	**Goal**	**Example**
Perplexity	Out-of-sample predictive fit	Lower	"Elbow"
Coherence	Within-topic semantic tightness	Higher	Mean NPMI: $0.2 - 0.3$
Exclusivity	Distinctness between topics	Higher	≥ 0.45
JSD	How much each topic's meaning drifts	Lower	≤ 0.3
Drifted Share	Fraction of topics whose top-10 words change substantially	Lower	max 40% topics drifted

To complement perplexity, I consider coherence, which evaluates how frequently top words within each topic co-occur across the documents. Coherence serves as a proxy for interpretability: higher coherence suggests that the top words in a topic form a semantically meaningful group, increasing confidence that the topic reflects a real innovation cluster (e.g., "chip," "circuit," "transistor" likely point to semiconductors). In particular, I will be using the NPMI-based coherence in evaluating the topic quality (see Mimno et al. (2011) for detailed discussion on the measurement). I optimize the topic model's hyperparameters to achieve a mean NPMI between 0.2 and 0.3, while also ensuring that the minimum NPMI remains around 0.1, to preserve the interpretability of even the weakest topics.

In addition to coherence, I include exclusivity, which assesses how uniquely a word belongs to a single topic, i.e., topic distinctiveness. A high exclusivity score indicates that top words of a topic do not overlap significantly with other topics, thus ensuring distinct innovation areas. While co-

herence favors commonality among words, exclusivity ensures separation between topics, both of which are desirable but often trade off (see Bischof and Airoldi (2012) for more discussions).

Beyond internal topic quality metrics, I also assess the semantic stability of topics over time. To evaluate this, I split the corpus into three time waves (e.g., 1990-1999, 2000-2009, and 2010–2019), which generally capture the standard narrative of technological evolution in the electronics industry, and compute the Jensen-Shannon divergence (JSD) between the topic-word distributions estimated in each wave. The JSD is a symmetric measure of how dissimilar the vocabulary of each topic is across time periods. In my implementation, JSD can take values in the range of zero and one. A low JSD indicates that topics retain similar meanings over time, which is desirable for maintaining a consistent mapping between topics and innovation areas. In particular, I am looking for the mean JSD below 0.3 when considering the optimal hyperparameters[22].

In addition, I define a measure called drifted share, which calculates the proportion of topics whose top keywords change substantially between time waves. That is, when the drifted share is low, most topics keep the same meaning across the three waves, and I can read each of them as the same innovation area throughout the sample. For example, there should be a maximum of 40% topics with share drifted more than 30% across period, when evaluating the optimal hyperparameter. Together, JSD and drifted share help identify topic models that are not only coherent and exclusive, but also temporally stable, which is essential for defining innovation areas and for the model implementation used in the next part of this paper[23]. One could also think of this as a way to mitigate the concerns raised in the dynamic, correlated topic model, as discussed by Blei and Lafferty (2006).

To find the optimal hyperparameters, I employ the following procedure (See Algorithm 1 in Appendix A.4, and Appendix A.5 for the fine-tuning result). First, I employ a 5-fold cross validation. That is, for each year, I randomly divide the patents into five parts. I will then fit the LDA for four parts and use the estimated model for the other part when evaluating and optimizing the hyperparameters. I repeat this exercise five times, each with different folds as training and validation sets. During this cross validation, I begin the iterative search[24]. In the first step, I grid search for the number of optimal topics K, while keeping other hyperparameters (τ and ω) fixed. Based on the evaluation metrics described above, I identify the optimal range for K. Next, I search over values of τ, again holding the other parameters fixed, and select its optimal range using the same set of metrics, then obtain the optimal τ and its bounds. I then repeat the process for ω. This procedure is iterated until further refinements to the hyperparameters do not yield significant improvements in the evaluation metrics.

As discussed earlier, the outputs of the LDA model are the document-topic and topic-word distributions. For each topic, I extract the top 25 most probable words and assign an interpretable label to the topic. To assist with labeling, I utilize two large language models (OpenAI's ChatGPT o3 and xAI's Grok 3 + Think) to propose candidate labels based on the top words associated with each topic. I then iterate between the two models until their suggested labels converge or are manually reconciled, using the agreed-upon label as the final topic label.

[22]See for example Lu et al. (2017). Their treatment of the extended JSD measure is applied to compare vast Twitter (now X) corpora across different time periods.

[23]I employ two procedures to match the distance matrix: greedy matching and Hungarian matching. Greedy matching does not require a 1-to-1 match between topics in one period to another, while Hungarian matching will enforce a 1-to-1 match. See Wang and McCallum (2006), Niekler and Jähnichen (2012), and Lu et al. (2017) for discussions.

[24]This algorithm is conceptually similar to the Expectation-Maximization (EM) algorithm in statistics, or to coordinate ascent and iterative grid search methods used in machine learning.

A.2 From topic proportions to area weights

The document-topic posterior derived above is the source of the patent-area weights used in Section 3. For a patent p filed by firm i in period t, the Dirichlet-multinomial posterior mean is a distribution over the J areas,

$$w_{pjt} \;=\; \frac{n_{pj} + \tau}{\sum_{j'=1}^{J}(n_{pj'} + \tau)},$$

where n_{pj} is the number of tokens in patent p assigned to area j, τ is the Dirichlet prior on patent-topic proportions, and $J = 20$ in the fitted model. By construction of the Dirichlet,

$$w_{pjt} \geq 0 \qquad \text{and} \qquad \sum_{j=1}^{J} w_{pjt} = 1.$$

A single patent can therefore assign positive weight to several areas at once, with the weights giving the patent's proportionate exposure to each area.

To construct the firm-year objects used in the estimation, I first set every patent weight below a threshold $\bar{w}$ to zero[25]. That is, if the fitted model gives a patent only a negligible share of its text in an area, the patent does not count toward that area. The remaining fitted weights are left as they are. The area-specific dollar value of the patent is then the product

$$A_{ipjt} \;=\; w_{pjt}\, A_{ipt},$$

where A_{ipt} is the market-implied patent value constructed in Appendix B.1. Summing over the firm's patents in the period gives the realized area valuation that enters the first-stage projection,

$$\sum_{p \in \mathcal{P}_{it}} A_{ipjt} \;=\; \sum_{p \in \mathcal{P}_{it}} w_{pjt}\, A_{ipt}.$$

The same weights define the firm-year entry indicator in Section 3: firm i is defined as entering area j in period t when the average patent weight in that area is at least higher or equal to the threshold $\bar{w}$[26],

$$d_{ijt} \;\equiv\; \mathbf{1}\left\{ \frac{1}{|\mathcal{P}_{it}|} \sum_{p \in \mathcal{P}_{it}} w_{pjt} \geq \bar{w} \right\}, \qquad \bar{w} = 0.01.$$

A.3 The entry threshold

The cutoff $\bar{w}$ is constructed from the patent-level weights w_{pjt}. First, I find that many of those weights are exact zeros, i.e., the fitted model assigns none of that patent's text to the area. On the strictly positive weights, $w_{pjt} > 0$, I take the natural logarithm $\ln(w_{pjt})$. Let f be the density of this logarithm. Figure 1 shows that f has two modes, which I denote by $m_L < m_R$. The low point between the modes is the logarithm at which positive weights are least common,

$$x^* \;=\; \arg \min_{x \in (m_L, m_R)} f(x),$$

[25] See Appendix A.3 for further details.
[26] See Appendix A.3 for further details.

and the share at that point is e^{x^*}. In these weights e^{x^*} is about $0.0094 \approx 0.01$ or $\bar{w} = 1\%$. I then define entry as the proportion of firm-year averages that are above this cutoff,

$$d_{ijt} \;=\; \mathbf{1}\left\{ \frac{1}{|\mathcal{P}_{it}|} \sum_{p \in \mathcal{P}_{it}} w_{pjt} \geq \bar{w} \right\}.$$

Figure 1 shows the density f. To draw it, I take the natural logarithm of every strictly positive patent-area weight and group these logarithms into bins of equal width. Each bar then shows the fraction of positive weights that falls into that bin. The lines denote the shares of 0.5%, 1%, 2%, and 5%. The solid red line is $\ln(0.01)$, i.e., where we find the lowest point between the two modes.

The figure shows two groups of positive weights. The group on the left consists of very small shares, mostly well below 1% of the text. These are leftovers: once a patent has placed most of its text in one or two areas, the fitted model often still gives a little weight to several other areas. For example, a patent on a flash memory cell may put most of its text in memory and in transistor device physics, and still leave a fraction of a percent in display. The group on the right consists of the shares that describe what the patent is about, from a few percent of the text up to nearly the whole patent. Shares of about 1% are the least common, so a cutoff there keeps the shares that make up the patent and drops the leftovers. In comparison, a cutoff of 0.5% would still keep some of the leftovers, while a cutoff of 5% would already drop some of the shares that describe the patent.

I also provide Figure 2 where it sorts the twenty weights inside each patent from largest to smallest. At each rank I plot the median and the mean across patents. The right panel begins at the third-largest weight, so that the same four lines can be read against the smaller ranks. In the median patent the largest weight is about 61% of the text, the second about 23%, the third about 8%, and the fourth just about 0.9%. The 1% line falls at that fourth weight.

Figure 1: Density of the natural logarithm of the positive patent-area weights. Exact zeros are omitted. The vertical lines mark shares of 0.5%, 1%, 2%, and 5% of the patent's text, and the solid line is the 1% cutoff.

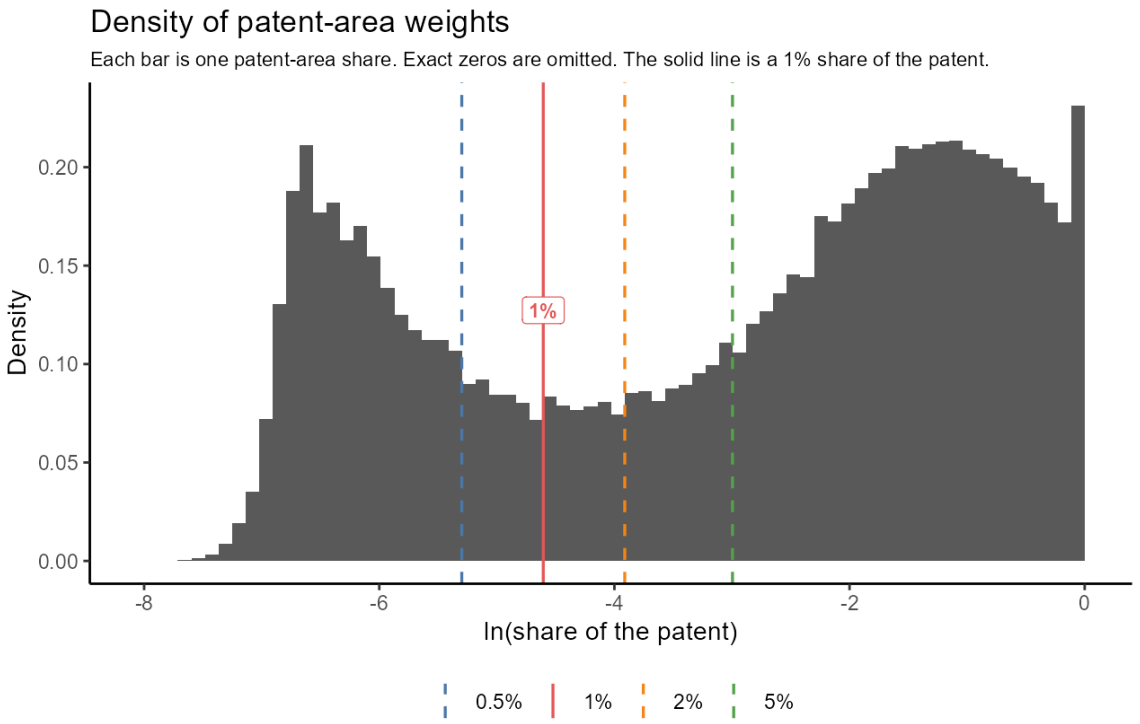
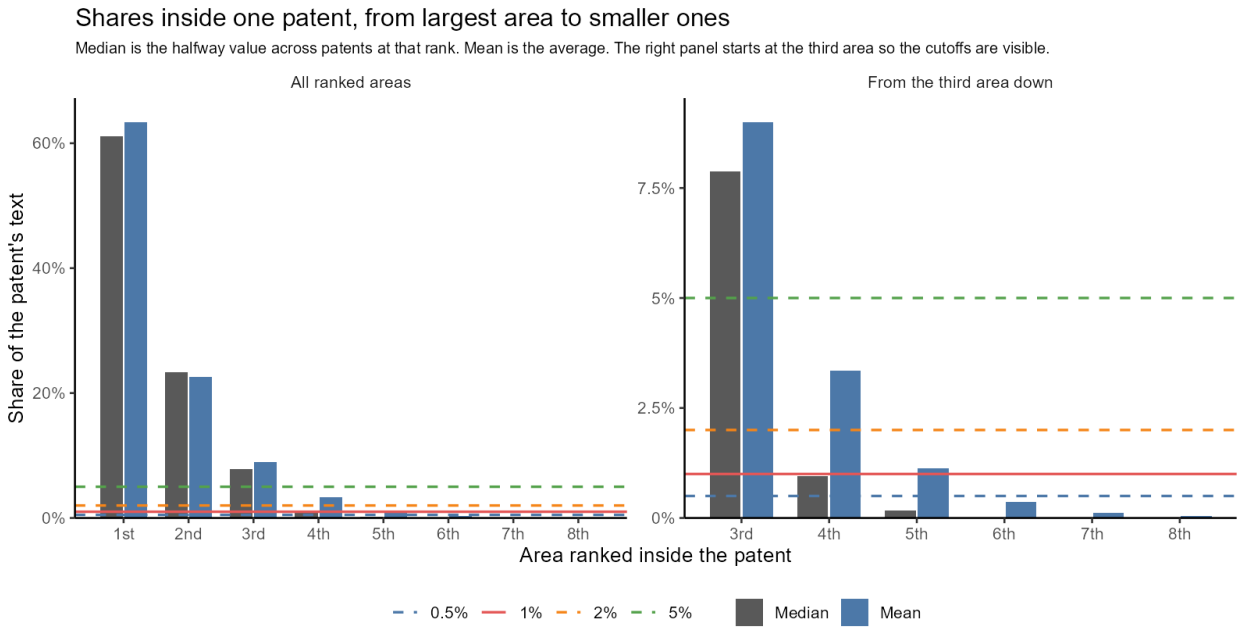


Figure 2: Patent-area weights ranked from largest to smallest within each patent. For each rank, the gray bar is the median across patents and the blue bar is the mean. The right panel starts at the third-largest weight, so that the four cutoff lines can be read against the smaller weights.



A.4 LDA Algorithm

Algorithm 1 Heuristic Hyperparameter Tuning Procedure for LDA

Input: Document-term matrix (DTM), Term-Co-occurence matrix (TCM), initial values for (K, τ, ω)
Set seed for the random number generator, e.g., 1234
Split data using 5-fold cross validation
Fit LDA on the set of training data
Cross-validate on set of test data
repeat
    Step 1: Tune K ▷ Grid search over K, fix τ, ω
        Evaluate perplexity, coherence, exclusivity, JSD, drifted share
        Update $K \leftarrow K^*$
    Step 2: Tune τ ▷ Grid search over τ, fix K, ω
        Evaluate same metrics
        Update $\tau \leftarrow \tau^*$
    Step 3: Tune ω ▷ Grid search over ω, fix K, τ
        Evaluate same metrics
        Update $\omega \leftarrow \omega^*$
until No significant improvement in evaluation metrics
Output: Optimal (K^*, τ^*, ω^*)

Important notions:

- Document-term matrix (DTM) will be used as the main input for fitting LDA. Dimension: $P \times V$, where P is the number of patents and V is the number of vocabs.
- Term Co-occurence matrix (TCM) is an important input to compute one of the evaluation metrics 'coherence' of a topic. Dimension $V \times V$. To construct TCM, I do the following:
  - First, convert DTM into a binary matrix B of shape $P \times V$.
  - Then, TCM is simply $B^T \cdot B$.

A.5 Fine-tuning LDA

As indicated in Algorithm 1, I first start with grid search of the number of topics K. I do this sequence in two parts; first, I sequence from 5 to 100, and qualitatively evaluate the metrics explained earlier in Section 2.2. Then, I sequence from 15 to 25. From here, I settle the number of topics at $K = 20$.

The second step is to perform a grid search for τ. First, I search for τ in this region, $\{0.01, 0.02, 0.05, 0.1, 0.5, 1\}$. Then, I sequence, with width of 0.01, from 0.01 to 0.15. For this step, I finally chose the hyperparameter τ to be at 0.05.

The final step is to perform a grid search for ω. I first search for ω in this region, (0.001, 0.002, 0.005, 0.007, 0.01, 0.02, 0.05, 0.07, 0.1, 0.2, 0.5, 0.7). Then, I sequence from 0.001 to 0.015 with a width of 0.001. From here, I chose the hyperparameter ω to be at 0.001.

To summarize the steps and the resulting optimal hyperparameters chosen in this paper, see Table 14 below.

Table 14: Summary of LDA finetuning

Hyperparameters	Steps and Results
Number of topics K	1. Grid search from 5 to 100, with width of 5 2. Grid search from 15 to 25, with width of 1 Optimal number of topics $J = 20$
Document-topic priors τ	1. Grid search in the region (0.01, 0.02, 0.05, 0.1, 0.5, 1) 2. Grid search from 0.01 to 0.15, with width of 0.01 Optimal $\tau^* = 0.05$
Topic-word priors ω	1. Grid search in the region (0.001, 0.002, 0.005, 0.007, 0.01, 0.02, 0.05, 0.07, 0.1, 0.2, 0.5, 0.7) 2. Grid search from 0.001 to 0.015, with width 0.001 Optimal $\omega^* = 0.001$

Let us now look at the coarse search over $K \in \{5, \ldots, 100\}$, which drives the choice of $K = 20$. Figure 3 shows that the decline in perplexity slows down after $K = 20$. At the same time, the mean NPMI stays inside the target band of 0.2 to 0.3, and the minimum NPMI is highest at $K = 20$. Figure 4 shows that the mean exclusivity remains above 0.75 around $K = 20$. As for the stability of the topics over time, Figures 5 and 6 show the Jensen-Shannon divergence and the drifted share from one decade to the next. Under Hungarian matching, the mean JSD stays below 0.3 for every K between 5 and 20. Combined with the searches over τ and ω described above, I therefore set $K = 20$, $\tau = 0.05$, and $\omega = 0.001$, and finer grids around these values lead to the same choice.

A.6 Topic labels and top words

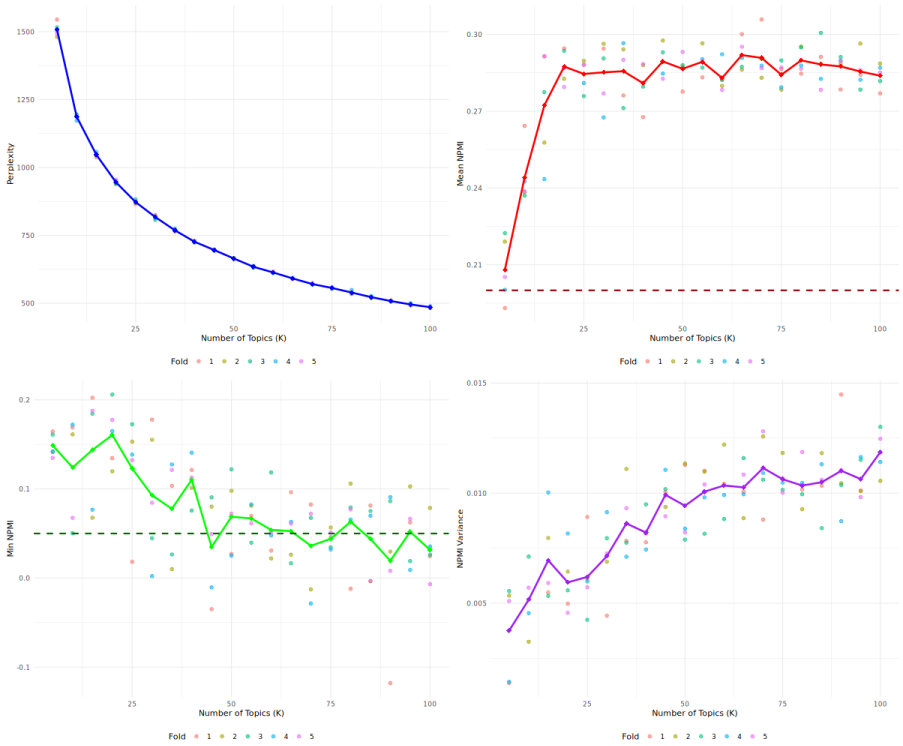
Table 15 reports the twenty labels and the highest-probability words. The main text lists labels only.

Table 15: Topics and Top Words

Topic	Label	Top Words
1	Semiconductor Memory Architecture	cell, memory_cell, line, row, array, bite_line, column, memory_device, memory_array, flash, word_line, nonvolatile, flash_memory, nonvolatile_memory, word, erase, cell_include, semiconductor_memory, bite, sram, memory_element, refresh, cell_array, transmission_line, dram
2	Audio and Speech-Signal Processing	value, estimate, symbol, calculate, speech, parameter, metric, measure, correlation, sequence, train, coefficient, sample, tone, set, noise, audio_signal, model, threshold, threshold_value, echo, audio, pilot, sound
3	Semiconductor Lithography and Fabrication Processes	dielectric, etch, dielectric_layer, mask, metal, pattern, silicon, metal_layer, deposit, layer_form, photoresist, material, barrier, nitride, expose, dielectric_material, first_dielectric, barrier_layer, mask_layer, oxide_layer, trench, remove, oxide, sacrificial
4	Integrated Circuit Design, Verification, and Testing	test, design, integrate, integrate_circuit, mask, logic, programmable, port, logic_device, ic, programmable_logic, layout, circuitry, circuit_design, output_port, pld, test_mode, method_test, device_test, test_signal, input_port, functional, test_circuit, architecture

Topic	Label	Top Words
5	PCB and Electronic Hardware with Connectors	connector, board, cable, circuit_board, house, module, end, print_circuit, assembly, electronic, conductor, mate, mount, antenna, 2_end, electronic_component, electrical_connector, first_end, socket, contact, print, receptacle, enclosure, component
6	Video Recording, Encoding, and Media Processing	record, video, stream, encode, reproduce, code, record_medium, block, picture, video_signal, decode, medium, frame, video_datum, reproduction, datum_stream, information_record, encoder, motion_vector, bit, compress, datum_record, datum_block, parity
7	Semiconductor Packaging and Interconnect Bonding	die, package, pad, bond, chip, conductive, solder, bump, first_surface, bond_pad, semiconductor_die, lead, semiconductor_package, semiconductor_chip, 2_surface, mems, microelectronic, ball, lead_frame, package_include, electrically, mold, interposer, leadframe
8	Thin-Film Materials, Deposition and Chemistry	composition, glass, temperature, particle, polymer, resin, gas, compound, film, coat, fuel, polish, ceramic, acid, mixture, reaction, electrolyte, carbon, catalyst, fuel_cell, melt, solution, hydrogen, powder
9	Wireless Base-Station and Mobile Communications	station, wireless, base_station, wireless_communication, transmission, ue, channel, radio, communication_system, uplink, user_equipment, downlink, mobile_station, equipment, resource, control_channel, equipment_ue, subframe, radio_communication, report, reception, wireless_device, communication_device
10	Clock Generation, Timing Circuits, and Signal Processing	clock, frequency, phase, clock_signal, delay, filter, analog, digital, oscillator, rf, output_signal, loop, input_signal, pulse, digital_signal, datum_signal, rf_signal, mixer, adc, pll, signal_2, signal_generate, gain, baseband
11	Power-Supply and Power-Management Electronics	power, power_supply, mode, switch, consumption, power_amplifier, supply, battery, power_source, low_power, transformer, ac, dc, power_consumption, power_converter, power_control, reduce_power, power_management, output_power, wireless_power, first_power, 2_power, primary, secondary
12	Networking Protocols and Data Infrastructure	network, packet, protocol, node, message, mode, link, network_node, communication_network, datum_packet, ip, service, traffic, network_device, access_point, subscriber, gateway, access_network, session, first_network, wireless_network, network_element, area_network, entity
13	Transistor Gate Structures and Semiconductor Device Physics	region, gate, fin, gate_electrode, drain, dope, sourcedrain, gate_structure, region_2, semiconductor_layer, conductivity_type, drain_region, source_drain, channel_region, region_form, conductivity, implant, region_first, epitaxial, sourcedrain_region, active_region, dopant, ptype, transistor
14	Imaging Sensors and Camera Pipelines	image, sensor, display, camera, image_datum, object, screen, information_process, image_process, capture, image_sensor, image_signal, process_apparatus, section, image_display, touch, color, display_unit, scene, capture_image, apparatus_image, image_capture, first_image, display_screen
15	Processor Architecture, Caches, Instruction Execution	processor, instruction, address, cache, bus, register, execution, command, store, host, memory_controller, interrupt, cpu, execute, memory_access, buff, thread, graphic, storage_device, controller, pipeline, entry, operand, transaction
16	Display Panel Tech (LCD/OLED)	electrode, liquid_crystal, panel, crystal, pixel, display_device, liquid, display, display_panel, first_electrode, 2_electrode, pixel_electrode, crystal_display, solidstate, organic, cpu, gate_line, organic_light, film_transistor, photoelectric, solidstate_image, electrode_2, lcd, plasma_display
17	Rotational Electromechanics and Magnetic Actuation	member, air, magnetic, head, shaft, rotate, roller, main_body, refrigerator, magnet, rotation, actuator, wash, motor, rotor, door, nozzle, cartridge, drum, inner, body, compressor, movable, laundry
18	Optics, Lasers, and Photonics	optical, light, lens, laser, fiber, beam, wavelength, light_source, optical_fiber, emit, laser_beam, waveguide, reflect, light_beam, optical_signal, refractive, incident, polarization, reflection, light_guide, illumination, grate, reflector, optical_system
19	Analog and Differential Amplifier Circuits	voltage, current, bias, differential, transistor, amplifier, resistor, reference_voltage, stage, current_source, input_terminal, output_terminal, voltage_level, first_voltage, supply_voltage, charge_pump, 2_voltage, terminal, output_node, driver, output_voltage, bias_voltage, voltage_first, voltage_2
20	Mobile UI and Content Services	user, key, content, server, user_interface, mobile_device, application, authentication, telephone, compute_device, item, computer, encrypt, mobile, security, client, encryption, location, database, computer_program, remote, phone, game, file

Figure 3: Perplexity and coherence. Coarse grid of K from 5 to 100.



A.7 Administrative-class composition

Table 16 reports how many class codes the examiners list on a grant in each decade. The number of codes grows over time: the mean rises from 5.1 codes in the 1990s to 8.0 in the 2010s, and the median from 3 to 6. However, the examiners only list the codes and do not say how much of the patent belongs to each of them, so the codes alone cannot give the weights that LDA gives.

Table 16: Listed Cooperative Patent Classification codes per grant

Decade	Grants	Mean listed codes	Median listed codes
1990s	109,872	5.1	3
2000s	250,567	6.1	4
2010s	270,930	8.0	6

Notes: Counts are the class codes listed on each grant in the electronics patent sample.

Table 17 shows, for each area, its top words next to the subclasses of the Cooperative Patent Classification (CPC) that receive most of the area's weight. What is interesting is that lithography, packaging, and transistor device physics all have the same main subclass, H01L. However, their top words describe different processes, i.e., etching and photoresist for lithography, dies and bond pads for packaging, and gates and drains for the transistor. A market defined by subclass H01L would therefore treat these three areas as one.

Figure 4: Perplexity and exclusivity. Coarse grid of K from 5 to 100.



Table 17: Innovation areas, top words, and administrative classes

Topic	Area	Top words	CPC subclasses
1	Semiconductor Memory Architecture	cell, memory_cell, line, row, array, bite_line, column, memory_device	G11C (48.0%); H01L (21.8%); G06F (8.7%)
2	Audio and Speech-Signal Processing	value, estimate, symbol, calculate, speech, parameter, metric, measure	H04L (16.9%); G06F (10.7%); H04N (8.8%)
3	Semiconductor Lithography and Fabrication Processes	dielectric, etch, dielectric_layer, mask, metal, pattern, silicon, metal_layer	H01L (74.9%); G03F (4.2%); H05K (1.6%)
4	Integrated Circuit Design, Verification, and Testing	test, design, integrate, integrate_circuit, mask, logic, programmable, port	G06F (20.1%); H01L (14.5%); G01R (8.4%)
5	PCB and Electronic Hardware with Connectors	connector, board, cable, circuit_board, house, module, end, print_circuit	H01R (15.6%); H01L (12.6%); H05K (9.5%)
6	Video Recording, Encoding, and Media Processing	record, video, stream, encode, reproduce, code, record_medium, block	H04N (33.7%); G11B (23.0%); H04L (9.3%)
7	Semiconductor Packaging and Interconnect Bonding	die, package, pad, bond, chip, conductive, solder, bump	H01L (70.5%); H05K (4.5%); G01R (1.9%)
8	Thin-Film Materials, Deposition and Chemistry	composition, glass, temperature, particle, polymer, resin, gas, compound	H01L (23.1%); H01M (5.9%); C08G (3.0%)

Topic	Area	Top words	CPC subclasses
9	Wireless Base-Station and Mobile Communications	station, wireless, base_station, wireless_communication, transmission, ue, channel, radio	H04W (45.3%); H04L (24.5%); H04B (11.9%)
10	Clock Generation, Timing Circuits, and Signal Processing	clock, frequency, phase, clock_signal, delay, filter, analog, digital	H04L (10.3%); H04B (9.6%); H03L (7.4%)
11	Power-Supply and Power-Management Electronics	power, power_supply, mode, switch, consumption, power_amplifier, supply, battery	H02M (12.4%); G06F (10.7%); H02J (7.8%)
12	Networking Protocols and Data Infrastructure	network, packet, protocol, node, message, mode, link, network_node	H04L (37.8%); H04W (32.7%); G06F (6.3%)
13	Transistor Gate Structures and Semiconductor Device Physics	region, gate, fin, gate_electrode, drain, dope, sourcedrain, gate_structure	H01L (82.9%); G11C (2.3%); G06F (1.5%)
14	Imaging Sensors and Camera Pipelines	image, sensor, display, camera, image_datum, object, screen, information_process	H04N (26.5%); G06F (16.8%); G06T (9.4%)
15	Processor Architecture, Caches, Instruction Execution	processor, instruction, address, cache, bus, register, execution, command	G06F (55.1%); G11C (9.2%); H04L (8.0%)
16	Display Panel Tech (LCD/OLED)	electrode, liquid_crystal, panel, crystal, pixel, display_device, liquid, display	H01L (28.2%); G02F (21.6%); G09G (17.3%)
17	Rotational Electromechanics and Magnetic Actuation	member, air, magnetic, head, shaft, rotate, roller, main_body	G11B (8.1%); H01L (7.0%); D06F (4.6%)
18	Optics, Lasers, and Photonics	optical, light, lens, laser, fiber, beam, wavelength, light_source	G02B (23.6%); H01L (10.2%); G11B (9.9%)
19	Analog and Differential Amplifier Circuits	voltage, current, bias, differential, transistor, amplifier, resistor, reference_voltage	H03K (14.1%); G11C (12.0%); H03F (10.8%)
20	Mobile UI and Content Services	user, key, content, server, user_interface, mobile_device, application, authentication	G06F (23.8%); H04L (15.5%); H04W (11.0%)

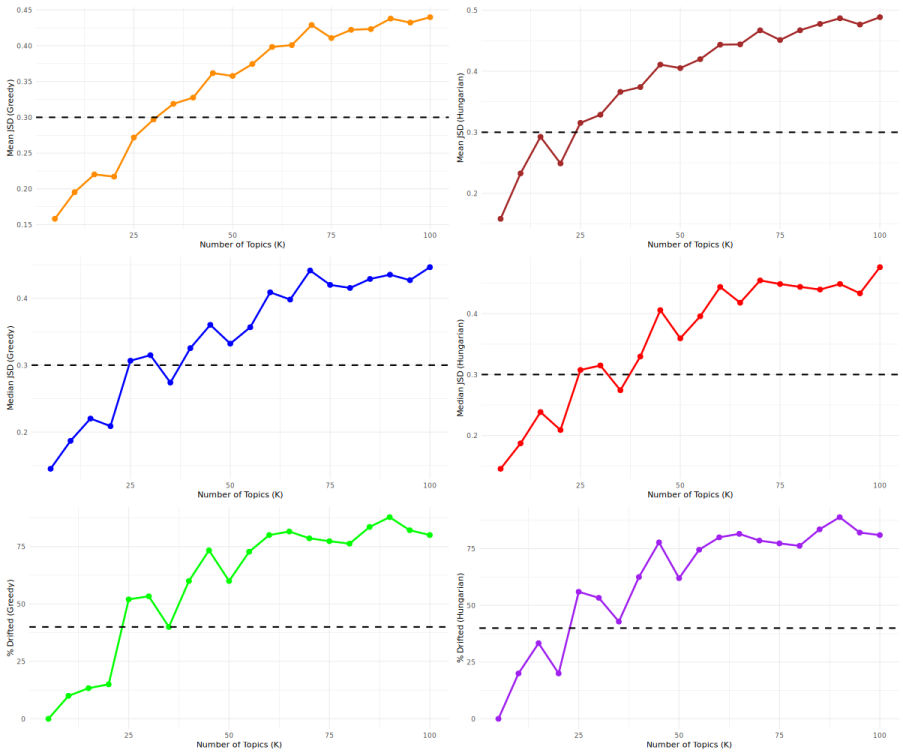
Notes: The top words are the eight words with the highest probability in each topic. The CPC subclasses are ranked by the weight they receive from the area, where I split the topic weight of each patent equally across the class codes listed on that grant. The ranks leave out section Y, which the CPC uses for tagging.

A.8 Competitor counts and valuations by decade

Tables 18 and 19 report the average rival counts and patent values by topic and decade, i.e., the levels behind the share tables in the main text.

Table 18 shows that every area has many rivals, but the numbers differ a lot across areas. Integrated circuit design and audio have the most, with about 165 and 157 rivals on average, while display has the fewest, with about 41. Memory is in between, with about 76. As for the changes over time, most areas are more crowded in the 2000s than in the 1990s, and less crowded in the 2010s than in the 2000s. For example, the number of rivals in memory falls from about 86 in the 2000s to about 56 in the 2010s, and in integrated circuit design from about 195 to about 134. Similarly, thin-film materials fall from about 131 to about 89, and rotational electromechanics from about 148 to about 113. Wireless and networking also decline from their peaks in the 2000s, but

Figure 5: Jensen-Shannon divergence and drifted share, 1990s to 2000s. Coarse grid of K from 5 to 100.



both stay above their counts in the 1990s.

These are the levels that enter the congestion term $h(N) = \text{asinh}(N)$, so one more rival matters more in display than in integrated circuit design (see Section 4 for the size of this difference). In the case study, the rival counts adjust to the new portfolios, and Appendix D reports the same exercises with the rival counts held at their observed values.

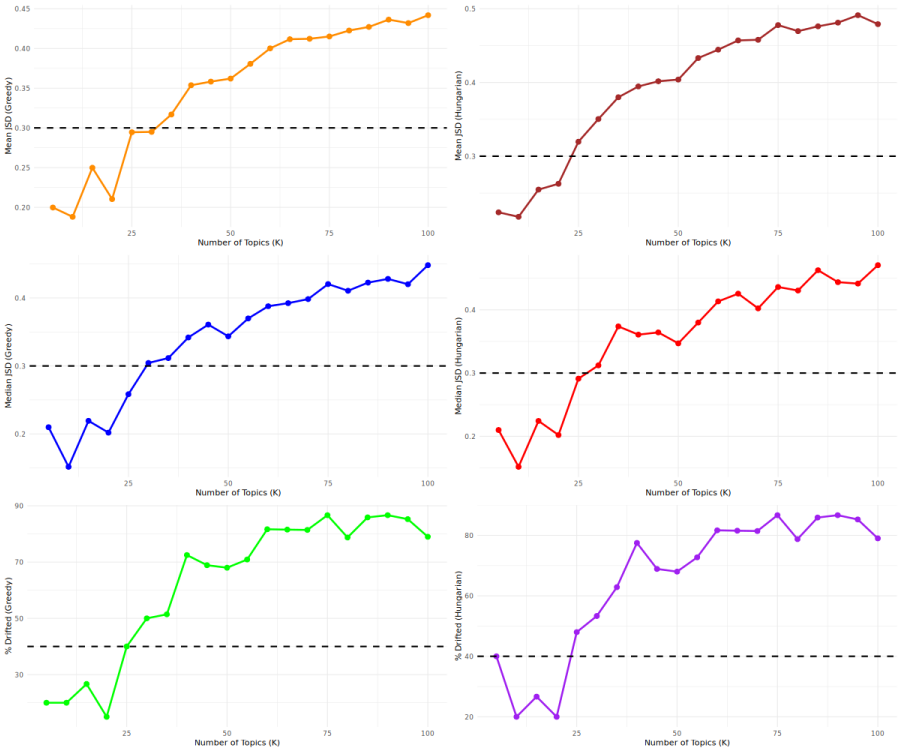
A.9 First-stage fit by area

Table 20 reports, for each of the twenty areas, the standard deviation of realized v_{ijt} and of fitted $\check{v}_{ijt}$, together with mean absolute deviation. On average across the areas, the standard deviation of the fitted value is 0.84, against 0.99 for the realized value, i.e., the projection removes part of the variation in realized value. The mean absolute deviation between the two is about 0.34 on average, and it ranges from about 0.28 in memory to about 0.41 in rotational electromechanics.

A.10 Persistence of area occupancy

The static game treats each year's portfolio as a new choice. To see how far this is from the data, I look at how much of a portfolio carries over from one year to the next (Table 21). First, I find that occupancy is persistent, but far from permanent. If a firm occupied an area in the previous year, it occupies the area again in 68.5 percent of the cases, i.e., about one third of occupied areas are vacated in the following year. If the firm did not occupy the area, it enters in 11.5 percent of the cases. Second, spells are short and returns are common. The median spell lasts one year, 58.4 percent of all spells last one year, and the mean spell lasts 2.94 years. Furthermore, 61 percent

Figure 6: Jensen-Shannon divergence and drifted share, 2000s to 2010s. Coarse grid of K from 5 to 100.



of entries are returns to an area that the firm occupied earlier in the sample. About half of these returns come after one year away (49.5 percent), 20.9 percent after two years, and 29.7 percent after three years or more. Third, among three-year windows, entries that are abandoned in the next year (10,330) outnumber entries that last at least two years (6,575). In 16.1 percent of the windows (31,328), the firm occupies the area in all three years, and in 57.6 percent (112,230), it stays out in all three years. Similarly, a firm that leaves an area is back in the next year in 5,276 windows and still out in 11,497. If entry were irreversible and sunk costs were large, spells would be long and returns would be rare. The pattern above is closer to a yearly decision in which part of the portfolio carries over.

As for the size of the portfolio, Table 21 shows that the firm occupies no area in 39.9 percent of the firm-years. In 39.0 percent of the firm-years, the firm occupies the same number of areas as in the previous year, and the correlation between this year's count and last year's count is 0.833. That is, a portfolio can keep a similar size from one year to the next while the areas inside it change.

Table 22 reports linear probability models of occupancy on last year's occupancy in the same area, the predicted valuation, the congestion term, and the number of other areas that the firm occupied in the previous year. With year effects only, last year's occupancy raises the probability of occupancy this year by 57 percentage points. Once I add firm-by-area effects, which absorb a firm's persistent advantage in an area, the coefficient falls to 9 percentage points, and adding area-by-year effects leaves it at that level. As for the other variables, predicted valuation raises the probability of occupancy by 7.6 percentage points with year effects and by 10.3 percentage points with firm-by-area effects. The coefficient on the inverse hyperbolic sine of the rival count is 11.0 percentage points with year effects and 3.9 percentage points with firm-by-area effects, and one

Table 18: Competitor counts by topic and decade

Topic	Label	1990s	2000s	2010s	Mean	SD
1	Semiconductor Memory Architecture	78.21	85.87	56.23	75.59	18.39
2	Audio and Speech-Signal Processing	144.49	182.38	135.87	156.65	31.26
3	Semiconductor Lithography and Fabrication Processes	90.13	110.33	81.42	95.58	19.25
4	Integrated Circuit Design, Verification, and Testing	155.05	194.85	134.22	164.87	35.76
5	PCB and Electronic Hardware with Connectors	142.59	164.38	127.97	147.16	23.77
6	Video Recording, Encoding, and Media Processing	76.85	94.41	63.95	80.25	17.79
7	Semiconductor Packaging and Interconnect Bonding	77.84	116.57	98.62	97.7	21.15
8	Thin-Film Materials, Deposition and Chemistry	128.76	130.93	89.09	119.62	24.89
9	Wireless Base-Station and Mobile Communications	75.97	110.2	89.96	92.42	20.77
10	Clock Generation, Timing Circuits, and Signal Processing	142.86	173.42	122.23	149.24	31.58
11	Power-Supply and Power-Management Electronics	123.27	155.58	119.92	134.65	25.09
12	Networking Protocols and Data Infrastructure	75.11	110.63	85	91.02	23.24
13	Transistor Gate Structures and Semiconductor Device Physics	74.05	95.88	71.04	81.55	16.07
14	Imaging Sensors and Camera Pipelines	64.76	83.04	80.26	75.56	13.67
15	Processor Architecture, Caches, Instruction Execution	106.24	139.03	105.6	118.48	25.26
16	Display Panel Tech (LCD/OLED)	38.63	45.8	36.6	40.83	8.18
17	Rotational Electromechanics and Magnetic Actuation	150.07	147.76	113.25	139.95	22.69
18	Optics, Lasers, and Photonics	78.65	96.04	74.21	84.11	14.26
19	Analog and Differential Amplifier Circuits	129.29	155.28	112.9	135.01	27.31
20	Mobile UI and Content Services	113.88	145.6	109.77	124.84	23.82

more other area occupied last year raises occupancy by 1.2 to 1.9 percentage points. Column (4) leaves out the rival count, because within an area-year the count differs across firms only through the firm's own occupancy.

The 9 percentage points that remain are what a sunk cost of entering an area, or a serially correlated unobserved shock, would have to explain. Note that this number likely understates the state dependence, since with firm-by-area effects and a short panel the coefficient on last year's occupancy is biased toward zero. Furthermore, the predicted valuation $\check{v}_{ijt}$ already carries firm-by-area and year-by-area effects, because I estimate the first-stage projection separately for each area with firm and year effects. Appendix B.5 asks how large a sunk cost of entering an area would have to be to account for the one-year spells and the quick returns.

B Model and empirical implementation

This appendix provides the details behind Section 3. I first restate the market-implied patent values of Kogan et al. (2017) in the notation of this paper (Appendix B.1). I then derive the characterization of the optimal bundle and the three families of moment inequalities (Appendix B.2), describe the estimation (Appendix B.3), show which objects the add, drop, and swap comparisons recover (Appendix B.4), and extend these comparisons by one period to separate a fixed cost from a sunk cost (Appendix B.5).

B.1 Market-implied patent values

For the value of each patent, I use the data of Kogan et al. (2017). That is, each patent p of firm i that is granted in period t has a dollar value A_{ipt}, which they construct from the stock market reaction

Table 19: Patent valuation by topic and decade

Topic	Label	1990s	2000s	2010s	Mean	SD
1	Semiconductor Memory Architecture	21.14	13.25	32.22	20.94	185.68
2	Audio and Speech-Signal Processing	59.98	67.45	164.38	89	946.12
3	Semiconductor Lithography and Fabrication Processes	37.22	27.85	57.59	38.79	334.05
4	Integrated Circuit Design, Verification, and Testing	48.72	44.95	99.48	60.03	511.19
5	PCB and Electronic Hardware with Connectors	33.69	36.43	116.4	55.48	702.7
6	Video Recording, Encoding, and Media Processing	21.46	19.92	43.85	26.5	263.56
7	Semiconductor Packaging and Interconnect Bonding	25.43	23.09	65.97	34.72	294.2
8	Thin-Film Materials, Deposition and Chemistry	57.21	53.77	63.61	57.52	563.75
9	Wireless Base-Station and Mobile Communications	32.71	37.62	159.88	66.48	1091.89
10	Clock Generation, Timing Circuits, and Signal Processing	56.24	46.29	85.24	59.75	421.75
11	Power-Supply and Power-Management Electronics	22.07	24.39	67.12	34.25	305.21
12	Networking Protocols and Data Infrastructure	32.29	39.73	90.41	49.69	499.55
13	Transistor Gate Structures and Semiconductor Device Physics	26.28	20.12	57.12	31.69	264.17
14	Imaging Sensors and Camera Pipelines	14.15	22.22	115.92	42.74	793.96
15	Processor Architecture, Caches, Instruction Execution	94.46	63.72	197.53	108.7	1345.49
16	Display Panel Tech (LCD/OLED)	4.44	6.15	27.71	10.93	185.31
17	Rotational Electromechanics and Magnetic Actuation	27.87	24.15	62.5	35.16	311.01
18	Optics, Lasers, and Photonics	17.74	18.64	39.9	23.64	265.96
19	Analog and Differential Amplifier Circuits	57.8	45.97	83	59.65	439.15
20	Mobile UI and Content Services	45.76	81.45	281.62	118.45	2037.15

to the grant. I take their series as given. In this subsection, I only restate their measurement model in the notation of this paper, i.e., for a patent of firm i instead of a generic grant j, so that it lines up with the area-level valuation v_{ijt} in Section 3.

The idea of Kogan et al. (2017) is the following. The USPTO announces grants on a known day, so the stock market learns on that day that the patent has been granted. However, before the announcement the market already expects the patent to be granted with some probability. If there is no other news on that day, the change in the firm's market value therefore only reflects the unexpected part of the patent's private value. Let ϖ_{ipt} be the market's ex-ante probability that the application is granted, and let ξ_{ipt} be the present value of the associated monopoly rents. Then

$$\Delta V_{ipt} \;=\; \left(1 - \varpi_{ipt}\right) \xi_{ipt}. \tag{4}$$

That is, the reaction on the announcement day understates the private value of the patent by the factor $1 - \varpi_{ipt}$. Note that ξ_{ipt} is the value of the patent itself, which is a different quantity from the value of the whole research program behind it. Following Kogan et al. (2017), I use a three-day window $[t, t+2]$ around the grant date. Even within this window, the return of the firm also reflects news that is unrelated to the patent. Let the market-adjusted return of firm i around the grant of patent p be

$$R_{ipt} \;=\; \nu_{ipt} + \varsigma_{ipt}, \tag{5}$$

where ν_{ipt} is the value of the patent as a fraction of the firm's market capitalization and ς_{ipt} is the residual return. Kogan et al. (2017) treat ν_{ipt} as a truncated normal, $\nu_{ipt} \sim N^{+}(0, \sigma^2_{\nu,it})$, and the

Table 20: Comparison of Predictive Performance Across Specifications

Topic	Label	$\mathrm{sd}(v_{ijt})$	$\mathrm{sd}(\hat{v}^{fe}_{ijt})$	$\mathrm{sd}(\hat{v}^{pool}_{ijt})$	MAD^{fe}	MAD^{pool}
1	Semiconductor Memory Architecture	0.8525	0.8356	0.8279	0.2795	0.2757
2	Audio and Speech-Signal Processing	1.1342	0.8356	0.8442	0.3828	0.3820
3	Semiconductor Lithography and Fabrication Processes	0.9727	0.8356	0.8342	0.3160	0.3156
4	Integrated Circuit Design, Verification, and Testing	1.1036	0.8356	0.8396	0.3661	0.3663
5	PCB and Electronic Hardware with Connectors	1.0636	0.8356	0.8416	0.4055	0.4048
6	Video Recording, Encoding, and Media Processing	0.8756	0.8356	0.8283	0.2832	0.2788
7	Semiconductor Packaging and Interconnect Bonding	0.9823	0.8356	0.8425	0.3080	0.3070
8	Thin-Film Materials, Deposition and Chemistry	1.0148	0.8356	0.8290	0.3691	0.3729
9	Wireless Base-Station and Mobile Communications	0.9366	0.8356	0.8390	0.2875	0.2884
10	Clock Generation, Timing Circuits, and Signal Processing	1.1375	0.8356	0.8389	0.4072	0.4072
11	Power-Supply and Power-Management Electronics	1.0299	0.8356	0.8417	0.3325	0.3354
12	Networking Protocols and Data Infrastructure	0.9842	0.8356	0.8365	0.3221	0.3231
13	Transistor Gate Structures and Semiconductor Device Physics	0.9542	0.8356	0.8343	0.3013	0.2995
14	Imaging Sensors and Camera Pipelines	0.8659	0.8356	0.8366	0.2827	0.2807
15	Processor Architecture, Caches, Instruction Execution	1.0744	0.8356	0.8365	0.3245	0.3254
16	Display Panel Tech (LCD/OLED)	0.6922	0.8356	0.8274	0.3289	0.3229
17	Rotational Electromechanics and Magnetic Actuation	1.0386	0.8356	0.8342	0.4109	0.4102
18	Optics, Lasers, and Photonics	0.8809	0.8356	0.8339	0.3230	0.3215
19	Analog and Differential Amplifier Circuits	1.1271	0.8356	0.8386	0.3987	0.4002
20	Mobile UI and Content Services	1.0503	0.8356	0.8428	0.3337	0.3350
	Mean	0.9886	0.8356	0.8364	0.3382	0.3376

Notes: v_{ijt} is realized patent value after asinh and a global standardization (the same left-hand side as Table 7). $\hat{v}^{fe}_{ijt}$ and $\hat{v}^{pool}_{ijt}$ are fitted values from Table 7 Model 2 (firm, year, and area effects) and Model 3 (firm and year-by-area effects), respectively. MAD = Mean Absolute Deviation between actual and fitted values on this common scale.

residual as Gaussian, $\varsigma_{ipt} \sim N(0, \sigma^2_{\varsigma,it})$, independent of ν_{ipt}. The filtered fraction is then

$$\mathbb{E}[\nu_{ipt} \mid R_{ipt}] \;=\; \delta_{it}\, R_{ipt} + \sqrt{\delta_{it}}\,\sigma_{\varsigma,it}\,\frac{\varphi\big(-\sqrt{\delta_{it}}\,R_{ipt}/\sigma_{\varsigma,it}\big)}{1-\Phi\big(-\sqrt{\delta_{it}}\,R_{ipt}/\sigma_{\varsigma,it}\big)}, \tag{6}$$

where φ and Φ are the standard normal density and distribution and

$$\delta_{it} \;=\; \frac{\sigma^2_{\nu,it}}{\sigma^2_{\nu,it}+\sigma^2_{\varsigma,it}} \tag{7}$$

is the signal-to-noise ratio. Let M_{it} be the firm's market capitalization on the day before the announcement, and let N_{ipt} be the number of patents granted to the firm on the same day. The dollar value that I take from their filtered series is

$$A_{ipt} \;\equiv\; \xi_{ipt} \;=\; \big(1-\bar{\omega}\big)^{-1}\frac{1}{N_{ipt}}\,\mathbb{E}[\nu_{ipt} \mid R_{ipt}]\,M_{it}. \tag{8}$$

The factor $(1-\bar{\omega})^{-1}$ corrects for the average grant probability in (4), and the factor $1/N_{ipt}$ splits the filtered value of a day equally across the patents that the firm receives on that day. I use their published dollar series, in both nominal and 1982 dollars, so the filter and the signal-to-noise ratio are exactly those of Kogan et al. (2017).

The last step is specific to this paper. The data of Kogan et al. (2017) give one dollar value per patent, while the entry model is defined on innovation areas. I therefore allocate the value of each

Table 21: Persistence of area occupancy

A. Year-to-year transitions	
Occupied last year, occupied this year (share)	0.685
Not occupied last year, entered this year (share)	0.115
Occupied areas this year that are new this year (share)	0.305
Occupied areas last year that are vacated this year (share)	0.315
B. Occupancy spells	
Number of spells	20,607
Median spell length (years); mean spell length	1; 2.94
Spells of one year (share)	0.584
Entries that return to an area the firm occupied earlier in the sample (share)	0.610
Years away before a return: 1 year (share of returns)	0.495
2 years (share of returns)	0.209
3 or more (share of returns)	0.297
C. Three-year windows	
Occupied in all three years (count and share of windows)	31,328 (0.161)
Entered at t and still occupied at $t+1$	6,575 (0.034)
Entered at t and vacated at $t+1$	10,330 (0.053)
Vacated at t and re-entered at $t+1$	5,276 (0.027)
Vacated at t and still out at $t+1$	11,497 (0.059)
Out at $t-1$ and t, entered at $t+1$	10,607 (0.054)
Out in all three years	112,230 (0.576)
D. Portfolio size	
Firm-years with no occupied area (share)	0.399
Firm-years whose number of occupied areas equals last year's (share)	0.390
Correlation between this year's and last year's number of occupied areas	0.833

Notes: The panel covers listed U.S. electronics firms from 1990 to 2019 and twenty innovation areas. As in the estimation, a firm occupies an area in a year when the average weight of its patents in that year on the area is at least 1%. I compute transitions and spells from adjacent calendar years only, and Panel C only counts firm-area observations for which the firm is observed in three consecutive years.

Table 22: Occupancy this year and occupancy last year

	(1)	(2)	(3)	(4)
Occupied last year	0.566	0.334	0.090	0.088
	(0.011)	(0.010)	(0.007)	(0.007)
Predicted valuation		0.076	0.103	0.097
		(0.005)	(0.010)	(0.010)
Rival count (inverse hyperbolic sine)		0.110	0.039	
		(0.004)	(0.012)	
Other areas occupied last year		0.012	0.018	0.019
		(0.001)	(0.001)	(0.001)
Firm-by-area effects	No	No	Yes	Yes
Area-by-year effects	No	No	No	Yes
Year effects	Yes	Yes	Yes	No
Observations	209,960	209,960	209,960	209,960
R-squared	0.328	0.421	0.524	0.525
Within R-squared			0.062	0.061

Notes: Linear probability models of occupancy on the firm-area-year panel, using adjacent years only. Standard errors clustered by firm are in parentheses. I use the coefficient on last year's occupancy only to describe persistence, and with firm-by-area effects and a short panel it is biased toward zero. Predicted valuation is the first-stage fitted value used in the moment inequalities. Column (4) leaves out the rival count, because within an area-year the number of rivals differs across firms only through the firm's own occupancy.

patent across areas with the LDA weights w_{pjt} of Appendix A.2,

$$A_{ipjt} \;=\; w_{pjt}\, A_{ipt}. \tag{9}$$

and add up these values over all patents of the firm in that period, which gives the realized total of the area,

$$\sum_{p \in \mathcal{P}_{it}} A_{ipjt} \;=\; \sum_{p \in \mathcal{P}_{it}} w_{pjt}\, A_{ipt}. \tag{10}$$

As in Section 3, the realized valuation of the area is then a monotone transformation $b(\cdot)$ of this total plus a mean-zero measurement error,

$$v_{ijt} \;=\; b\left(\sum_{p \in \mathcal{P}_{it}} w_{pjt}\, A_{ipt} \right) + e_{ijt}. \tag{11}$$

In the empirical implementation, I use the inverse hyperbolic sine for $b(\cdot)$, and the valuation that enters the inequalities is the predicted value from the first-stage projection in Section 4. The error e_{ijt} contains the noise from the filter in A_{ipt}, the noise in the LDA weights, and any error from adding up the patents to the firm, year, and area. Since firms do not observe e_{ijt} when they choose their areas, $\mathbb{E}[e_{ijt} \mid \mathcal{F}_{ijt}] = 0$.

B.2 Derivation of the moment inequalities

In this subsection, I first show how Result 2 follows from Result 1, and I then derive Results 3–5 from those two results. Throughout, I take the equilibrium rival counts $N_{jt,-i}$ as given, and the

number of areas is finite. As in Section 3, the unobservable ϵ_{ijt} is a composite ex-post forecast error with $\mathbb{E}[\epsilon_{ijt} \mid \mathcal{F}_{ijt}] = 0$, the instruments Z are a subset of $\mathcal{F}_{ijt}$, and $l(\cdot)$ maps them to a nonnegative scalar. Therefore, all the objects in the deterministic indexes below are measurable with respect to $\mathcal{F}_{ijt}$.

Let $S = \{j : d_{ijt} = 1\}$ be the observed bundle, with $k_{it} = |S|$, and let

$$\zeta^e_{ijt} \;=\; \breve{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa$$

be the expected profitability index of area j. From Result 1, expected profit is

$$\mathbb{E}[\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; \sum_{j \in S} \zeta^e_{ijt} + \beta\, g(k_{it}). \tag{12}$$

In a Nash equilibrium, no unilateral deviation can raise the firm's expected profit. That is, for any other bundle S',

$$\mathbb{E}\big[\Pi_{it}(S') - \Pi_{it}(S) \;\big|\; \mathcal{F}_{ijt}\big] \;\le\; 0. \tag{13}$$

Characterization of the optimal bundle. The portfolio function g depends on the bundle only through the number of areas in it. Given the rival counts, the firm can therefore rank the areas by a single index and then decide how many of the top-ranked areas to enter. I show this in three steps, using only Result 1 and this property of g. First, Lemma 1 computes what a swap does to expected profit. Second, Lemma 2 shows that, for a given number of areas, the best bundles are those that never leave a better area out while keeping a worse one in. Third, Lemma 3 shows that the firm then only has to compare the best bundles of each of the $J + 1$ possible sizes.

Lemma 1 (Swap identity) *Let S be a bundle of size k, and let $j \in S$ and $m \notin S$. The swapped bundle $S' = (S \setminus \{j\}) \cup \{m\}$ has the same size, and*

$$\mathbb{E}\big[\Pi_{it}(S') - \Pi_{it}(S) \;\big|\; \mathcal{F}_{ijt}\big] \;=\; \zeta^e_{imt} - \zeta^e_{ijt}. \tag{14}$$

Proof. From (12), expected profit is the sum of the indexes of the entered areas, plus βg evaluated at the number of entered areas. The swap removes j from the bundle and adds m, so the sums of the indexes on the two bundles differ only by $\zeta^e_{imt} - \zeta^e_{ijt}$. Since both bundles have k areas, the term $\beta g(k)$ is the same on both and cancels. ■

Lemma 2 (Fixed portfolio size) *Fix a size $k \in \{0, \dots, J\}$. A bundle S of size k maximizes expected profit among all bundles of size k if and only if it satisfies the threshold property*

$$\min_{j \in S} \zeta^e_{ijt} \;\ge\; \max_{m \notin S} \zeta^e_{imt}, \tag{15}$$

where the minimum over the empty set is $+\infty$ and the maximum over the empty set is $-\infty$. Every such bundle attains

$$U^e_k \;=\; \sum_{r=1}^{k} \zeta^e_{(r)} + \beta\, g(k), \tag{16}$$

with $\zeta^e_{(1)} \ge \cdots \ge \zeta^e_{(J)}$ the decreasing rearrangement of the indexes. That k-set is unique if and only if $\zeta^e_{(k)} > \zeta^e_{(k+1)}$, reading $\zeta^e_{(0)} = +\infty$ and $\zeta^e_{(J+1)} = -\infty$.

Proof. Note first that the empty bundle ($k = 0$) and the full set of areas ($k = J$) satisfy (15) by

construction, since there is either no area inside or no area outside the bundle. Let us therefore look at any other size k.

First, suppose that S violates (15). In this case, the firm has entered some area j whose index is strictly below the index of some area m that it left out. By Lemma 1, replacing j with m then raises expected profit without changing the size of the bundle. That is, a bundle that maximizes expected profit among bundles of size k must satisfy the threshold property.

Second, suppose that S satisfies (15), and let T be any other bundle of the same size. Since S and T have the same size, the areas that are in S but not in T can be paired one for one with the areas that are in T but not in S. Each area of the first group is in S, so by (15) its index is at least as large as the index of any area outside S, and each area of the second group is outside S. Summing over the pairs gives

$$\sum_{j \in S \setminus T} \zeta^e_{ijt} \;\geq\; \sum_{m \in T \setminus S} \zeta^e_{imt}. \tag{17}$$

The areas that S and T have in common contribute the same to both bundles, and the portfolio term $\beta g(k)$ is also the same. As such, expected profit on S is at least as large as on T, i.e., every bundle of size k that satisfies the threshold property is optimal for that size.

Third, a bundle of size k that satisfies the threshold property contains every area whose index is strictly above the k-th largest index $\zeta^e_{(k)}$, and it fills the remaining places with areas whose index equals that cutoff. All such bundles add up the same k largest indexes, so they give the same expected profit U^e_k in (16). If $\zeta^e_{(k)} > \zeta^e_{(k+1)}$, there is no tie at the cutoff, and the bundle is unique. If several areas tie at the cutoff, any choice among them that brings the bundle to size k is optimal for that k. ■

Lemma 3 (Optimal size) *The $J+1$ numbers $U^e_0, \ldots, U^e_J$ attain a maximum. Write k^* for any maximizer. A bundle S maximizes expected profit over the power set if and only if $|S| = k^*$ for some such k^* and S satisfies* (15)*. In particular, every optimal bundle is a set of k^* areas with the largest indexes.*

Proof. Every bundle has a size k between 0 and J. By Lemma 2, the best expected profit among bundles of size k is U^e_k, and only the bundles of size k that satisfy the threshold property attain it. The firm therefore only has to compare the $J+1$ numbers $U^e_0, \ldots, U^e_J$. Since this is a finite list, at least one best size k^* exists. The best expected profit over all bundles is then $\max_k U^e_k$, and the bundles that attain it are the bundles of a best size that satisfy the threshold property. ■

Together, Lemmas 1–3 give Result 2. Note that several sizes can attain the same largest value of U^e_k, in which case each of them is an optimal k^*.

Adding M areas. Fix an exogenous subset $\mathcal{M}$ of size M, and let $D^0_{\mathcal{M},it} \equiv \prod_{j \in \mathcal{M}} (1 - d_{ijt})$ indicate that the firm entered none of these areas. On the event $\{D^0_{\mathcal{M},it} = 1\}$, the subset and the bundle do not overlap, $\mathcal{M} \cap S = \emptyset$, so the firm can deviate to $S' = S \cup \mathcal{M}$, which has $k_{it} + M$ areas. From (12),

$$\begin{aligned}
\mathbb{E}[\Pi_{it}(S') \mid \mathcal{F}_{ijt}] &= \sum_{j \in S} \zeta^e_{ijt} + \sum_{j \in \mathcal{M}} \zeta^e_{ijt} + \beta\, g(k_{it} + M), \\
\mathbb{E}[\Pi_{it}(S) \mid \mathcal{F}_{ijt}] &= \sum_{j \in S} \zeta^e_{ijt} + \beta\, g(k_{it}).
\end{aligned}$$

The expected profit difference is therefore

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; \sum_{j\in\mathcal{M}} \zeta^{e}_{ijt} + \beta\big(g(k_{it}+M) - g(k_{it})\big) \;=\; \eta^{+}_{i,\mathcal{M},t}, \tag{18}$$

where $\Delta g^{+}_{M}(k_{it}) \equiv g(k_{it}+M) - g(k_{it})$ and

$$\eta^{+}_{i,\mathcal{M},t} \;=\; \sum_{j\in\mathcal{M}} \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) + \beta\, \Delta g^{+}_{M}(k_{it}).$$

By revealed preference (13), $\eta^{+}_{i,\mathcal{M},t} \le 0$ on the event $\{D^{0}_{\mathcal{M},it} = 1\}$, and the product $D^{0}_{\mathcal{M},it}\,\eta^{+}_{i,\mathcal{M},t}$ is zero off that event. As such,

$$D^{0}_{\mathcal{M},it}\,\eta^{+}_{i,\mathcal{M},t} \;\le\; 0 \tag{19}$$

almost surely. Since $l(Z_{i,\mathcal{M},t}) \ge 0$, multiplying by the instrument keeps the product nonpositive, and taking expectations gives the unconditional moment

$$\mathbf{M}_1(Z_{i,\mathcal{M},t};\theta) \;=\; \mathbb{E}\big[D^{0}_{\mathcal{M},it}\,\eta^{+}_{i,\mathcal{M},t}\,l(Z_{i,\mathcal{M},t})\big] \;\le\; 0, \tag{20}$$

which is Result 3.

Dropping M areas. For the same exogenous $\mathcal{M}$, let $D^{1}_{\mathcal{M},it} \equiv \prod_{j\in\mathcal{M}} d_{ijt}$ indicate that the firm entered every area in the subset. On the event $\{D^{1}_{\mathcal{M},it} = 1\}$, the subset is part of the bundle, $\mathcal{M} \subset S$, so $k_{it} \ge M$, and the firm can deviate to $S' = S \setminus \mathcal{M}$, which has $k_{it} - M$ areas. Expanding (12) in the same way gives

$$\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] \;=\; -\sum_{j\in\mathcal{M}} \zeta^{e}_{ijt} - \beta\big(g(k_{it}) - g(k_{it}-M)\big) \;=\; -\eta^{-}_{i,\mathcal{M},t}, \tag{21}$$

where $\Delta g^{-}_{M}(k_{it}) \equiv g(k_{it}) - g(k_{it}-M)$ and

$$\eta^{-}_{i,\mathcal{M},t} \;=\; \sum_{j\in\mathcal{M}} \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) + \beta\, \Delta g^{-}_{M}(k_{it}).$$

By revealed preference, $-\eta^{-}_{i,\mathcal{M},t} \le 0$ on the event $\{D^{1}_{\mathcal{M},it} = 1\}$, so

$$D^{1}_{\mathcal{M},it}\,\big(-\eta^{-}_{i,\mathcal{M},t}\big) \;\le\; 0 \tag{22}$$

almost surely. Multiplying by $l(Z_{i,\mathcal{M},t}) \ge 0$ and taking expectations then gives

$$\mathbf{M}_2(Z_{i,\mathcal{M},t};\theta) \;=\; \mathbb{E}\big[D^{1}_{\mathcal{M},it}\,\big(-\eta^{-}_{i,\mathcal{M},t}\big)\,l(Z_{i,\mathcal{M},t})\big] \;\le\; 0, \tag{23}$$

which is Result 4.

Pairwise swaps. Let (j,m) be an ordered pair with $j \in S$ and $m \notin S$, so that $D^{\text{swap}}_{j,m,it} \equiv d_{ijt}(1 - d_{imt}) = 1$, and consider the deviation that replaces j with m. Since the number of areas does not

change, $\Delta g = 0$, and the expected profit difference is just the difference of the two indexes,

$$\begin{aligned}\mathbb{E}[\Delta\Pi_{it} \mid \mathcal{F}_{ijt}] &= \zeta^e_{imt} - \zeta^e_{ijt} \\ &= \big(\check{v}_{imt} + \alpha\, h(N_{mt,-i}) - \kappa\big) - \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) - \kappa\big) \\ &= \big(\check{v}_{imt} + \alpha\, h(N_{mt,-i})\big) - \big(\check{v}_{ijt} + \alpha\, h(N_{jt,-i})\big) \;=\; \mu_{ijmt}(\theta).\end{aligned} \tag{24}$$

Note that the occupancy cost κ also cancels. By revealed preference, $\mu_{ijmt}(\theta) \le 0$ on the event $\{D^{\text{swap}}_{j,m,it} = 1\}$, so

$$D^{\text{swap}}_{j,m,it}\, \mu_{ijmt}(\theta) \;\le\; 0 \tag{25}$$

almost surely, and the unconditional moment is

$$\mathbf{M}_3(Z_{ijmt}; \theta) \;=\; \mathbb{E}\big[D^{\text{swap}}_{j,m,it}\, \mu_{ijmt}(\theta)\, l(Z_{ijmt})\big] \;\le\; 0, \tag{26}$$

which is Result 5. Since the portfolio term drops out, $\mathbf{M}_3$ depends on congestion α but not on the scale of $g(\cdot)$.

From pathwise inequalities to unconditional moments. The three products (19), (22), and (25) are nonpositive for every realization. To go from these products to the unconditional moments, I only need that multiplying by a nonnegative instrument does not change their sign. Lemma 4 states this formally.

Lemma 4 (Nonnegative measurable weights) *Let X be integrable and $\mathcal{F}_{ijt}$-measurable, with $X \le 0$ almost surely. Let W be nonnegative and $\mathcal{F}_{ijt}$-measurable, and suppose WX is integrable. Then $WX \le 0$ almost surely, and*

$$\mathbb{E}[WX] \;\le\; 0. \tag{27}$$

Proof. The event $\{W \ge 0\} \cap \{X \le 0\}$ has probability one, and on this event the product WX is never strictly positive, so $WX \le 0$ almost surely. Now, for any integrable random variable Y,

$$\mathbb{E}[Y] \;=\; \mathbb{E}[Y\mathbf{1}_{\{Y \le 0\}}] + \mathbb{E}[Y\mathbf{1}_{\{Y > 0\}}].$$

If $P(Y > 0) = 0$, the second term is zero, and the first term is the expectation of a nonpositive integrable function, which is at most zero. Setting $Y = WX$ gives (27). ■

Since X is $\mathcal{F}_{ijt}$-measurable, $\mathbb{E}[X \mid \mathcal{F}_{ijt}] = X$, and by the law of iterated expectations,

$$\mathbb{E}[WX] \;=\; \mathbb{E}\big[W\, \mathbb{E}[X \mid \mathcal{F}_{ijt}]\big]. \tag{28}$$

The right-hand side is the usual way to build an unconditional moment from a conditional moment $\mathbb{E}[X \mid \mathcal{F}_{ijt}] \le 0$. In this model, however, the complete-information assumption already gives the bound $X \le 0$ for every realization, so I can apply Lemma 4 directly.

I apply this lemma three times, each time with $W = l(Z)$, a nonnegative function of instruments in $\mathcal{F}_{ijt}$.

For adding M areas, let $X = D^0_{\mathcal{M},it}\, \eta^+_{i,\mathcal{M},t}$. The indicator $D^0_{\mathcal{M},it}$ depends on the observed bundle. Under the information structure of Section 3, the firm has no private type, so, given the rivals' actions, the bundle is a function of $\mathcal{F}_{ijt}$. The index $\eta^+_{i,\mathcal{M},t}$ depends on $\check{v}$, N, k_{it}, and θ, where the first three are in $\mathcal{F}_{ijt}$ or are conditioned on, and θ is a constant. Hence X is $\mathcal{F}_{ijt}$-measurable, and

(19) gives $X \leq 0$. Applying Lemma 4 to (W, X) then gives Result 3.

The same argument works for dropping M areas, with $X = D^{1}_{\mathcal{M},it}(-\eta^{-}_{i,\mathcal{M},t})$. The index $\eta^{-}_{i,\mathcal{M},t}$ depends on the same objects as in the add case, and (22) gives $X \leq 0$, so Lemma 4 gives Result 4.

Similarly, for a pairwise swap, let $X = D^{\text{swap}}_{j,m,it}\,\mu_{ijmt}(\theta)$. The indicator depends on (d_{ijt}, d_{imt}), and $\mu_{ijmt}(\theta)$ depends on the two area indexes. Together with (25), Lemma 4 then gives Result 5.

That is, the inequalities only require that $l(\cdot)$ is nonnegative and $\mathcal{F}_{ijt}$-measurable, and that the product $l(Z)X$ is integrable. In the estimates I report, $l(Z)$ is a set of binary firm-year indicators that split the sample by lagged R&D, assets, and patent flow. Since these instruments are bounded, $l(Z)X$ is integrable whenever X is.

Note that the algebra above holds for any portfolio function $g(\cdot)$. In the baseline specification, I evaluate the inequalities on exogenous pairs and triplets of areas ($M \in \{2, 3\}$) together with the swap pairs, as described in Section 3, and the estimates in Section 4 use $g(k) = \binom{k}{2}$.

B.3 Empirical implementation

This subsection gives the details of the estimation that Section 3 only summarizes.

Subset sampling. With $J = 20$ areas, there are many pairs, triplets, and ordered swaps, and the inequalities only need a subset of them. I therefore rank the subsets by their predicted values $\check{v}_{ijt}$, before looking at the realized d_{ijt}, and keep the first $|M|$ subsets of that ranking, with $|M| \in \{10, 20, 40\}$ on the instrument grid in Appendix C. The joint deviations add or drop pairs and triplets of areas ($M \in \{2, 3\}$), so that they change how many areas enter the portfolio term (see Appendix B.4 for the profit gaps). As discussed in Section 3, for add moments I take the subsets with the highest average predicted value, and for drop moments those with the lowest. For swaps, I rank the 380 ordered pairs of areas by the difference in $\check{v}$ and draw from the top, the bottom, and the middle of the ranking. The baseline specification uses $|M| = 40$ for each type of deviation.

Test statistic and profiling. What I estimate is a confidence set for congestion and for the occupancy-to-scope ratio. The test statistic $T(\theta)$ is the max-type studentized sample moment of Chernozhukov et al. (2019), and the critical value $c(1 - a, \theta)$ is their two-step self-normalized critical value at $a = 0.01$. To find parameter values inside the set, I search a bounded parameter region by simulated annealing (Grieco, 2014) and keep the best draws as starting values. If the test statistic of the best draw still exceeds the critical value, I first minimize the difference between the two without constraints and use the result as a starting value. The profiled bounds for α and for $\rho = \kappa/\beta$ then solve

$$\min \alpha, \quad \max \alpha \qquad \text{and} \qquad \min \rho, \quad \max \rho \qquad \text{subject to} \qquad T(\theta) \leq c(1 - a, \theta).$$

Static game. As discussed in Section 3 and in the conclusion, the static game answers how a firm composes its portfolio in a given year and how the rival counts in that year shift its expected profit, and it leaves out learning, option value, preemption, and irreversible research outlays. Two further points matter for reading the estimates. First, if entry is sticky, α mixes the crowding in the current year with the shadow cost of a persistent stock of rivals. Second, the firm effects

and the year-by-area effects in the first stage absorb a capability level that does not change over time and the conditions that are common to an area in a given year. A sunk cost of entering an area can be bounded with the same revealed-preference argument, by comparing the observed path with a path that changes the portfolio in a single year and then returns to the observed path (Morales et al., 2019; Pakes et al., 2015). In this comparison, the continuation values cancel except for the sunk cost that is saved or brought forward, so I do not need an equilibrium-selection rule or a transition for the rival counts. Appendix A.10 reports how persistent occupancy is, and Appendix B.5 carries out the one-period comparison.

B.4 Identification of congestion α and occupancy-to-scope ratio ρ

In this subsection, I show which objects the add, drop, and swap comparisons of Section 3 can recover, i.e., why Section 4 reports sets for α and for the ratio ρ instead of β and κ separately.

First, a swap keeps the number of occupied areas fixed. As explained after Result 5, the portfolio term and the occupancy cost are then the same on both sides of the comparison, and only expected value and congestion differ between the two areas. This is why the swaps are informative about congestion, and why Table 9 reports a set for α.

Second, an add or a drop of M areas changes how many areas the firm occupies. Let A be the sum of expected valuations and congestion over the areas that enter or leave the bundle. The profit index is then

$$A + \beta\,\Delta g - M\kappa, \tag{29}$$

where Δg is the change in the portfolio function. Since I treat occupancy as a cost, κ and β have the same sign when I form the ratio below.

Pairwise interactions. In the baseline, the portfolio function is $g(k) = \binom{k}{2}$. If a firm that already occupies k areas adds M areas, then $\Delta g^{+} = Mk + \binom{M}{2}$, or $M\big(k + \frac{M-1}{2}\big)$. Substituting this into (29) and writing $\rho = \kappa/\beta$ gives

$$A + M\beta\Big(k + \tfrac{M-1}{2} - \rho\Big). \tag{30}$$

Since k varies across firms and $M \in \{2, 3\}$, these comparisons are informative about ρ. Under pairwise interactions, ρ is the number of areas at which one more area covers its own occupancy cost. To see this, take $M = 1$. Adding one area then creates k new pairs, so the extra scope benefit is βk and the extra occupancy cost is κ, and the two balance when $k = \rho$. For example, for the firm in Section 4 that occupies five areas, a sixth area exactly covers its cost when $\rho = 5$. However, β itself remains a scale in front of the parenthesis, since expected value enters the profit with a coefficient of one. I therefore report a profiled 99 percent set for ρ. In the baseline specification, this set keeps a ratio of zero, rejects negative ratios, and is bounded above. With lagged patents as the only instrument, the set is also bounded above, but it is not bounded away from zero.

Portfolio size times its inverse hyperbolic sine. When $g(k) = k\,\mathrm{asinh}(k)$, the increment is $\Delta g = (k + M)\,\mathrm{asinh}(k + M) - k\,\mathrm{asinh}(k)$, and (29) becomes $A + \beta(\Delta g - M\rho)$. The comparisons then recover the same ratio ρ, but in the units of this increment. As such, the upper end of the set in Table 10 cannot be compared directly with the upper end under pairwise interactions.

Linear portfolio. When $g(k) = k$, the increment is $\Delta g = M$, and (29) becomes $A + M\delta$ with $\delta = \beta - \kappa$. In this case, the comparisons recover the difference between the scope benefit and

the occupancy cost of one more area, and the last column on the linear row of Table 10 is a 99 percent set for δ. That is, a positive δ means that one more area brings more scope benefit than it costs.

Appendix B.3 describes how I choose the subsets and how I construct the 99 percent sets.

B.5 Fixed and sunk costs

In the static game, the firm pays a single cost κ for each area it occupies. To see how large a sunk cost the panel can support, I now let the firm pay a fixed cost κ_f in every year it occupies an area and, in addition, a sunk cost κ_s in the first year of each spell. To separate the two costs, I extend the add, drop, and swap comparisons by one period, following Morales et al. (2019) and Pakes et al. (2015), and I keep the observed portfolios. I go through the comparison in seven steps.

Step 1: the observable flow. For firm i, area j, and year t, let the observable part of the payoff from occupying the area be

$$\bar{f}_{ijt} = \check{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta\, \Delta g,$$

which contains the same value, congestion, and scope terms as the static add and drop moments, with Δg the change in the portfolio function for that comparison. As before, expected value enters with a coefficient of one. What changes in this extension is the cost, which I describe in the next step.

Step 2: two costs. Let $d_{ijt} \in \{0, 1\}$ be occupancy. The firm pays

$$C_{ijt} = \kappa_f\, d_{ijt} + \kappa_s\, \mathbf{1}\{d_{ij,t-1} = 0,\, d_{ijt} = 1\}.$$

That is, the firm pays the fixed cost in every year it occupies the area, and it pays the sunk cost only when it enters an area that it did not occupy in the previous year. The static game is the special case with $\kappa_s = 0$.

As in the static case, I measure both costs in pairs' worth of scope, i.e., $\rho_f = \kappa_f/\beta$ and $\rho_s = \kappa_s/\beta$. Under pairwise interactions, a firm that already occupies k areas creates k new pairs when it adds one more area, so the extra scope term is βk, and the extra fixed cost is κ_f. These two terms balance when $k = \rho_f$, and the sunk cost balances the same new pairs when $k = \rho_s$. For example, for the firm with five areas in Section 4, a sixth area covers its fixed cost when $\rho_f = 5$. The net payoff of occupying the area is then

$$\check{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta\big(\Delta g - \rho_f\big) - \beta\rho_s\, \mathbf{1}\{d_{ij,t-1} = 0,\, d_{ijt} = 1\}.$$

As in the static model, β remains a scale in front of the parenthesis, since expected value enters with a coefficient of one, and the variation in the number of areas that firms already occupy is what makes the comparisons informative about ρ_f and ρ_s. In each row of the results, I hold congestion at one value from the static set and report the pairs (ρ_f, ρ_s) that the inequalities accept at that value. Note that, when $\rho_s = 0$, the yearly ratio ρ_f is the occupancy-to-scope ratio of Appendix B.4.

Step 3: a one-period deviation. Now consider a path that differs from the observed one in year t only, i.e., it flips occupancy in that year, $d'_{ijt} = 1 - d_{ijt}$, and then follows the observed choices from year $t + 1$ on. From year $t + 2$ onward, the two paths have the same occupancy history,

so their continuation values cancel. In year $t+1$, occupancy is also the same on both paths, so next year's flow $\bar{f}_{ij,t+1}$ cancels as well. The only term that can still differ in $t+1$ is the sunk cost, because occupancy in year t is d_{ijt} on the observed path and $1-d_{ijt}$ on the deviation. I discount this difference in $t+1$ at $\delta = 0.95$. By revealed preference, the observed path must be at least as good as the deviation,

$$V^{\text{obs}} - V^{\text{dev}} = \big(d_{ijt}\bar{f}_{ijt} - C^{\text{obs}}_{ijt}\big) - \big(d'_{ijt}\bar{f}_{ijt} - C^{\text{dev}}_{ijt}\big) + \delta\big(C^{\text{dev}}_{ij,t+1} - C^{\text{obs}}_{ij,t+1}\big) \geq 0.$$

In this difference, I use the observed occupancy in year $t+1$ instead of the firm's forecast of it, which is valid as long as the forecast errors are uncorrelated with the instruments.

Step 4: the firm is out, and the deviation is to enter. Take first a firm that does not occupy the area in year t, $d_{ijt} = 0$, so that it pays no cost in that year. The deviation occupies the area and pays $\kappa_f + \kappa_s \mathbf{1}\{d_{ij,t-1} = 0\}$. In $t+1$, the observed path starts from an empty area and therefore pays the sunk cost if $d_{ij,t+1} = 1$, while the deviation already occupies the area and does not pay it. That is, the deviation saves $\delta\kappa_s d_{ij,t+1}$ in $t+1$. For the observed choice to be better than entering, it must be that

$$\bar{f}_{ijt} \leq \kappa_f + \kappa_s\big(\mathbf{1}\{d_{ij,t-1} = 0\} - \delta d_{ij,t+1}\big).$$

A firm outside the area in both $t-1$ and t reveals $\bar{f}_{ijt} \leq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1})$. A firm that has just left reveals $\bar{f}_{ijt} \leq \kappa_f - \delta\kappa_s d_{ij,t+1}$.

Step 5: the firm is in, and the deviation is to leave. Take now a firm that occupies the area in year t, so that on the observed path it pays $\kappa_f + \kappa_s \mathbf{1}\{d_{ij,t-1} = 0\}$ in that year, while the deviation pays nothing. However, if the firm leaves in year t and comes back in $t+1$, it has to pay the sunk cost again, so the deviation costs an extra $\delta\kappa_s d_{ij,t+1}$ in $t+1$. For staying to be better than leaving, it must be that

$$\bar{f}_{ijt} \geq \kappa_f + \kappa_s\big(\mathbf{1}\{d_{ij,t-1} = 0\} - \delta d_{ij,t+1}\big).$$

A firm that has just entered reveals $\bar{f}_{ijt} \geq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1})$. A firm that stayed reveals $\bar{f}_{ijt} \geq \kappa_f - \delta\kappa_s d_{ij,t+1}$.

The four groups are therefore

$$\begin{aligned}
(d_{ij,t-1}, d_{ijt}) = (0,0) \quad &\Rightarrow \quad \bar{f}_{ijt} \leq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1}),\\
(d_{ij,t-1}, d_{ijt}) = (0,1) \quad &\Rightarrow \quad \bar{f}_{ijt} \geq \kappa_f + \kappa_s(1 - \delta d_{ij,t+1}),\\
(d_{ij,t-1}, d_{ijt}) = (1,1) \quad &\Rightarrow \quad \bar{f}_{ijt} \geq \kappa_f - \delta\kappa_s d_{ij,t+1},\\
(d_{ij,t-1}, d_{ijt}) = (1,0) \quad &\Rightarrow \quad \bar{f}_{ijt} \leq \kappa_f - \delta\kappa_s d_{ij,t+1}.
\end{aligned}$$

Next year's occupancy enters these conditions because it shows whether the deviation pays the sunk cost earlier or later. For example, take a firm that is out of the area this year and in the area next year. If it entered today, it would pay the sunk cost today instead of next year, and it would save $\delta\kappa_s$ next year. Without next year's occupancy, the sunk cost would only enter through $1 - \delta = 0.05$, and the comparisons would say very little about it.

Write

$$w_{ijt} = \mathbf{1}\{d_{ij,t-1} = 0\} - \delta\, d_{ij,t+1}$$

and

$$L_{ijt} = \breve{v}_{ijt} + \alpha\, h(N_{jt,-i}) + \beta\big(\Delta g - \rho_f - \rho_s w_{ijt}\big).$$

Then a firm that is out of the area in year t reveals $L_{ijt} \leq 0$, and a firm that is in the area reveals $L_{ijt} \geq 0$. These two restrictions give the four groups above, with $w_{ijt} = 1 - \delta d_{ij,t+1}$ when the firm was out in $t-1$ and $w_{ijt} = -\delta d_{ij,t+1}$ when it was in.

Step 6: swaps. In a swap, the firm leaves area j and enters area m in the same year, and then returns to the observed path. Since the number of areas does not change, $\Delta g = 0$, and since the firm pays the fixed cost on m instead of j, ρ_f cancels. Each area keeps its sunk-cost weight from Steps 4 and 5, so the expected profit of the swap is

$$\left(\check{v}_{imt} - \check{v}_{ijt}\right) + \alpha\left(h(N_{mt,-i}) - h(N_{jt,-i})\right) - \beta\rho_s\left(w_{imt} - w_{ijt}\right).$$

If $\rho_s = 0$, this is the static swap, which is informative about congestion. Once ρ_s is free, however, congestion and the sunk-cost ratio enter the same expression. This is why each row of the results holds congestion at one value from the static set and asks which pairs (ρ_f, ρ_s) the inequalities accept at that value. I pool the swap comparisons across the four groups, and each firm-year keeps its own sunk-cost weights. Since consecutive years of the same firm enter the same comparison, I cluster the inference by firm.

Step 7: the sets the inequalities recover. I use the test statistic and the self-normalized critical value of Section 3, with forty subsets per deviation family and all three instruments, at the one percent level. Unlike in the static estimation, I sum the moments within firm before studentizing. For each row, I fix α at one value from the static set in Table 9. At that value, and for a candidate pair (ρ_f, ρ_s), I minimize the criterion over the scale β, and I keep the pair when the minimized criterion is non-positive. Table 23 reports the smallest and largest accepted value of each ratio. When $\rho_s = 0$, Steps 4 and 5 reduce to the static add and drop comparisons, still split by last year's status and still clustered by firm, so that a sunk-cost ratio of zero is the cost structure of the static game. Finally, I compute the accepted pairs twice: once dropping a firm that is absent from the sample in the following year, and once treating that absence as an exit.

One concern is that the comparisons may not separate ρ_s from ρ_f very sharply. The reason is that the first-stage projection predicts realized value, which is zero in most firm-area-years in which the firm does not occupy the area. As such, predicted valuation rises with how often a firm has occupied an area, and firms that have just entered and firms that have just left receive nearly the same prediction. Note that I only carry out this check for the pairwise baseline of Table 9.

First, I find that every row of Table 23 keeps a sunk-cost ratio of zero, and that every row is bounded above. At the static midpoint of congestion in Panel A, the fixed-cost ratio ρ_f runs from 0 to 9.2 and the sunk-cost ratio ρ_s from 0 to 7.4. Second, the ranges are about the same at strong congestion and at the interior point, where ρ_f goes up to 8.9 and 9 and ρ_s up to 7.6 and 7.5. Under weak congestion, $\alpha = -0.1847$, the fixed-cost ratio starts at 3.5 and runs to 9.4, and the sunk-cost ratio runs from 0 to 6.8. That is, when an extra rival reduces expected profit only a little, the data require a larger fixed cost. Third, Panel B, which treats a firm that is absent in the next year as having left, gives upper ends within one ratio point of those in Panel A.

On every row, the inequality that rejects the next larger sunk-cost ratio comes from the firms that have just entered an area. Such a firm pays both the fixed and the sunk cost when it enters, so its choice reveals that its predicted value and the new pairs from the entry are large enough to cover both costs. This is what bounds ρ_s from above. Since a sunk-cost ratio of zero stays inside the set on every row, the cost structure of the static game, which the case study uses when it cuts ρ, is

Table 23: Fixed-cost and sunk-cost ratios accepted by the one-period inequalities

Held fixed	α	Fixed-cost ratio ρ_f	Sunk-cost ratio ρ_s	Accepted pairs	Group that rejects the next larger sunk-cost ratio
Panel A: firms absent from the sample next year are dropped					
Static midpoint	-0.65035	$[0, 9.2]$	$[0, 7.4]$	4,644/12,221	firms that have just entered
Strong congestion	-1.116	$[0, 8.9]$	$[0, 7.6]$	4,540/12,221	firms that have just entered
Weak congestion	-0.1847	$[3.5, 9.4]$	$[0, 6.8]$	2,240/12,221	firms that have just entered
Static interior point	-1.03	$[0, 9]$	$[0, 7.5]$	4,548/12,221	firms that have just entered
Panel B: firms absent from the sample next year are treated as having left					
Static midpoint	-0.65035	$[0, 9.1]$	$[0, 7.7]$	4,765/12,221	firms that have just entered
Strong congestion	-1.116	$[0, 8.9]$	$[0, 7.9]$	4,705/12,221	firms that have just entered
Weak congestion	-0.1847	$[3.7, 9.3]$	$[0, 7.5]$	2,172/12,221	firms that have just entered
Static interior point	-1.03	$[0, 8.9]$	$[0, 7.9]$	4,779/12,221	firms that have just entered

Notes: Each row holds congestion at one value from the static set in Table 9 and reports the fixed-cost ratio ρ_f and the sunk-cost ratio ρ_s that the one-period inequalities of Appendix B.5 accept at the one percent level. The static midpoint is the midpoint of that set for congestion, strong and weak congestion are its two ends, and the static interior point is another value inside it. For each pair of ratios, I minimize the criterion over the scale in front of the portfolio term, and I keep the pair when the minimized criterion is non-positive. The grid has 121 values of the fixed-cost ratio, from 0 to 12, and 101 values of the sunk-cost ratio, from 0 to 10, so there are $121 \times 101 = 12{,}221$ pairs per row, and the intervals are the smallest and largest accepted values on that grid. For inference, I use the self-normalized critical value with forty subsets per deviation family and all three instruments, and I sum the moments within firm before studentizing (763 firms). I define the comparison groups by occupancy in the previous and the current year, and occupancy in the following year enters the payoff with a discount factor of 0.95. I pool the swap comparisons over the groups, and each firm-year keeps its own sunk-cost terms. The last column shows the comparison group whose inequality rejects the smallest sunk-cost ratio above the accepted range. For a row with no accepted pair, it shows the group that rejects at the pair closest to acceptance.

consistent with these comparisons.

C Robustness of the identified sets

The baseline confidence sets in Section 4 use the pairwise-interaction portfolio function and add and drop deviations of size two and three. Table 24 reports the full instrument and subset-size grid. Table 25 repeats that grid after dropping size-three add and drop moments. Table 10 in the main text reports the portfolio-function comparison at the largest subset sample. This appendix reports the pairwise-interaction grid.

C.1 Alternative portfolio functions

Under pairwise interactions, a positive β means that the scope benefit grows with the number of pairs among the occupied areas, so one more area is worth more to a firm that already occupies many. Table 10 compares this portfolio function with the alternatives in the baseline specification. The last column is a 99% set for $\rho = \kappa/\beta$ on the pairwise and $k \operatorname{asinh}(k)$ rows, and for $\delta = \beta - \kappa$ on the linear row (see Appendix B.4 for the profit gaps).

First, I find that the 99% confidence set is empty when the portfolio function is the inverse hyperbolic sine, the square root, or the logarithm of portfolio size. With these concave functions, each extra area would add less scope than the one before. That is, no combination of congestion, scope,

Table 24: Estimated parameter bounds under different instrument setups

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1084, -0.1111]$	—
RD_{it-1}	20	$[-1.1077, -0.2144]$	—
AT_{it-1}	20	$[-1.1088, -0.1444]$	—
PAT_{it-1}	20	$[-1.1106, -0.0966]$	$[0, 12.6250]$
RD_{it-1}, AT_{it-1}	20	$[-1.1109, -0.2052]$	—
RD_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1992]$	$[0, 12.3750]$
AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1453]$	$[0, 12.8750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	10	$[-1.1112, 0.1287]^{*}$	$[0, 15.8750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1992]$	$[0, 12.8750]^{\dagger}$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1160, -0.1847]$	$[0, 12.3750]$

Notes: The table reports 99 percent confidence sets under pairwise interactions among entered areas, add and drop deviations of size two and three, and swap moments, on the full instrument and subset-size grid. The add and drop moments depend on the occupancy cost and on the scope benefit only through their ratio $\rho = \kappa/\beta$. The swap moments do not depend on this ratio, and they are informative about congestion. The table therefore reports α and a set for ρ. For congestion α, the table reports the lowest and the highest value among the jointly feasible points. For ρ, I fix the ratio at each value, search over the remaining parameters, and keep the value when the criterion is non-positive. A dash means that ratio is not identified. Where the ratio is identified, the set includes zero and is bounded above. When the set for the ratio starts at zero, the ratio of zero is kept and negative ratios are rejected. On all instrumented rows except the one with all three instruments and $|M| = 10$, I reject congestion equal to zero. On the same rows, except the one with lagged R&D alone, I also reject positive congestion. The row with all three instruments and $|M| = 10$ keeps both zero and positive congestion. A dagger marks a confidence set that omits a single evaluated point in the interior (12.5 at RD_{it-1}, AT_{it-1}, PAT_{it-1} with $|M| = 20$). An asterisk marks an interval that contains zero. Table 9 reports the no-instrument, lagged-patent, and all-instrument largest-subset rows. Table 25 repeats this grid with add and drop deviations of size two only. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

and occupancy cost satisfies the inequalities at the 99% critical value when the gain from an extra area shrinks with the size of the portfolio.

Second, with a linear portfolio function, one more area always adds the same scope benefit, and the comparisons recover the gap $\delta = \beta - \kappa$ between that benefit and the occupancy cost. In the baseline specification, the set for this gap is $[1.6250, 3.3750]$.

Third, when I multiply portfolio size by its inverse hyperbolic sine, the set is not empty, congestion stays negative, $\alpha < 0$, and $\rho \in [0, 4.1250]$. This upper end cannot be compared directly with the pairwise upper end of 12.3750, because one more area changes $g(k)$ by a different amount under the two functions. I keep pairwise interactions as the baseline, because it is the portfolio function of Section 3 and because its set for ρ in the baseline specification is bounded above.

To conclude, congestion stays negative under every portfolio function for which the set is not empty, and the set is empty under every concave portfolio function.

The case study in Section 5 uses the pairwise row. In this row, the value of one more area rises with the number of areas the firm already occupies, the set for ρ runs from zero to 12.3750, and congestion is negative. The linear row answers a different question, i.e., whether one more area brings more scope benefit than it costs, and since $\delta = \beta - \kappa$ lies in $[1.6250, 3.3750]$, it always does on that row. The simulations, however, change a ratio ρ and a pair value b, and these two objects belong to the pairwise portfolio function, in which the extra scope from one more area is β times the number of areas already occupied.

Table 25: Estimated parameter bounds, add and drop deviations of size two only

Setup	$\lvert M\rvert$	α	$\rho = \kappa/\beta$
No IV	20	$[-1.1050, -0.1106]$	—
RD_{it-1}	20	$[-1.1077, -0.2199]$	—
AT_{it-1}	20	$[-1.1059, -0.1541]$	—
PAT_{it-1}	20	$[-1.1091, -0.1016]$	$[0] \cup [8.1250, 12.6250]$
RD_{it-1}, AT_{it-1}	20	$[-1.1083, -0.2177]$	—
RD_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.2054]$	$[0, 12.3750]$
AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.1449]$	$[0, 12.8750]^{\dagger}$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	10	$[-1.1081, 0.1209]^{*}$	$[0, 15.3750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	20	$[-1.1106, -0.2054]$	$[0, 12.3750]$
RD_{it-1}, AT_{it-1}, PAT_{it-1}	40	$[-1.1128, -0.1964]$	$[0, 12.3750]$

Notes: The table uses the same pairwise-interaction portfolio function and the same instrument and subset-size grid as Table 24, with add and drop deviations of size two only. The add and drop moments depend on the occupancy cost and on the scope benefit only through their ratio $\rho = \kappa/\beta$. The swap moments do not depend on this ratio, and they are informative about congestion. The table therefore reports α and a set for ρ. For congestion α, the table reports the lowest and the highest value among the jointly feasible points. For ρ, I fix the ratio at each value, search over the remaining parameters, and keep the value when the criterion is non-positive. A dash means that ratio is not identified. Where the ratio is identified, the set includes zero and is bounded above. When the set for the ratio starts at zero, the ratio of zero is kept and negative ratios are rejected. On the lagged-patent row, the ratio of zero is kept, negative ratios are rejected, and the positive ratios that are kept form the interval printed beside zero. On all instrumented rows except the one with all three instruments and $|M| = 10$, I reject both zero and positive congestion. That row keeps both zero and positive congestion. The row without instruments also rejects both zero and positive congestion. A dagger marks a confidence set that omits a single evaluated point in the interior (12.5 at AT_{it-1}, PAT_{it-1} with $|M| = 20$). An asterisk marks an interval that contains zero. Table 9 reports the no-instrument, lagged-patent, and all-instrument largest-subset rows under add and drop deviations of size two and three. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

The portfolio function $k\,\text{asinh}(k)$ also gives a non-empty set, with ρ up to 4.1250. However, take again the firm that already occupies five areas. Under this function, a sixth area adds about 3.39β of scope, against five new pairs under $g(k) = \binom{k}{2}$, so a 25 percent cut in ρ removes a different amount of scope under the two functions. For this reason, I build the portfolios in Appendix D from the pairwise increment, $g(k+1) - g(k) = k$. The concave functions are not used in the simulations, since their sets are empty.

C.2 Alternative entry thresholds

As explained in the main text and previous Appendix, I define entry when the firm's text exceeds the threshold $\bar{w} \approx 1\%$ (see Appendix A.3). For the sensitivity analysis, I re-estimate the specification in the last row of Table 9 at $\tilde{w} = 0.5\%$ and at $\tilde{w} = 2\%$. Then, I rebuild rival counts from the new entry thresholds and keep the predicted valuations, the instruments, and the firm-year sample of the main specification. Table 26 reports the 99% sets and the average number of areas a firm occupies in a year.

The first row repeats the last row of Table 9 which uses $\bar{w} = 1\%$. The second row uses $\tilde{w} = 0.5\%$, and in this case the average firm occupies 6.19 areas in a year. This cutoff $\tilde{w} = 0.5\%$ falls among the small leftover shares on the left of Figure 1, so these leftover shares now count toward entry. I still find that an extra rival lowers expected profit. Congestion α lies in $[-1.3366, -0.3148]$, and from further profiling I reject the possibility that $\alpha \geq 0$. However, the occupancy-to-scope ratio ρ is not identified. With those leftover shares now counted, the 99% set also contains values at which

Table 26: Estimated parameter bounds by entry threshold

Threshold $\bar{w}$	Mean k_{it}	α	$\rho = \kappa/\beta$
1%	5.37	$[-1.1160, -0.1847]$	$[0, 12.3750]$
0.5%	6.19	$[-1.3366, -0.3148]$	—
2%	4.34	—	—

Notes: The table reports 99% confidence sets for congestion α and occupancy-to-scope ratio ρ in the specification of the last row of Table 9: pairwise interactions among entered areas, add and drop deviations of size two and three, swap moments, all three instruments, and the largest subset sample ($|M| = 40$). Each row changes the weight threshold $\bar{w}$ in both places where it enters the entry indicator. Patent weights below $\bar{w}$ are set to zero, and a firm enters an area when the average of those weights over its patents in that year is at least $\bar{w}$. Rival counts are rebuilt from the new entry indicators. The firm-year sample, the predicted valuations, and the instruments are those of the main specification. The first row repeats Table 9, and the other two thresholds are the lines at 0.5 and 2 percent in Figure 1. Mean k_{it} is the average number of areas a firm occupies in a year in the estimation sample. On the rows at 1 percent, I further profile each parameter and find that I reject the possibility of $\alpha \geq 0$ and $\rho < 0$. At the 0.5 percent threshold, a negative ratio is kept. A dash in the last column means that ratio is not identified. At the 2 percent threshold, the inequalities found no jointly feasible point, so both columns are empty. Inference uses the self-normalized two-step procedure of Chernozhukov et al. (2019) at significance level $a = 0.01$.

the gain from occupying more areas at once, β, and the cost of one more area, κ, are both close to zero. Near such values, a small change in either one moves their ratio far in either direction. The set therefore keeps negative ratios as well as positive ones, and the negative ratios I find pair a cost just above zero with a gain just below zero.

The third row uses $\bar{w} = 2\%$, and the average firm now only occupies 4.34 areas in a year. This threshold $\bar{w} = 2\%$ line lies above the fourth-largest share of the median patent, about 0.9 percent, and below its third-largest share, about 8 percent (see Figure 2). At this threshold, notice that the fourth area of the median patent does not count toward entry. As for the result, I find that there is no combination of congestion, scope, and occupancy cost that satisfies the inequalities at the 99% critical value, so both columns of the row are empty.

D Details on Numerical Simulation

This appendix provides notes on how I build the portfolios in Section 5 and how I read the cells in the tables. It then details the area score under each exercise. In that section the rival counts are those implied by the portfolios. The same exercises with the observed counts are at the end of this appendix. Let $\mathcal{B}$ be the four semiconductor areas named in that section: memory, lithography and fabrication, packaging, and the transistor.

D.1 Parameters

The exercises use the 99 percent confidence set in the third row of Table 9. That row is the specification with all three instruments and the largest subset sample. The comparisons are the add, drop, and swap moments, and the portfolio term counts pairwise interactions among entered areas. Throughout, $h(N) = \operatorname{asinh}(N)$ and $g(k) = \binom{k}{2}$, as in Section 3. The portfolio coefficient is a pair value $b = \beta$. The per-area cost in an ordinary area is then $\kappa = b\rho$. I evaluate $b \in \{0.25, 1, 4\}$.

For each pair value b, I only use the pairs (α, ρ) that the inequalities accept at $\theta = (\alpha, b, b\rho)$. To find them, I evaluate the criterion on a grid, in which congestion runs from -1.50 up to just below zero in steps of 0.05, and the occupancy-to-scope ratio runs from 0 to 12.375 in steps of 0.25. The grid

also includes the endpoints of the sets in Table 9, -1.1160, -0.1847, and 12.375. As in that table, I keep a pair when the criterion at that θ is zero or negative at the 99 percent level. This leaves 9 pairs at $b = 0.25$, 9 pairs at $b = 1$, and 29 pairs at $b = 4$. Note that these pairs do not fill the rectangle formed by the two intervals in Table 9, since a value of α and a value of ρ that are each inside their interval are not always accepted together.

In the fixed-size columns of Tables 29 and 11, only congestion matters, so I evaluate α from -1.1160 to -0.1847 in steps of 0.02. The extra-area columns, and both margins of Table 12, use the accepted pairs at the pair value of the column. When I halve congestion, I map each accepted α to $\alpha/2$, even when $\alpha/2$ lies outside the confidence set. In every column, the reported range is the minimum and the maximum over the parameters used in that column. Tables 27 and 28 use the same pairs but hold the rival counts at their observed values.

D.2 Building the portfolio

For a bundle S with $k = |S|$ areas, expected profit is

$$\mathbb{E}[\Pi(S)] = \sum_{j \in S} \big(\check{v}_j + \alpha\, h(N_j) - \kappa_j\big) + b\, g(k).$$

I build the bundle in two ways. First, I hold k at the number of areas that the firm occupies in the data. In this case, the portfolio term $b\, g(k)$ is the same for every bundle of that size, and a cost that is the same in every area is paid the same number of times, so the firm simply occupies the k areas with the largest area score $\zeta_j = \check{v}_j + \alpha\, h(N_j) - \kappa_j$. Second, I start from the bundle that the firm actually occupies and allow k to rise. Adding one empty area m then changes expected profit by

$$\Delta_m = \big(\check{v}_m + \alpha\, h(N_m) - \kappa_m\big) + b\, k,$$

since $g(k+1) - g(k) = k$. In an ordinary area, i.e., an area whose cost the policy does not change, $\kappa_m = b\rho$, so $\Delta_m = \check{v}_m + \alpha\, h(N_m) + b(k - \rho)$. The firm adds the empty area with the largest positive increment, raises k by one, and repeats this until no empty area has a positive increment. Areas that the firm already occupies stay occupied, and when two empty areas tie, I add the one with the smaller topic number.

D.3 Rival counts

In the tables of Section 5, the rival counts agree with the simulated portfolios. I compute them year by year. Let $d_{ijt} = 1$ when firm i occupies area j in the current assignment, and let F_{jt} be the number of firms that appear in the patent counts for that area and year but are outside the sample. I keep these outside firms where they are in the data. The rival count of firm i is then the number of these outside firms plus the other sample firms that occupy the area,

$$N_{jt,-i} = F_{jt} + \sum_{i' \neq i} d_{i'jt}.$$

That is, when one firm moves, the rival counts of every firm scored after that move change as well.

The algorithm works as follows. I start from the portfolios in the data and take the firms in the sample one at a time, in the order of the sample. For each firm, I recompute $N_{jt,-i}$ from the current assignment and score the areas again with the area score of the previous subsection. Under the

fixed-size rule, I replace the portfolio with the top k areas if this raises the sum of the area scores, where k is the number of areas in the data. Under the extra-area rule, I rebuild the portfolio from the observed bundle with the one-area increments above, and I replace the current portfolio if the rebuilt one is different. In both cases, the areas that the firm occupies in the data stay occupied. If the firm moves, I update the assignment before I score the next firm, and I repeat the passes until a full pass leaves every portfolio unchanged.

I run this algorithm twice, first with the scores before the policy, starting from the observed portfolios, and then with the policy scores, starting from the assignment that the first run delivers. Each cell of the tables is the change from the first assignment to the second.

To illustrate the algorithm, take three firms in one year, each of which occupies one area, and suppose that there are only two areas, memory and display. In the tables, the penalty from rivals is $\alpha\, h(N_{jt,-i})$. In this example, I set that penalty at 3 for each other firm in the area, and $F_{jt} = 0$. The values and the starting areas are the following.

	Value in memory	Value in display	Starting area
Firm 1	10	6	Memory
Firm 2	10	4	Memory
Firm 3	15	8	Display

In the first pass, firm 1 faces one other firm in memory and one in display, so memory scores $10 - 3 = 7$ and display scores $6 - 3 = 3$, and firm 1 stays in memory. Firm 2 faces the same counts, with memory scoring $10 - 3 = 7$ and display $4 - 3 = 1$, so it also stays in memory. Firm 3, however, faces two other firms in memory and none in display. Memory then scores $15 - 6 = 9$ against 8 for display, so firm 3 moves to memory.

In the second pass, memory contains all three firms. Firm 1 now faces two other firms in memory and none in display, so memory scores $10 - 6 = 4$ against 6 for display, and firm 1 moves to display. After this move, firm 2 faces one other firm in each area, with memory scoring $10 - 3 = 7$ and display $4 - 3 = 1$, so it stays in memory. Firm 3 also faces one other firm in each area, with memory scoring $15 - 3 = 12$ and display $8 - 3 = 5$, so it stays in memory as well. A third pass leaves every portfolio unchanged. In the end, memory contains firms 2 and 3, each of which faces one other firm there, and display contains firm 1, which faces no other firm.

That is, if I scored each firm only once against the starting counts, memory would end up with all three firms. Since the algorithm updates the rival counts after every move, firm 1 moves to display, and memory ends up with firms 2 and 3.

D.4 Reading the cells

Each cell reports, in percentage points, how much an area's share of all occupied slots changes when I move from the situation before the policy to the policy, holding the parameters fixed. The last row reports, also in percentage points, the change in the share of firm-years that add at least one area, which is zero by construction in the fixed-size columns. Note that, in the extra-area columns, occupied areas stay occupied. A negative cell there therefore means that the new slots go to other areas, so that the area accounts for a smaller share of a larger total, even though no firm leaves it.

D.5 A subsidy based on predicted value

The scores in this subsection and in the next one apply both to the tables in Section 5, where the rival count comes from the current portfolios of the other firms, and to Tables 27 and 28, where it is the observed count.

For this policy, I add a quarter of predicted value in every area of $\mathcal{B}$, so that expected profit becomes

$$\mathbb{E}[\Pi'] = \sum_{j \in S} (\check{v}_j + \alpha\, h(N_j) - \kappa_j) + \sum_{j \in S \cap \mathcal{B}} 0.25\, \check{v}_j + b\, g(k).$$

The fixed-size column uses the first construction above, with α on the interval in Table 9. Since k is fixed, the pair value cancels, and the firm ranks the areas by

$$\zeta_j = \begin{cases} 1.25\, \check{v}_j + \alpha\, h(N_j) & j \in \mathcal{B}, \\ \check{v}_j + \alpha\, h(N_j) & j \notin \mathcal{B}, \end{cases}$$

and keeps the top k.

The extra-area columns use the second construction, with the accepted pairs at the pair value b of each column. The firm then adds area m when

$$\Delta_m = \begin{cases} 1.25\, \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \in \mathcal{B}, \\ \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \notin \mathcal{B}. \end{cases}$$

Before the policy, every area uses the second line.

D.6 A lower occupancy cost in semiconductor areas

For this policy, I replace ρ with 0.75ρ in the areas of $\mathcal{B}$ only. That is, the occupancy cost of a semiconductor area falls from $b\rho$ to $0.75\, b\rho$, which raises the score of the area by $0.25\, b\rho$.

In the fixed-size columns, which use the first construction above, the firm ranks the areas by

$$\zeta_j = \begin{cases} \check{v}_j + \alpha\, h(N_j) + 0.25\, b\rho & j \in \mathcal{B}, \\ \check{v}_j + \alpha\, h(N_j) & j \notin \mathcal{B}, \end{cases}$$

and keeps the top k. In the extra-area columns, which use the second construction, the firm adds area m when

$$\Delta_m = \begin{cases} \check{v}_m + \alpha\, h(N_m) + b(k - \rho) + 0.25\, b\rho & m \in \mathcal{B}, \\ \check{v}_m + \alpha\, h(N_m) + b(k - \rho) & m \notin \mathcal{B}. \end{cases}$$

Before the policy, every area uses the second line, and both constructions use the accepted pairs at the pair value of the column. Unlike in the previous case, the fixed-size columns now depend on b, since the cost cut $0.25\, b\rho$ scales with the pair value.

D.7 Halving rival congestion

For this exercise, I replace α with $\alpha/2$ and leave the costs unchanged. Here, each firm is scored only once, against the observed rival counts.

In the fixed-size column, which uses the first construction above, the firm ranks the twenty areas by

$$\zeta_j = \check{v}_j + \frac{\alpha}{2} h(N_j)$$

and keeps the top k, where k is the number of areas in the data. Before the policy, the same ranking uses α. Note that the pair value cancels, since $b(k - \rho)$ is the same for every area once k is fixed.

In the extra-area columns, which use the second construction with the accepted pairs at each b, the firm adds the empty area m when

$$\Delta_m = \check{v}_m + \frac{\alpha}{2} h(N_m) + b(k - \rho) > 0,$$

raises k by one, and repeats. Before the policy, the same condition uses α, and in both cases occupied areas stay occupied.

First, I find that, with the number of areas held fixed, firms move toward the more crowded areas, since these areas gain the most when the congestion penalty is cut in half. The changes are about one to two percentage points. "Display Panel Tech (LCD/OLED)" loses the most, between 1.25 and 1.86 percentage points, while "Integrated Circuit Design, Verification, and Testing" gains the most, between 0.56 and 1.37. Memory, video, and imaging also lose share, and audio and board-level hardware gain share.

The main reason is that the fixed-size column only reshuffles the observed number of areas. Cutting the penalty in half adds $(\alpha/2 - \alpha)\, h(N) = (-\alpha/2)\, h(N)$ to the score of each area, and this addition is larger where $h(N)$ is larger, i.e., where there are more rivals. Display, which is the thinnest area in Table 18, therefore loses the most. Memory, imaging, video, and the transistor lose between 0.24 and 1.37 percentage points, and optics between 0.17 and 0.75. On the other side of the ranking, audio, board-level hardware, clock circuits, and power-supply electronics gain between 0.30 and 1.10 percentage points, while lithography and packaging barely move. That is, since the firm reorders its areas by $\check{v}_j + (\alpha/2)\, h(N_j)$ with k held at the observed count, the gain of one area is the loss of another.

Second, when firms can add areas, the same halving leads to a large expansion when a pair is cheap, and to little expansion when a pair is expensive. The share of firm-years that add at least one area rises by 16.13 to 24.56 percentage points at $b = 0.25$, by 6.08 to 9.40 at $b = 1$, and by 0.18 to 2.39 at $b = 4$. The main reason is that a cheap pair makes the one-area increment positive for many empty areas, and the areas that the firm adds are those that a steep congestion penalty held back. At $b = 0.25$, display gains 1.10 to 1.48 percentage points, and memory, video, and imaging gain between 0.56 and 0.88. Meanwhile, integrated circuit design, audio, and board-level hardware account for a smaller share of the larger total, with losses between 0.58 and 1.28 percentage points.

D.8 A lower occupancy cost in every area

For this exercise, I replace ρ with 0.75ρ in every area, so that $\kappa_j = 0.75\, b\rho$ everywhere, and I leave congestion unchanged. As in the halving exercise, each firm is scored only once, against the observed rival counts.

For two bundles S and S' of the same size, the difference in expected profit after the cut is

$$\mathbb{E}[\Pi'(S)] - \mathbb{E}[\Pi'(S')] = \sum_{j \in S} \big(\check{v}_j + \alpha\, h(N_j)\big) - \sum_{j \in S'} \big(\check{v}_j + \alpha\, h(N_j)\big).$$

Since the new cost is the same in every area, the top-k bundle does not change, and neither do the shares when the number of areas is fixed. For this reason, Table 30 only has extra-area columns.

With the accepted pairs at each b, the firm adds area m when

$$\Delta_m = \check{v}_m + \alpha\, h(N_m) + b\big(k - 0.75\rho\big) > 0.$$

Before the policy, the same condition uses ρ.

First, I find that the expansion is small when a pair is cheap and larger when a pair is expensive. The share of firm-years that add at least one area rises by 0.00 to 1.33 percentage points at $b = 0.25$, by 6.72 to 8.31 at $b = 1$, and by 10.00 to 12.80 at $b = 4$. The main reason is that the relief in every area is $0.25\, b\rho$, which is large when the pair value is high. In the halving exercise, a cheap pair gives the large expansion instead, since halving α does not scale with b.

Second, area shares move by less than one percentage point. At $b = 0.25$, every area share moves by at most 0.09, and at $b = 1$, display gains 0.48 to 0.61. At $b = 4$, display again gains the most, 0.65 to 0.83 percentage points, and memory, video, imaging, and the transistor gain between 0.31 and 0.49. Meanwhile, integrated circuit design, audio, and clock circuits lose between 0.42 and 0.78 percentage points. The areas that gain are those whose score $\check{v} + \alpha\, h(N)$ clears the lower threshold $b(k - 0.75\rho)$, and since occupied areas stay occupied, the three areas that lose share only do so because the new slots go elsewhere. That is, because the cut is the same in every area, two bundles of equal size keep the same ranking, and every change in Table 30 is expansion from the observed portfolio.

D.9 Holding rival counts fixed

Tables 27 and 28 repeat the two semiconductor exercises of Section 5, but score each firm only once, against the rival counts in the data. The halving exercise and the uniform cut in the occupancy cost above use the same construction.

Table 27: Case I: a 25% subsidy based on predicted value, with rival counts held fixed

Area	Fixed size	Extra areas		
		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.84, 0.96]	[0.05, 0.07]	[0.02, 0.02]	[0.00, 0.01]
Audio and Speech-Signal Processing	[−0.16, −0.07]	[−0.03, −0.03]	[−0.03, −0.02]	[−0.02, −0.00]
Semiconductor Lithography and Fabrication Processes	[0.70, 0.86]	[0.03, 0.05]	[0.01, 0.01]	[−0.00, 0.01]
Integrated Circuit Design, Verification, and Testing	[−0.23, −0.18]	[−0.04, −0.03]	[−0.03, −0.02]	[−0.02, −0.00]
PCB and Electronic Hardware with Connectors	[−0.27, −0.15]	[−0.03, −0.02]	[−0.02, −0.02]	[−0.01, 0.00]
Video Recording, Encoding, and Media Processing	[−0.28, −0.19]	[−0.00, 0.01]	[0.01, 0.01]	[−0.00, 0.01]
Semiconductor Packaging and Interconnect Bonding	[0.75, 0.89]	[0.04, 0.05]	[0.01, 0.01]	[0.00, 0.01]
Thin-Film Materials, Deposition and Chemistry	[−0.24, −0.15]	[−0.01, −0.01]	[−0.01, −0.00]	[−0.00, 0.00]
Wireless Base-Station and Mobile Communications	[−0.31, −0.23]	[−0.00, 0.00]	[0.01, 0.02]	[0.00, 0.01]
Clock Generation, Timing Circuits, and Signal Processing	[−0.15, −0.10]	[−0.03, −0.03]	[−0.02, −0.01]	[−0.02, −0.00]
Power-Supply and Power-Management Electronics	[−0.24, −0.18]	[−0.02, −0.02]	[−0.02, −0.02]	[−0.01, −0.00]
Networking Protocols and Data Infrastructure	[−0.20, −0.16]	[−0.01, 0.01]	[0.01, 0.02]	[−0.00, 0.01]
Transistor Gate Structures and Semiconductor Device Physics	[0.70, 0.80]	[0.06, 0.06]	[0.02, 0.02]	[−0.00, 0.01]
Imaging Sensors and Camera Pipelines	[−0.36, −0.24]	[0.01, 0.02]	[0.01, 0.02]	[0.00, 0.01]
Processor Architecture, Caches, Instruction Execution	[−0.15, −0.10]	[−0.02, −0.01]	[−0.00, 0.00]	[−0.01, 0.00]
Display Panel Tech (LCD/OLED)	[−0.22, −0.16]	[0.01, 0.02]	[0.03, 0.03]	[0.00, 0.02]
Rotational Electromechanics and Magnetic Actuation	[−0.24, −0.20]	[−0.02, −0.01]	[−0.02, −0.02]	[−0.00, 0.00]
Optics, Lasers, and Photonics	[−0.33, −0.19]	[−0.01, −0.00]	[−0.00, 0.01]	[0.00, 0.01]
Analog and Differential Amplifier Circuits	[−0.32, −0.23]	[−0.03, −0.02]	[−0.02, −0.01]	[−0.01, −0.00]
Mobile UI and Content Services	[−0.19, −0.12]	[−0.02, −0.02]	[−0.01, −0.01]	[−0.01, −0.00]
Firm-years adding an area	0	[0.83, 0.96]	[0.33, 0.38]	[0.07, 0.21]

Notes: Each area row reports the change, in percentage points, in that area's share of occupied slots, and the last row reports the change in the share of firm-years that add at least one area. Under "Fixed size", every firm keeps the number of areas it occupies in the data, while under "Extra areas" a firm keeps the areas it already occupies and may add new ones. Rival counts stay at their observed values.

Table 28: Case II: a 25% occupancy-to-scope ratio reduction, with rival counts held fixed

Area	Fixed size			Extra areas		
	$b = 0.25$	$b = 1$	$b = 4$	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 2.22]	[4.48, 5.28]	[4.57, 7.10]	[0.00, 0.05]	[0.29, 0.36]	[0.47, 0.60]
Audio and Speech-Signal Processing	[−0.24, 0.00]	[−0.92, −0.72]	[−1.56, −0.75]	[−0.02, 0.00]	[−0.31, −0.26]	[−0.55, −0.47]
Semiconductor Lithography and Fabrication Processes	[0.00, 2.18]	[6.00, 6.16]	[6.15, 6.37]	[−0.00, 0.02]	[0.15, 0.20]	[0.31, 0.41]
Integrated Circuit Design, Verification, and Testing	[−0.33, 0.00]	[−1.18, −0.89]	[−2.01, −0.91]	[−0.02, 0.00]	[−0.33, −0.28]	[−0.61, −0.53]
PCB and Electronic Hardware with Connectors	[−0.25, 0.00]	[−1.11, −0.95]	[−1.69, −0.97]	[−0.01, 0.00]	[−0.23, −0.19]	[−0.35, −0.30]
Video Recording, Encoding, and Media Processing	[−1.32, 0.00]	[−2.11, −2.06]	[−2.13, −1.60]	[−0.00, 0.01]	[0.13, 0.14]	[0.19, 0.22]
Semiconductor Packaging and Interconnect Bonding	[0.00, 2.39]	[6.00, 6.12]	[6.10, 6.31]	[−0.00, 0.03]	[0.11, 0.16]	[0.22, 0.31]
Thin-Film Materials, Deposition and Chemistry	[−0.34, 0.00]	[−0.90, −0.74]	[−1.41, −0.78]	[−0.00, 0.00]	[−0.02, −0.01]	[−0.02, 0.00]
Wireless Base-Station and Mobile Communications	[−0.97, 0.00]	[−1.74, −1.71]	[−1.81, −1.73]	[−0.00, 0.00]	[0.05, 0.06]	[0.05, 0.08]
Clock Generation, Timing Circuits, and Signal Processing	[−0.26, 0.00]	[−1.00, −0.80]	[−1.45, −0.82]	[−0.02, 0.00]	[−0.26, −0.21]	[−0.45, −0.39]
Power-Supply and Power-Management Electronics	[−0.33, 0.00]	[−1.31, −1.11]	[−1.99, −1.13]	[−0.01, 0.00]	[−0.20, −0.17]	[−0.34, −0.31]
Networking Protocols and Data Infrastructure	[−0.91, 0.00]	[−1.60, −1.56]	[−1.64, −1.59]	[0.00, 0.01]	[0.08, 0.09]	[0.10, 0.12]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 2.42]	[4.95, 5.63]	[5.04, 6.88]	[0.00, 0.03]	[0.23, 0.29]	[0.40, 0.51]
Imaging Sensors and Camera Pipelines	[−1.72, 0.00]	[−2.83, −2.62]	[−2.85, −1.79]	[0.00, 0.01]	[0.14, 0.15]	[0.20, 0.24]
Processor Architecture, Caches, Instruction Execution	[−0.33, 0.00]	[−1.08, −0.88]	[−1.47, −0.91]	[−0.01, 0.00]	[−0.07, −0.05]	[−0.16, −0.11]
Display Panel Tech (LCD/OLED)	[−0.26, 0.00]	[−2.85, −2.70]	[−2.91, −2.20]	[−0.00, 0.01]	[0.29, 0.35]	[0.53, 0.62]
Rotational Electromechanics and Magnetic Actuation	[−0.30, 0.00]	[−1.14, −0.95]	[−1.72, −0.99]	[−0.01, 0.00]	[−0.16, −0.13]	[−0.25, −0.21]
Optics, Lasers, and Photonics	[−1.09, 0.00]	[−1.82, −1.67]	[−1.84, −1.20]	[−0.01, 0.00]	[0.06, 0.08]	[0.18, 0.21]
Analog and Differential Amplifier Circuits	[−0.26, 0.00]	[−1.06, −0.86]	[−1.66, −0.90]	[−0.02, 0.00]	[−0.17, −0.14]	[−0.27, −0.24]
Mobile UI and Content Services	[−0.28, 0.00]	[−0.96, −0.80]	[−1.51, −0.83]	[−0.01, 0.00]	[−0.13, −0.10]	[−0.25, −0.19]
Firm-years adding an area	0	0	0	[0.00, 0.50]	[4.54, 5.69]	[8.86, 10.67]

Notes: Each area row reports the change, in percentage points, in that area's share of occupied slots, and the last row reports the change in the share of firm-years that add at least one area. Under "Fixed size", every firm keeps the number of areas it occupies in the data, while under "Extra areas" a firm keeps the areas it already occupies and may add new ones. Rival counts stay at their observed values.

Table 29: Halving rival congestion

Area	Fixed size	Extra areas		
		$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[−1.31, −0.41]	[0.66, 0.88]	[0.26, 0.38]	[0.01, 0.10]
Audio and Speech-Signal Processing	[0.36, 1.10]	[−1.06, −0.77]	[−0.51, −0.34]	[−0.14, −0.01]
Semiconductor Lithography and Fabrication Processes	[−0.08, 0.03]	[0.20, 0.33]	[0.10, 0.17]	[−0.00, 0.05]
Integrated Circuit Design, Verification, and Testing	[0.56, 1.37]	[−1.28, −0.90]	[−0.62, −0.40]	[−0.16, −0.02]
PCB and Electronic Hardware with Connectors	[0.37, 1.01]	[−0.83, −0.58]	[−0.39, −0.26]	[−0.10, −0.01]
Video Recording, Encoding, and Media Processing	[−1.10, −0.24]	[0.56, 0.79]	[0.25, 0.38]	[0.01, 0.10]
Semiconductor Packaging and Interconnect Bonding	[−0.17, 0.04]	[0.07, 0.20]	[0.05, 0.08]	[0.00, 0.03]
Thin-Film Materials, Deposition and Chemistry	[0.22, 0.44]	[−0.18, −0.08]	[−0.07, −0.05]	[−0.03, 0.01]
Wireless Base-Station and Mobile Communications	[−0.28, −0.09]	[0.31, 0.43]	[0.15, 0.21]	[0.00, 0.07]
Clock Generation, Timing Circuits, and Signal Processing	[0.37, 0.78]	[−0.90, −0.66]	[−0.40, −0.29]	[−0.10, −0.01]
Power-Supply and Power-Management Electronics	[0.30, 0.90]	[−0.65, −0.47]	[−0.30, −0.21]	[−0.08, −0.00]
Networking Protocols and Data Infrastructure	[−0.30, −0.16]	[0.41, 0.56]	[0.21, 0.27]	[0.00, 0.08]
Transistor Gate Structures and Semiconductor Device Physics	[−1.01, −0.32]	[0.40, 0.61]	[0.19, 0.30]	[0.01, 0.06]
Imaging Sensors and Camera Pipelines	[−1.37, −0.41]	[0.56, 0.73]	[0.24, 0.36]	[0.01, 0.10]
Processor Architecture, Caches, Instruction Execution	[0.16, 0.47]	[−0.11, −0.09]	[−0.05, −0.03]	[−0.03, −0.00]
Display Panel Tech (LCD/OLED)	[−1.86, −1.25]	[1.10, 1.48]	[0.45, 0.67]	[0.02, 0.17]
Rotational Electromechanics and Magnetic Actuation	[0.32, 0.78]	[−0.60, −0.40]	[−0.27, −0.19]	[−0.07, −0.00]
Optics, Lasers, and Photonics	[−0.75, −0.17]	[0.33, 0.52]	[0.13, 0.22]	[0.01, 0.06]
Analog and Differential Amplifier Circuits	[0.26, 0.74]	[−0.66, −0.52]	[−0.32, −0.22]	[−0.07, −0.01]
Mobile UI and Content Services	[0.24, 0.63]	[−0.24, −0.15]	[−0.11, −0.06]	[−0.02, −0.00]
Firm-years adding an area	0	[16.13, 24.56]	[6.08, 9.40]	[0.18, 2.39]

Table 30: A lower occupancy cost in every area

Area	Extra areas		
	$b = 0.25$	$b = 1$	$b = 4$
Semiconductor Memory Architecture	[0.00, 0.06]	[0.29, 0.34]	[0.39, 0.49]
Audio and Speech-Signal Processing	[−0.05, 0.00]	[−0.45, −0.37]	[−0.63, −0.52]
Semiconductor Lithography and Fabrication Processes	[−0.00, 0.01]	[0.12, 0.15]	[0.20, 0.24]
Integrated Circuit Design, Verification, and Testing	[−0.06, 0.00]	[−0.55, −0.44]	[−0.78, −0.66]
PCB and Electronic Hardware with Connectors	[−0.04, 0.00]	[−0.34, −0.28]	[−0.45, −0.35]
Video Recording, Encoding, and Media Processing	[0.00, 0.05]	[0.28, 0.33]	[0.34, 0.42]
Semiconductor Packaging and Interconnect Bonding	[−0.00, 0.00]	[0.06, 0.07]	[0.11, 0.16]
Thin-Film Materials, Deposition and Chemistry	[−0.00, 0.00]	[−0.07, −0.03]	[−0.08, −0.05]
Wireless Base-Station and Mobile Communications	[−0.00, 0.01]	[0.16, 0.20]	[0.19, 0.24]
Clock Generation, Timing Circuits, and Signal Processing	[−0.05, 0.00]	[−0.37, −0.31]	[−0.52, −0.42]
Power-Supply and Power-Management Electronics	[−0.04, 0.00]	[−0.28, −0.23]	[−0.38, −0.32]
Networking Protocols and Data Infrastructure	[−0.00, 0.03]	[0.21, 0.25]	[0.25, 0.31]
Transistor Gate Structures and Semiconductor Device Physics	[0.00, 0.03]	[0.21, 0.26]	[0.31, 0.39]
Imaging Sensors and Camera Pipelines	[0.00, 0.05]	[0.26, 0.33]	[0.34, 0.41]
Processor Architecture, Caches, Instruction Execution	[−0.03, 0.00]	[−0.05, −0.03]	[−0.09, −0.06]
Display Panel Tech (LCD/OLED)	[0.00, 0.09]	[0.48, 0.61]	[0.65, 0.83]
Rotational Electromechanics and Magnetic Actuation	[−0.03, 0.00]	[−0.25, −0.21]	[−0.33, −0.25]
Optics, Lasers, and Photonics	[0.00, 0.02]	[0.14, 0.20]	[0.28, 0.33]
Analog and Differential Amplifier Circuits	[−0.04, 0.00]	[−0.29, −0.23]	[−0.35, −0.31]
Mobile UI and Content Services	[−0.01, 0.00]	[−0.09, −0.07]	[−0.17, −0.13]
Firm-years adding an area	[0.00, 1.33]	[6.72, 8.31]	[10.00, 12.80]